Chapter 5

NUMERICAL METHODS IN HEAT CONDUCTION

Why Numerical Methods

5-1C Analytical solution methods are limited to *highly simplified problems* in *simple geometries*. The geometry must be such that its entire surface can be described mathematically in a coordinate system by setting the variables equal to constants. Also, heat transfer problems can not be solved analytically if the *thermal conditions* are not sufficiently simple. For example, the consideration of the variation of thermal conductivity with temperature, the variation of the heat transfer coefficient over the surface, or the radiation heat transfer on the surfaces can make it impossible to obtain an analytical solution. Therefore, analytical solutions are limited to problems that are simple or can be simplified with reasonable approximations.

5-2C The *analytical solutions* are based on (1) driving the governing differential equation by performing an energy balance on a differential volume element, (2) expressing the boundary conditions in the proper mathematical form, and (3) solving the differential equation and applying the boundary conditions to determine the integration constants. The *numerical solution* methods are based on replacing the *differential equations* by *algebraic equations*. In the case of the popular *finite difference* method, this is done by replacing the *derivatives* by *differences*. The analytical methods are simple and they provide solution functions applicable to the entire medium, but they are limited to simple problems in simple geometries. The numerical methods are usually more involved and the solutions are obtained at a number of points, but they are applicable to any geometry subjected to any kind of thermal conditions.

5-3C The *energy balance method* is based on *subdividing* the medium into a sufficient number of volume elements, and then applying an *energy balance* on each element. The formal *finite difference method* is based on replacing derivatives by their finite difference approximations. For a specified nodal network, these two methods will result in the same set of equations.

5-4C In practice, we are most likely to use a software package to solve heat transfer problems even when analytical solutions are available since we can do parametric studies very easily and present the results graphically by the press of a button. Besides, once a person is used to solving problems numerically, it is very difficult to go back to solving differential equations by hand.

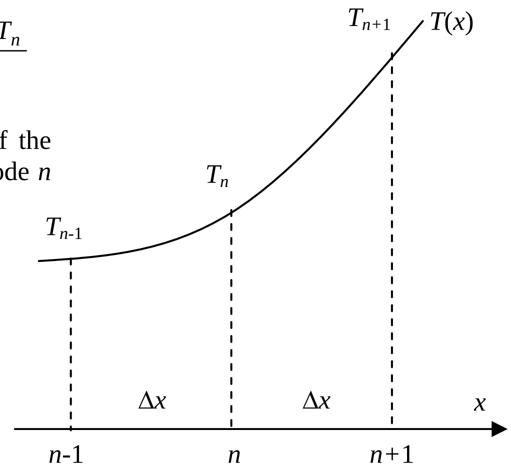
5-5C The experiments will most likely prove engineer B right since an approximate solution of a more realistic model is more accurate than the exact solution of a crude model of an actual problem.

Finite Difference Formulation of Differential Equations

5-6C A point at which the finite difference formulation of a problem is obtained is called a *node*, and all the nodes for a problem constitute the *nodal network*. The region about a node whose properties are represented by the property values at the nodal point is called the *volume element*. The distance between two consecutive nodes is called the *nodal spacing*, and a differential equation whose derivatives are replaced by differences is called a *difference equation*.

5-7 We consider three consecutive nodes n-1, n, and n+1 in a plain wall. Using Eq. 5-6, the first derivative of temperature dT/dx at the midpoints n - 1/2 and n + 1/2 of the sections surrounding the node n can be expressed as

$$\left.\frac{dT}{dx}\right|_{n-\frac{1}{2}} \cong \frac{T_n - T_{n-1}}{\Delta x} \quad \text{and} \quad \left.\frac{dT}{dx}\right|_{n+\frac{1}{2}} \cong \frac{T_{n+1} - T_n}{\Delta x}$$



Noting that second derivative is simply the derivative of the first derivative, the second derivative of temperature at node n can be expressed as

$$\left.\frac{d^2T}{dx^2}\right|_n \cong \frac{\left.\frac{dT}{dx}\right|_{n+\frac{1}{2}} - \left.\frac{dT}{dx}\right|_{n-\frac{1}{2}}}{\Delta x} = \frac{\frac{T_{n+1} - T_n}{\Delta x} - \frac{T_n - T_{n-1}}{\Delta x}}{\Delta x} = \frac{T_{n-1} - 2T_n + T_{n+1}}{\Delta x^2}$$

which is the *finite difference representation* of the *second derivative* at a general internal node n. Note that the second derivative of temperature at a node n is expressed in terms of the temperatures at node n and its two neighboring nodes

5-8 The finite difference formulation of steady two-dimensional heat conduction in a medium with heat generation and constant thermal conductivity is given by

$$\frac{T_{m-1,n} - 2T_{m,n} + T_{m+1,n}}{\Delta x^2} + \frac{T_{m,n-1} - 2T_{m,n} + T_{m,n+1}}{\Delta y^2} + \frac{\dot{g}_{m,n}}{k} = 0$$

in rectangular coordinates. This relation can be modified for the three-dimensional case by simply adding another index j to the temperature in the z direction, and another difference term for the z direction as

$$\frac{T_{m-1,n,j} - 2T_{m,n,j} + T_{m+1,n,j}}{\Delta x^2} + \frac{T_{m,n-1,j} - 2T_{m,n,j} + T_{m,n+1,j}}{\Delta y^2} + \frac{T_{m,n,j-1} - 2T_{m,n,j} + T_{m,n,j+1}}{\Delta z^2} + \frac{\dot{g}_{m,n,j}}{k} = 0$$

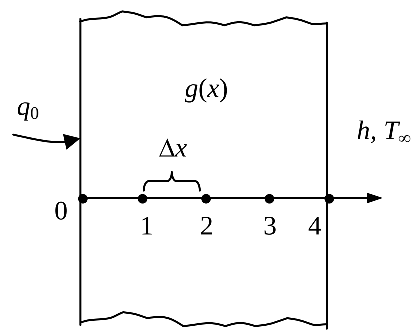
5-9 A plane wall with variable heat generation and constant thermal conductivity is subjected to uniform heat flux $\dot{q}_0$ at the left (node 0) and convection at the right boundary (node 4). Using the finite difference form of the 1st derivative, the finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is steady since there is no indication of change with time. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Thermal conductivity is constant and there is nonuniform heat generation in the medium. **4** Radiation heat transfer is negligible.

Analysis The boundary conditions at the left and right boundaries can be expressed analytically as

At $x = 0$: $$-k\frac{dT(0)}{dx} = q_0$$

At $x = L$: $$-k\frac{dT(L)}{dx} = h[T(L) - T_\infty]$$



Replacing derivatives by differences using values at the closest nodes, the finite difference form of the 1st derivative of temperature at the boundaries (nodes 0 and 4) can be expressed as

$$\left.\frac{dT}{dx}\right|_{\text{left, m} = 0} \cong \frac{T_1 - T_0}{\Delta x} \quad \text{and} \quad \left.\frac{dT}{dx}\right|_{\text{right, m} = 4} \cong \frac{T_4 - T_3}{\Delta x}$$

Substituting, the finite difference formulation of the boundary nodes become

At $x = 0$: $$-k\frac{T_1 - T_0}{\Delta x} = q_0$$

At $x = L$: $$-k\frac{T_4 - T_3}{\Delta x} = h[T_4 - T_\infty]$$

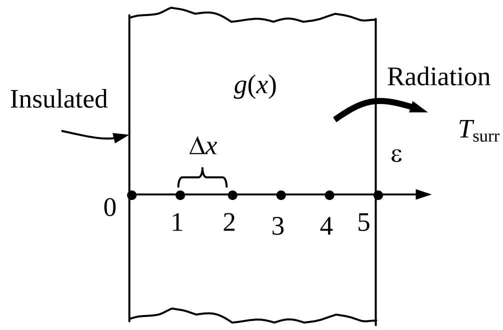
5-10 A plane wall with variable heat generation and constant thermal conductivity is subjected to insulation at the left (node 0) and radiation at the right boundary (node 5). Using the finite difference form of the 1st derivative, the finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is steady since there is no indication of change with time. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Thermal conductivity is constant and there is nonuniform heat generation in the medium. **4** Convection heat transfer is negligible.

Analysis The boundary conditions at the left and right boundaries can be expressed analytically as

At $x = 0$: $$-k\frac{dT(0)}{dx} = 0 \quad \text{or} \quad \frac{dT(0)}{dx} = 0$$

At $x = L$: $$-k\frac{dT(L)}{dx} = \varepsilon\sigma[T^4(L) - T_{surr}^4]$$



Replacing derivatives by differences using values at the closest nodes, the finite difference form of the 1st derivative of temperature at the boundaries (nodes 0 and 5) can be expressed as

$$\left.\frac{dT}{dx}\right|_{\text{left, m = 0}} \cong \frac{T_1 - T_0}{\Delta x} \quad \text{and} \quad \left.\frac{dT}{dx}\right|_{\text{right, m = 5}} \cong \frac{T_5 - T_4}{\Delta x}$$

Substituting, the finite difference formulation of the boundary nodes become

At $x = 0$: $$-k\frac{T_1 - T_0}{\Delta x} = 0 \quad \text{or} \quad T_1 = T_0$$

At $x = L$: $$-k\frac{T_5 - T_4}{\Delta x} = \varepsilon\sigma[T_5^4 - T_{surr}^4]$$

One-Dimensional Steady Heat Conduction

5-11C The finite difference form of a heat conduction problem by the *energy balance method* is obtained by *subdividing* the medium into a sufficient number of volume elements, and then applying an *energy balance* on each element. This is done by first *selecting* the nodal points (or nodes) at which the temperatures are to be determined, and then *forming elements* (or control volumes) over the nodes by drawing lines through the midpoints between the nodes. The properties *at the node* such as the temperature and the rate of heat generation represent the *average* properties of the element. The temperature is assumed to vary *linearly* between the nodes, especially when expressing heat conduction between the elements using Fourier's law.

5-12C In the energy balance formulation of the finite difference method, it is recommended that all heat transfer at the boundaries of the volume element be assumed to be *into* the volume element even for steady heat conduction. This is a valid recommendation even though it seems to violate the conservation of energy principle since the assumed direction of heat conduction at the surfaces of the volume elements has no effect on the formulation, and some heat conduction terms turn out to be negative.

5-13C In the finite difference formulation of a problem, an insulated boundary is best handled by replacing the insulation by a mirror, and treating the node on the boundary as an *interior* node. Also, a thermal symmetry line and an insulated boundary are treated the same way in the finite difference formulation.

5-14C A node on an insulated boundary can be treated as an interior node in the finite difference formulation of a plane wall by replacing the insulation on the boundary by a *mirror*, and considering the reflection of the medium as its extension. This way the node next to the boundary node appears on both sides of the boundary node because of symmetry, converting it into an interior node.

5-15C In a medium in which the finite difference formulation of a general interior node is given in its simplest form as

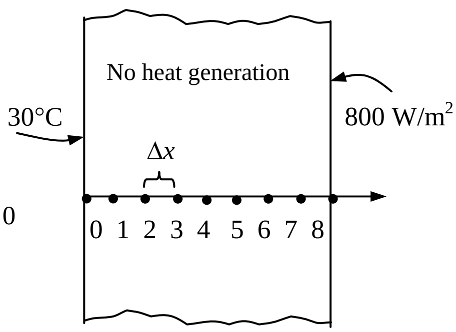
$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0$$

(*a*) heat transfer in this medium is **steady**, (*b*) it is **one-dimensional**, (*c*) there **is** heat generation, (*d*) the nodal spacing is **constant**, and (*e*) the thermal conductivity is **constant**.

5-16 A plane wall with no heat generation is subjected to specified temperature at the left (node 0) and heat flux at the right boundary (node 8). The finite difference formulation of the boundary nodes and the finite difference formulation for the rate of heat transfer at the left boundary are to be determined.

Assumptions **1** Heat transfer through the wall is given to be steady, and the thermal conductivity to be constant. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** There is no heat generation in the medium.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become



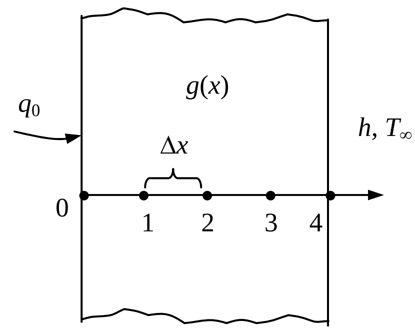
Left boundary node: $T_0 = 30$

Right boundary node: $kA\frac{T_7 - T_8}{\Delta x} + \dot{q}_0 A = 0$ or $k\frac{T_7 - T_8}{\Delta x} + 800 = 0$

Heat transfer at left surface: $\dot{Q}_{\text{left surface}} + kA\frac{T_1 - T_0}{\Delta x} = 0$

5-17 A plane wall with variable heat generation and constant thermal conductivity is subjected to uniform heat flux $\dot{q}_0$ at the left (node 0) and convection at the right boundary (node 4). The finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is given to be steady, and the thermal conductivity to be constant. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Radiation heat transfer is negligible.



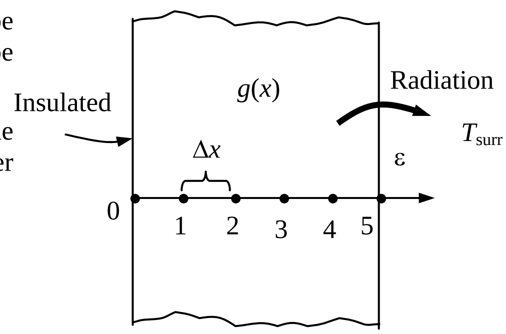
Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

Left boundary node: $$\dot{q}_0 A + kA\frac{T_1 - T_0}{\Delta x} + \dot{g}_0(A\Delta x/2) = 0$$

Right boundary node: $$kA\frac{T_3 - T_4}{\Delta x} + hA(T_\infty - T_4) + \dot{g}_4(A\Delta x/2) = 0$$

5-18 A plane wall with variable heat generation and constant thermal conductivity is subjected to insulation at the left (node 0) and radiation at the right boundary (node 5). The finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer is negligible.

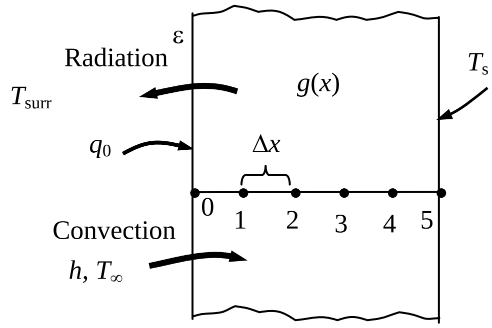


Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

Left boundary node: $$kA\frac{T_1 - T_0}{\Delta x} + \dot{g}_0(A\Delta x/2) = 0$$

Right boundary node: $$\varepsilon\sigma A(T_{surr}^4 - T_5^4) + kA\frac{T_4 - T_5}{\Delta x} + \dot{g}_5(A\Delta x/2) = 0$$

5-19 A plane wall with variable heat generation and constant thermal conductivity is subjected to combined convection, radiation, and heat flux at the left (node 0) and specified temperature at the right boundary (node 5). The finite difference formulation of the left boundary node (node 0) and the finite difference formulation for the rate of heat transfer at the right boundary (node 5) are to be determined.



Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional. **2** The thermal conductivity is given to be constant.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

Left boundary node (all temperatures are in K):

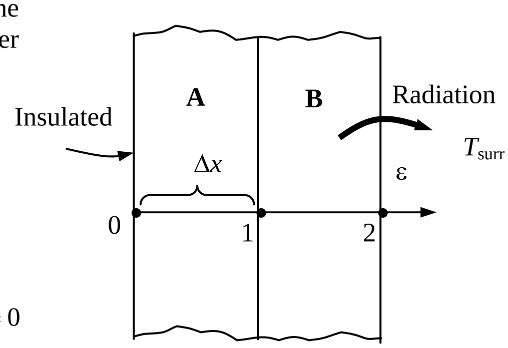
$$\varepsilon\sigma A(T_{surr}^4 - T_0^4) + hA(T_\infty - T_0) + kA\frac{T_1 - T_0}{\Delta x} + \dot{q}_0 A + \dot{g}_0 (A\Delta x/2) = 0$$

Heat transfer at right surface: $\dot{Q}_{right\ surface} + kA\dfrac{T_4 - T_5}{\Delta x} + \dot{g}_5 (A\Delta x/2) = 0$

5-20 A composite plane wall consists of two layers A and B in perfect contact at the interface where node 1 is. The wall is insulated at the left (node 0) and subjected to radiation at the right boundary (node 2). The complete finite difference formulation of this problem is to be obtained.

Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer is negligible. **3** There is no heat generation.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

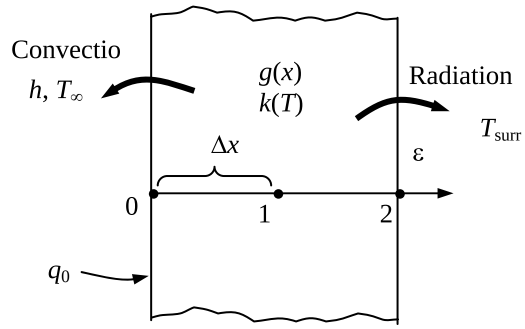


Node 0 (at left boundary): $k_A A\dfrac{T_1 - T_0}{\Delta x} = 0 \;\rightarrow\; T_1 = T_0$

Node 1 (at the interface): $k_A A\dfrac{T_0 - T_1}{\Delta x} + k_B A\dfrac{T_2 - T_1}{\Delta x} = 0$

Node 2 (at right boundary): $\varepsilon\sigma A(T_{surr}^4 - T_2^4) + k_B A\dfrac{T_1 - T_2}{\Delta x} = 0$

5-21 A plane wall with variable heat generation and variable thermal conductivity is subjected to specified heat flux $\dot{q}_0$ and convection at the left boundary (node 0) and radiation at the right boundary (node 5). The complete finite difference formulation of this problem is to be obtained.



Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional, and the thermal conductivity and heat generation to be variable. **2** Convection heat transfer at the right surface is negligible.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

Node 0 (at left boundary): $$\dot{q}_0 A + hA(T_\infty - T_0) + k_0 A \frac{T_1 - T_0}{\Delta x} + \dot{g}_0 (A\Delta x / 2) = 0$$

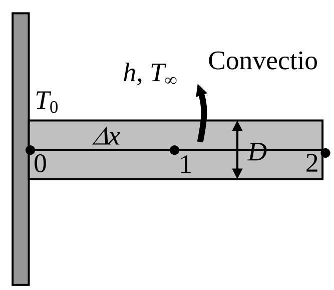
Node 1 (at the mid plane): $$k_1 A \frac{T_0 - T_1}{\Delta x} + k_1 A \frac{T_2 - T_1}{\Delta x} + \dot{g}_1 (A\Delta x / 2) = 0$$

Node 2 (at right boundary): $$\varepsilon\sigma A(T_{surr}^4 - T_2^4) + k_2 A \frac{T_1 - T_2}{\Delta x} + \dot{g}_2 (A\Delta x / 2) = 0$$

5-22 A pin fin with negligible heat transfer from its tip is considered. The complete finite difference formulation for the determination of nodal temperatures is to be obtained.

Assumptions **1** Heat transfer through the pin fin is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer coefficient is constant and uniform. **3** Radiation heat transfer is negligible. **4** Heat loss from the fin tip is given to be negligible.

Analysis The nodal network consists of 3 nodes, and the base temperature T_0 at node 0 is specified. Therefore, there are two unknowns T_1 and T_2, and we need two equations to determine them. Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become



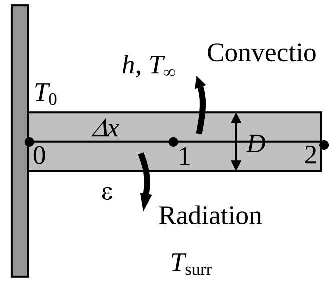
Node 1 (at midpoint): $$kA \frac{T_0 - T_1}{\Delta x} + kA \frac{T_2 - T_1}{\Delta x} + hp\Delta x(T_\infty - T_1) = 0$$

Node 2 (at fin tip): $$kA \frac{T_1 - T_2}{\Delta x} + h(p\Delta x / 2)(T_\infty - T_2) = 0$$

where $A = \pi D^2 / 4$ is the cross-sectional area and $p = \pi D$ is the perimeter of the fin.

5-23 A pin fin with negligible heat transfer from its tip is considered. The complete finite difference formulation for the determination of nodal temperatures is to be obtained.

Assumptions **1** Heat transfer through the pin fin is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer coefficient is constant and uniform. **3** Heat loss from the fin tip is given to benegligible.



Analysis The nodal network consists of 3 nodes, and the base temperature T_0 at node 0 is specified. Therefore, there are two unknowns T_1 and T_2, and we need two equations to determine them. Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

Node 1 (at midpoint): $$kA\frac{T_0 - T_1}{\Delta x} + kA\frac{T_2 - T_1}{\Delta x} + h(p\Delta x/2)(T_\infty - T_1) + \varepsilon\sigma A(T_{\text{surr}}^4 - T_1^4) = 0$$

Node 2 (at fin tip): $$kA\frac{T_1 - T_2}{\Delta x} + h(p\Delta x/2)(T_\infty - T_2) + \varepsilon\sigma(p\Delta x/2)(T_{\text{surr}}^4 - T_2^4) = 0$$

where $A = \pi D^2/4$ is the cross-sectional area and $p = \pi D$ is the perimeter of the fin.

5-24 A uranium plate is subjected to insulation on one side and convection on the other. The finite difference formulation of this problem is to be obtained, and the nodal temperatures under steady conditions are to be determined.

Assumptions **1** Heat transfer through the wall is steady since there is no indication of change with time. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Thermal conductivity is constant. **4** Radiation heat transfer is negligible.

Properties The thermal conductivity is given to be k = 28 W/m·°C.

Analysis The number of nodes is specified to be M = 6. Then the nodal spacing Δx becomes

$$\Delta x = \frac{L}{M-1} = \frac{0.05\text{ m}}{6\text{-}1} = 0.01\text{ m}$$

This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations to determine them uniquely. Node 0 is on insulated boundary, and thus we can treat it as an interior note by using the mirror image concept. Nodes 1, 2, 3, and 4 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0, \quad \text{for } m = 0, 1, 2, 3, \text{and } 4$$

Finally, the finite difference equation for node 5 on the right surface subjected to convection is obtained by applying an energy balance on the half volume element about node 5 and taking the direction of all heat transfers to be towards the node under consideration:

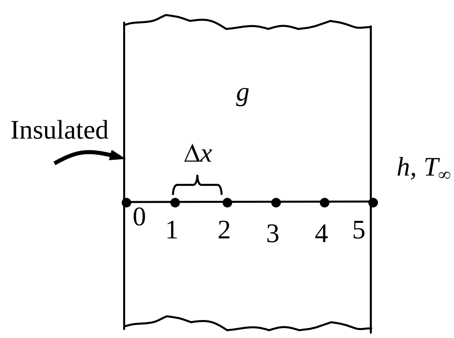
Node 0 (Left surface - insulated): $\dfrac{T_1 - 2T_0 + T_1}{\Delta x^2} + \dfrac{\dot{g}}{k} = 0$

Node 1 (interior): $\dfrac{T_0 - 2T_1 + T_2}{\Delta x^2} + \dfrac{\dot{g}}{k} = 0$

Node 2 (interior): $\dfrac{T_1 - 2T_2 + T_3}{\Delta x^2} + \dfrac{\dot{g}}{k} = 0$

Node 3 (interior): $\dfrac{T_2 - 2T_3 + T_4}{\Delta x^2} + \dfrac{\dot{g}}{k} = 0$

Node 4 (interior): $\dfrac{T_3 - 2T_4 + T_5}{\Delta x^2} + \dfrac{\dot{g}}{k} = 0$



Node 5 (right surface - convection): $h(T_\infty - T_5) + k\dfrac{T_4 - T_5}{\Delta x} + \dot{g}(\Delta x/2) = 0$

where $\Delta x = 0.01\text{ m}$, $\dot{g} = 6\times10^5\text{ W/m}^3$, $k = 28\text{ W/m}\cdot°\text{C}$, $h = 60\text{ W/m}^2\cdot°\text{C}$, and $T_\infty = 30°\text{C}$. This system of 6 equations with six unknown temperatures constitute the finite difference formulation of the problem.

(*b*) The 6 nodal temperatures under steady conditions are determined by solving the 6 equations above simultaneously with an equation solver to be

$\boldsymbol{T_0}$ **=556.8°C,** $\boldsymbol{T_1}$ **=555.7°C,** $\boldsymbol{T_2}$ **=552.5°C,** $\boldsymbol{T_3}$ **=547.1°C,** $\boldsymbol{T_4}$ **=539.6°C, and** $\boldsymbol{T_5}$ **=530.0°C**

Discussion This problem can be solved analytically by solving the differential equation as described in Chap. 2, and the analytical (exact) solution can be used to check the accuracy of the numerical solution above.

5-25 A long triangular fin attached to a surface is considered. The nodal temperatures, the rate of heat transfer, and the fin efficiency are to be determined numerically using 6 equally spaced nodes.

Assumptions **1** Heat transfer along the fin is given to be steady, and the temperature along the fin to vary in the x direction only so that $T = T(x)$. **2** Thermal conductivity is constant.

Properties The thermal conductivity is given to be k = 180 W/m·°C. The emissivity of the fin surface is 0.9.

Analysis The fin length is given to be L = 5 cm, and the number of nodes is specified to be M = 6. Therefore, the nodal spacing Δx is

$$\Delta x = \frac{L}{M-1} = \frac{0.05\text{ m}}{6\text{-}1} = 0.01\text{ m}$$

The temperature at node 0 is given to be T_0 = 200°C, and the temperatures at the remaining 5 nodes are to be determined. Therefore, we need to have 5 equations to determine them uniquely. Nodes 1, 2, 3, and 4 are interior nodes, and the finite difference formulation for a *general interior node m* is obtained by applying an energy balance on the volume element of this node. Noting that heat transfer is steady and there is no heat generation in the fin and assuming heat transfer to be into the medium from all sides, the energy balance can be expressed as

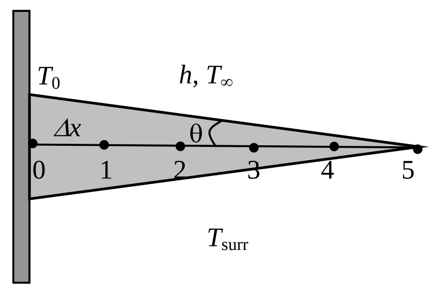
$$\sum_{\text{all sides}} \dot{Q} = 0 \quad \rightarrow \quad kA_{\text{left}} \frac{T_{m-1} - T_m}{\Delta x} + kA_{\text{right}} \frac{T_{m+1} - T_m}{\Delta x} + hA_{\text{conv}}(T_\infty - T_m) + \varepsilon\sigma A_{\text{surface}}[T_{\text{surr}}^4 - (T_m + 273)^4\} = 0$$

Note that heat transfer areas are different for each node in this case, and using geometrical relations, they can be expressed as

$$A_{\text{left}} = (\text{Height} \times \text{width})_{@\, m-1/2} = 2w\left[L - (m - 1/2)\Delta x\right]\tan\theta$$

$$A_{\text{right}} = (\text{Height} \times \text{width})_{@\, m+1/2} = 2w\left[L - (m + 1/2)\Delta x\right]\tan\theta$$

$$A_{\text{surface}} = 2 \times \text{Length} \times \text{width} = 2w(\Delta x / \cos\theta)$$



Substituting,

$$2kw[L - (m - 0.5)\Delta x]\tan\theta \frac{T_{m-1} - T_m}{\Delta x} + 2kw[L - (m + 0.5)\Delta x]\tan\theta \frac{T_{m+1} - T_m}{\Delta x}$$
$$+ 2w(\Delta x / \cos\theta)\{h(T_\infty - T_m) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_m + 273)^4]\} = 0$$

Dividing each term by $2kwL\tan\theta / \Delta x$ gives

$$\left[1 - (m - 1/2)\frac{\Delta x}{L}\right](T_{m-1} - T_m) + \left[1 - (m + 1/2)\frac{\Delta x}{L}\right](T_{m+1} - T_m) + \frac{h(\Delta x)^2}{kL\sin\theta}(T_\infty - T_m) + \frac{\varepsilon\sigma(\Delta x)^2}{kL\sin\theta}[T_{\text{surr}}^4 - (T_m + 273)^4] = 0$$

Substituting,

m = 1: $$\left[1 - 0.5\frac{\Delta x}{L}\right](T_0 - T_1) + \left[1 - 1.5\frac{\Delta x}{L}\right](T_2 - T_1) + \frac{h(\Delta x)^2}{kL\sin\theta}(T_\infty - T_1) + \frac{\varepsilon\sigma(\Delta x)^2}{kL\sin\theta}[T_{\text{surr}}^4 - (T_1 + 273)^4] = 0$$

m = 2: $$\left[1 - 1.5\frac{\Delta x}{L}\right](T_1 - T_2) + \left[1 - 2.5\frac{\Delta x}{L}\right](T_3 - T_2) + \frac{h(\Delta x)^2}{kL\sin\theta}(T_\infty - T_2) + \frac{\varepsilon\sigma(\Delta x)^2}{kL\sin\theta}[T_{\text{surr}}^4 - (T_2 + 273)^4] = 0$$

m = 3: $$\left[1 - 2.5\frac{\Delta x}{L}\right](T_2 - T_3) + \left[1 - 3.5\frac{\Delta x}{L}\right](T_4 - T_3) + \frac{h(\Delta x)^2}{kL\sin\theta}(T_\infty - T_3) + \frac{\varepsilon\sigma(\Delta x)^2}{kL\sin\theta}[T_{\text{surr}}^4 - (T_3 + 273)^4] = 0$$

m = 4: $$\left[1 - 3.5\frac{\Delta x}{L}\right](T_3 - T_4) + \left[1 - 4.5\frac{\Delta x}{L}\right](T_5 - T_4) + \frac{h(\Delta x)^2}{kL\sin\theta}(T_\infty - T_4) + \frac{\varepsilon\sigma(\Delta x)^2}{kL\sin\theta}[T_{\text{surr}}^4 - (T_4 + 273)^4] = 0$$

An energy balance on the 5$^{\text{th}}$ node gives the 5$^{\text{th}}$ equation,

m = 5: $$2k\frac{\Delta x}{2}\tan\theta\frac{T_4 - T_5}{\Delta x} + 2h\frac{\Delta x/2}{\cos\theta}(T_\infty - T_5) + 2\varepsilon\sigma\frac{\Delta x/2}{\cos\theta}[T_{\text{surr}}^4 - (T_5 + 273)^4] = 0$$

Solving the 5 equations above simultaneously for the 5 unknown nodal temperatures gives

$$\boldsymbol{T_1 = 177.0°C,\quad T_2 = 174.1°C,\quad T_3 = 171.2°C,\quad T_4 = 168.4°C,\quad \text{and} \quad T_5 = 165.5°C}$$

(*b*) The total rate of heat transfer from the fin is simply the sum of the heat transfer from each volume element to the ambient, and for w = 1 m it is determined from

$$\dot{Q}_{\text{fin}} = \sum_{m=0}^{5} \dot{Q}_{\text{element},\,m} = \sum_{m=0}^{5} hA_{\text{surface},\,m}(T_m - T_\infty) + \sum_{m=0}^{5} \varepsilon\sigma A_{\text{surface},\,m}[(T_m + 273)^4 - T_{\text{surr}}^4]$$

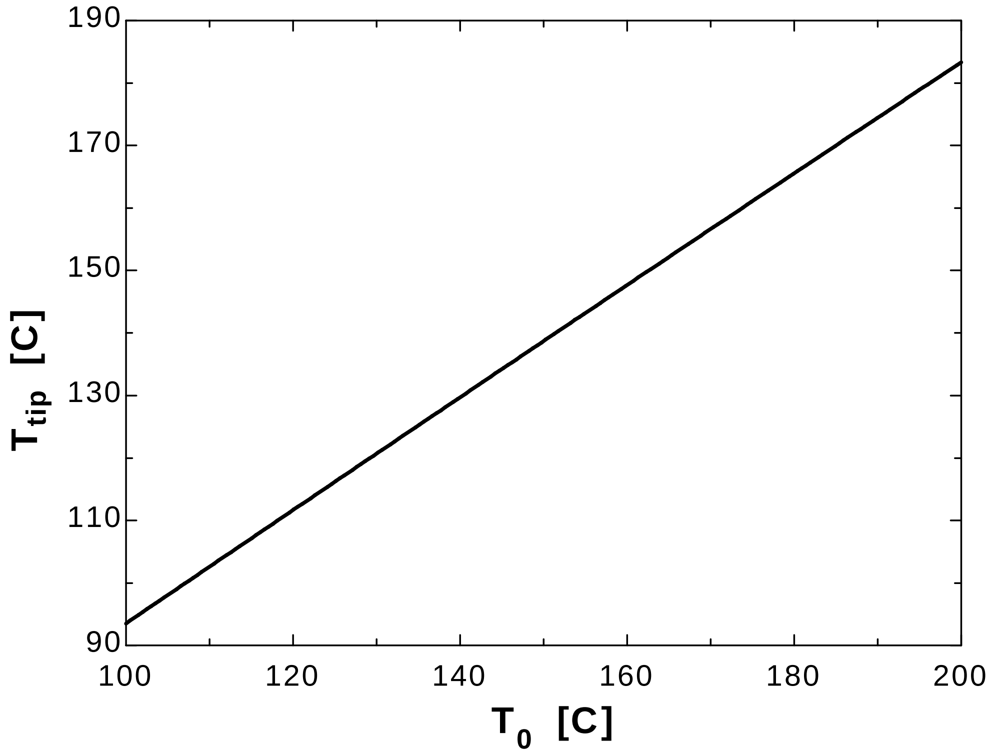
Noting that the heat transfer surface area is $w\Delta x/\cos\theta$ for the boundary nodes 0 and 5, and twice as large for the interior nodes 1, 2, 3, and 4, we have

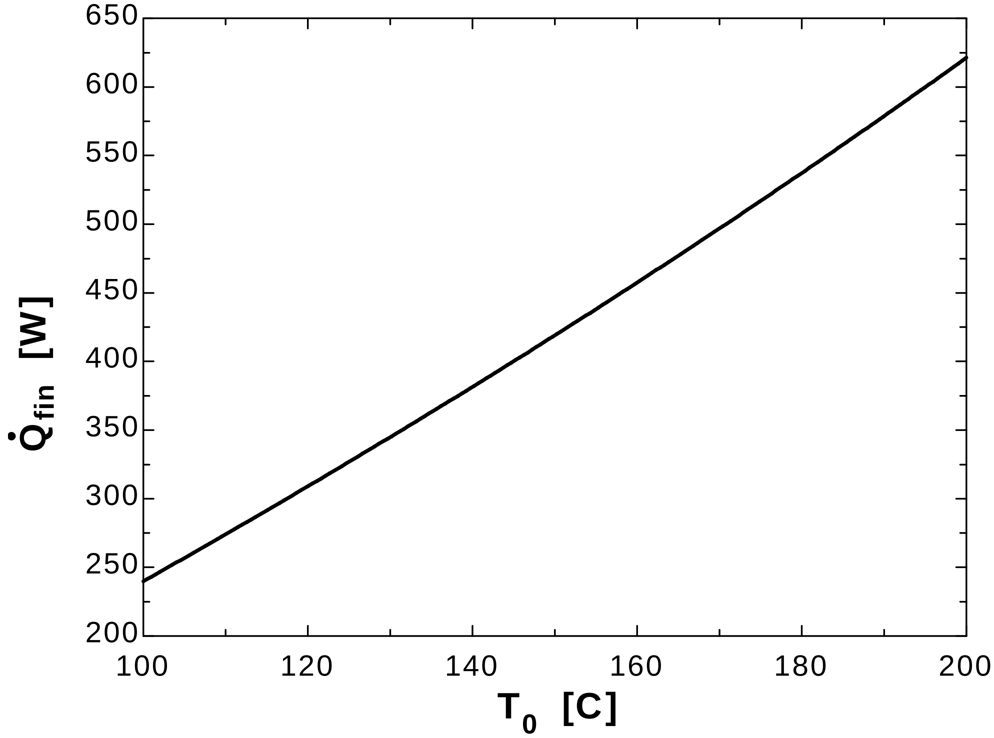
$$\begin{aligned}
\dot{Q}_{\text{fin}} &= h\frac{w\Delta x}{\cos\theta}\left[(T_0 - T_\infty) + 2(T_1 - T_\infty) + 2(T_2 - T_\infty) + 2(T_3 - T_\infty) + 2(T_4 - T_\infty) + (T_5 - T_\infty)\right] \\
&\quad + \varepsilon\sigma\frac{w\Delta x}{\cos\theta}\{[(T_0 + 273)^4 - T_{\text{surr}}^4] + 2[(T_1 + 273)^4 - T_{\text{surr}}^4] + 2[(T_2 + 273)^4 - T_{\text{surr}}^4] + 2[(T_3 + 273)^4 - T_{\text{surr}}^4] \\
&\quad + 2[(T_4 + 273)^4 - T_{\text{surr}}^4] + [(T_5 + 273)^4 - T_{\text{surr}}^4]\} \\
&= \mathbf{533\ W}
\end{aligned}$$

5-26 "!PROBLEM 5-26"

```
"GIVEN"
k=180 "[W/m-C]"
L=0.05 "[m]"
b=0.01 "[m]"
w=1 "[m]"
"T_0=180 [C], parameter to be varied"
T_infinity=25 "[C]"
h=25 "[W/m^2-C]"
T_surr=290 "[K]"
M=6
epsilon=0.9
tan(theta)=(0.5*b)/L
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"

"ANALYSIS"
"(a)"
DELTAx=L/(M-1)
"Using the finite difference method, the five equations for the temperatures at 5 nodes are
determined to be"
(1-0.5*DELTAx/L)*(T_0-T_1)+(1-1.5*DELTAx/L)*(T_2-
T_1)+(h*DELTAx^2)/(k*L*sin(theta))*(T_infinity-
T_1)+(epsilon*sigma*DELTAX^2)/(k*L*sin(theta))*(T_surr^4-(T_1+273)^4)=0 "for mode 1"

(1-1.5*DELTAx/L)*(T_1-T_2)+(1-2.5*DELTAx/L)*(T_3-
T_2)+(h*DELTAx^2)/(k*L*sin(theta))*(T_infinity-
T_2)+(epsilon*sigma*DELTAX^2)/(k*L*sin(theta))*(T_surr^4-(T_2+273)^4)=0 "for mode 2"

(1-2.5*DELTAx/L)*(T_2-T_3)+(1-3.5*DELTAx/L)*(T_4-
T_3)+(h*DELTAx^2)/(k*L*sin(theta))*(T_infinity-
T_3)+(epsilon*sigma*DELTAX^2)/(k*L*sin(theta))*(T_surr^4-(T_3+273)^4)=0 "for mode 3"

(1-3.5*DELTAx/L)*(T_3-T_4)+(1-4.5*DELTAx/L)*(T_5-
T_4)+(h*DELTAx^2)/(k*L*sin(theta))*(T_infinity-
T_4)+(epsilon*sigma*DELTAX^2)/(k*L*sin(theta))*(T_surr^4-(T_4+273)^4)=0 "for mode 4"

2*k*DELTAx/2*tan(theta)*(T_4-T_5)/DELTAx+2*h*(0.5*DELTAx)/cos(theta)*(T_infinity-
T_5)+2*epsilon*sigma*(0.5*DELTAx)/cos(theta)*(T_surr^4-(T_5+273)^4)=0 "for mode 5"

T_tip=T_5

"(b)"
Q_dot_fin=C+D "where"

C=h*(w*DELTAx)/cos(theta)*((T_0-T_infinity)+2*(T_1-T_infinity)+2*(T_2-T_infinity)+2*(T_3-
T_infinity)+2*(T_4-T_infinity)+(T_5-T_infinity))

D=epsilon*sigma*(w*DELTAx)/cos(theta)*(((T_0+273)^4-T_surr^4)+2*((T_1+273)^4-
T_surr^4)+2*((T_2+273)^4-T_surr^4)+2*((T_3+273)^4-T_surr^4)+2*((T_4+273)^4-
T_surr^4)+((T_5+273)^4-T_surr^4))
```

T_0 [C]	**T_{tip} [C]**	**Q_{fin} [W]**
100	93.51	239.8
105	98.05	256.8
110	102.6	274
115	107.1	291.4
120	111.6	309
125	116.2	326.8
130	120.7	344.8
135	125.2	363.1
140	129.7	381.5
145	134.2	400.1
150	138.7	419
155	143.2	438.1
160	147.7	457.5
165	152.1	477.1
170	156.6	496.9
175	161.1	517
180	165.5	537.3
185	170	557.9
190	174.4	578.7
195	178.9	599.9
200	183.3	621.2





650
600
550
500
450
400
350
300
250
200
100
120
140
160
180
200
$\dot{Q}_{fin}$ [W]
T_0 [C]

5-27 A plate is subjected to specified temperature on one side and convection on the other. The finite difference formulation of this problem is to be obtained, and the nodal temperatures under steady conditions as well as the rate of heat transfer through the wall are to be determined.

Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional. **2** Thermal conductivity is constant. **3** There is no heat generation. **4** Radiation heat transfer is negligible.

Properties The thermal conductivity is given to be k = 2.3 W/m·°C.

Analysis The nodal spacing is given to be Δx=0.1 m. Then the number of nodes M becomes

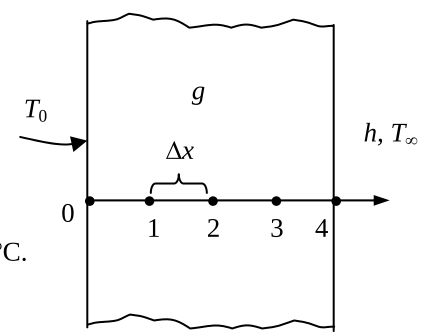
$$M = \frac{L}{\Delta x} + 1 = \frac{0.4\text{ m}}{0.1\text{ m}} + 1 = 5$$

The left surface temperature is given to be T_0 =80°C. This problem involves 4 unknown nodal temperatures, and thus we need to have 4 equations to determine them uniquely. Nodes 1, 2, and 3 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0 \rightarrow T_{m-1} - 2T_m + T_{m+1} = 0 \quad (\text{since } \dot{g} = 0), \quad \text{for } m = 0, 1, 2, \text{ and } 3$$

The finite difference equation for node 4 on the right surface subjected to convection is obtained by applying an energy balance on the half volume element about node 4 and taking the direction of all heat transfers to be towards the node under consideration:

$$\begin{aligned}
&\text{Node 1 (interior):} && T_0 - 2T_1 + T_2 = 0 \\
&\text{Node 2 (interior):} && T_1 - 2T_2 + T_3 = 0 \\
&\text{Node 3 (interior):} && T_2 - 2T_3 + T_4 = 0 \\
&\text{Node 4 (right surface - convection):} && h(T_\infty - T_4) + k\frac{T_3 - T_4}{\Delta x} = 0
\end{aligned}$$



where $\Delta x = 0.1\text{ m},\ k = 2.3\text{ W/m}\cdot\text{°C},\ h = 24\text{ W/m}^2\cdot\text{°C, and } T_\infty = 15\text{°C}.$

The system of 4 equations with 4 unknown temperatures constitute the finite difference formulation of the problem.

(b) The nodal temperatures under steady conditions are determined by solving the 4 equations above simultaneously with an equation solver to be

$$\boldsymbol{T_1 = 66.9°C, \quad T_2 = 53.8°C, \quad T_3 = 40.7°C, \text{ and } \quad T_4 = 27.6°C}$$

(c) The rate of heat transfer through the wall is simply convection heat transfer at the right surface,

$$\dot{Q}_{\text{wall}} = \dot{Q}_{\text{conv}} = hA(T_4 - T_\infty) = (24\text{ W/m}^2.\text{°C})(20\text{ m}^2)(27.56\text{-}15)\text{°C} = \mathbf{6029\ W}$$

Discussion This problem can be solved analytically by solving the differential equation as described in Chap. 2, and the analytical (exact) solution can be used to check the accuracy of the numerical solution above.

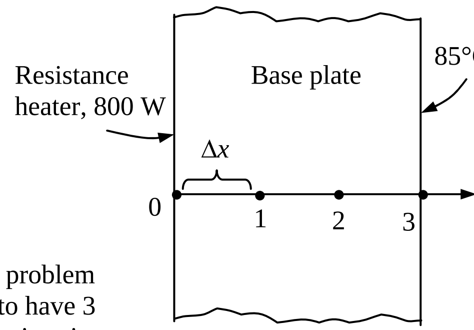
5-28 A plate is subjected to specified heat flux on one side and specified temperature on the other. The finite difference formulation of this problem is to be obtained, and the unknown surface temperature under steady conditions is to be determined.

Assumptions **1** Heat transfer through the base plate is given to be steady. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** There is no heat generation in the plate. **4** Radiation heat transfer is negligible. **5** The entire heat generated by the resistance heaters is transferred through the plate.

Properties The thermal conductivity is given to be k = 20 W/m·°C.

Analysis The nodal spacing is given to be Δx=0.2 cm. Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{0.6\text{ cm}}{0.2\text{ cm}} + 1 = 4$$



The right surface temperature is given to be T_3 =85°C. This problem involves 3 unknown nodal temperatures, and thus we need to have 3 equations to determine them uniquely. Nodes 1, 2, and 3 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0 \rightarrow T_{m-1} - 2T_m + T_{m+1} = 0 \quad (\text{since } \dot{g} = 0), \quad \text{for } m = 1 \text{ and } 2$$

The finite difference equation for node 0 on the left surface subjected to uniform heat flux is obtained by applying an energy balance on the half volume element about node 0 and taking the direction of all heat transfers to be towards the node under consideration:

$$\begin{aligned}
&\text{Node 4 (right surface - convection)}: && \dot{q}_0 + k\frac{T_1 - T_0}{\Delta x} = 0 \\
&\text{Node 1 (interior)}: && T_0 - 2T_1 + T_2 = 0 \\
&\text{Node 2 (interior)}: && T_1 - 2T_2 + T_3 = 0
\end{aligned}$$

where $\Delta x = 0.2\text{ cm},\ k = 20\text{ W/m}\cdot°\text{C},\ T_3 = 85°\text{C, and } \dot{q}_0 = \dot{Q}_0 / A = (800\text{W})/(0.0160\text{ m}^2) = 50{,}000\text{ W/m}^2$.

The system of 3 equations with 3 unknown temperatures constitute the finite difference formulation of the problem.

(*b*) The nodal temperatures under steady conditions are determined by solving the 3 equations above simultaneously with an equation solver to be

$\boldsymbol{T_0}$ **=100°C**, T_1 =95°C, and T_2 =90°C

Discussion This problem can be solved analytically by solving the differential equation as described in Chap. 2, and the analytical (exact) solution can be used to check the accuracy of the numerical solution above.

5-29 A plate is subjected to specified heat flux and specified temperature on one side, and no conditions on the other. The finite difference formulation of this problem is to be obtained, and the temperature of the other side under steady conditions is to be determined.

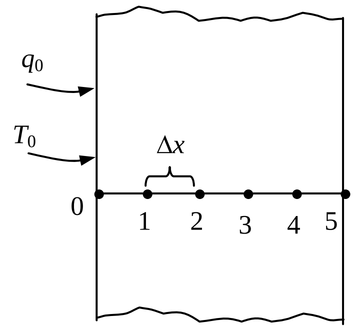
Assumptions **1** Heat transfer through the plate is given to be steady and one-dimensional. **2** There is no heat generation in the plate.

Properties The thermal conductivity is given to be k = 2.5 W/m·°C.

Analysis The nodal spacing is given to be Δx=0.06 m.

Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{0.3\text{ m}}{0.06\text{ m}} + 1 = 6$$



Nodes 1, 2, 3, and 4 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0 \rightarrow T_{m+1} - 2T_m + T_{m-1} = 0 \quad (\text{since } \dot{g} = 0), \quad \text{for } m = 1, 2, 3, \text{and } 4$$

The finite difference equation for node 0 on the left surface is obtained by applying an energy balance on the half volume element about node 0 and taking the direction of all heat transfers to be towards the node under consideration,

$$\dot{q}_0 + k\frac{T_1 - T_0}{\Delta x} = 0 \quad \longrightarrow \quad 700\text{ W/m}^2 + (2.5\text{ W/m}\cdot°\text{C})\frac{T_1 - 60°\text{C}}{0.06\text{ m}} = 0 \quad \longrightarrow \quad T_1 = 43.2°\text{C}$$

Other nodal temperatures are determined from the general interior node relation as follows:

$$\begin{aligned}
m &= 1: & T_2 &= 2T_1 - T_0 = 2\times 43.2 - 60 = 26.4°\text{C} \\
m &= 2: & T_3 &= 2T_2 - T_1 = 2\times 26.4 - 43.2 = 9.6°\text{C} \\
m &= 3: & T_4 &= 2T_3 - T_2 = 2\times 9.6 - 26.4 = -7.2°\text{C} \\
m &= 4: & T_5 &= 2T_4 - T_3 = 2\times(-7.2) - 9.6 = \mathbf{-24°C}
\end{aligned}$$

Therefore, the temperature of the other surface will be –24°C

Discussion This problem can be solved analytically by solving the differential equation as described in Chap. 2, and the analytical (exact) solution can be used to check the accuracy of the numerical solution above.

5-30E A large plate lying on the ground is subjected to convection and radiation. Finite difference formulation is to be obtained, and the top and bottom surface temperatures under steady conditions are to be determined.

Assumptions **1** Heat transfer through the plate is given to be steady and one-dimensional. **2** There is no heat generation in the plate and the soil. **3** Thermal contact resistance at plate-soil interface is negligible.

Properties The thermal conductivity of the plate and the soil are given to be k_{plate} = 7.2 Btu/h·ft·°F and k_{soil} = 0.49 Btu/h·ft·°F.

Analysis The nodal spacing is given to be Δx_1=1 in. in the plate, and be Δx_2=0.6 ft in the soil. Then the number of nodes becomes

$$M = \left(\frac{L}{\Delta x}\right)_{\text{plate}} + \left(\frac{L}{\Delta x}\right)_{\text{soil}} + 1 = \frac{5\text{ in}}{1\text{ in}} + \frac{3\text{ ft}}{0.6\text{ ft}} + 1 = 11$$

The temperature at node 10 (bottom of thee soil) is given to be T_{10} =50°F. Nodes 1, 2, 3, and 4 in the plate and 6, 7, 8, and 9 in the soil are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0 \rightarrow T_{m-1} - 2T_m + T_{m+1} = 0 \quad (\text{since } \dot{g} = 0)$$

The finite difference equation for node 0 on the left surface and node 5 at the interface are obtained by applying an energy balance on their respective volume elements and taking the direction of all heat transfers to be towards the node under consideration:

Node 0 (top surface): $h(T_\infty - T_0) + \varepsilon\sigma[T_{sky}^4 - (T_0 + 460)^4] + k_{\text{plate}} \dfrac{T_1 - T_0}{\Delta x_1} = 0$

Node 1 (interior): $T_0 - 2T_1 + T_2 = 0$

Node 2 (interior): $T_1 - 2T_2 + T_3 = 0$

Node 3 (interior): $T_2 - 2T_3 + T_4 = 0$

Node 4 (interior): $T_3 - 2T_4 + T_5 = 0$

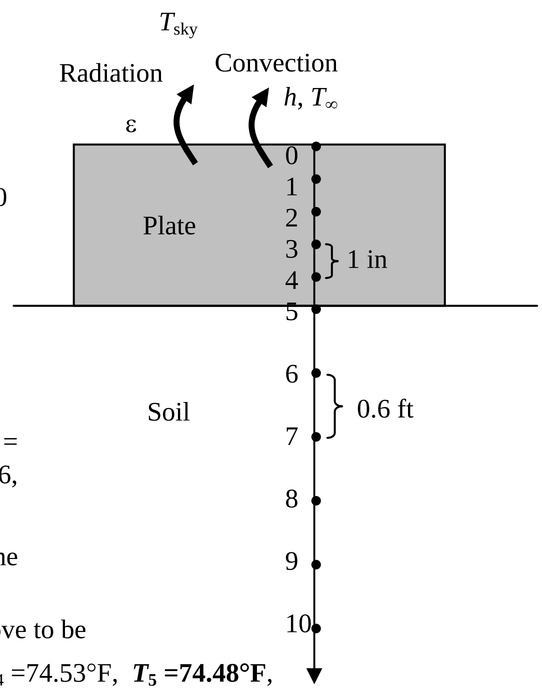
Node 5 (interface): $k_{\text{plate}} \dfrac{T_4 - T_5}{\Delta x_1} + k_{\text{soil}} \dfrac{T_6 - T_5}{\Delta x_2} = 0$

Node 6 (interior): $T_5 - 2T_6 + T_7 = 0$

Node 7 (interior): $T_6 - 2T_7 + T_8 = 0$

Node 8 (interior): $T_7 - 2T_8 + T_9 = 0$

Node 9 (interior): $T_8 - 2T_9 + T_{10} = 0$



where Δx_1=1/12 ft, Δx_2=0.6 ft, k_{plate} = 7.2 Btu/h·ft·°F, k_{soil} = 0.49 Btu/h·ft·°F, h = 3.5 Btu/h·ft^2·°F, T_{sky} =510 R, ε = 0.6, $T_\infty = 80$°F, and T_{10} =50°F.

This system of 10 equations with 10 unknowns constitute the finite difference formulation of the problem.

(*b*) The temperatures are determined by solving equations above to be

T_0 = 74.71°F, T_1 =74.67°F, T_2 =74.62°F, T_3 =74.58°F, T_4 =74.53°F, **T_5 =74.48°F**,

T_6 =69.6°F, T_7 =64.7°F, T_8 =59.8°F, T_9 =54.9°F

Discussion Note that the plate is essentially isothermal at about 74.6°F. Also, the temperature in each layer varies linearly and thus we could solve this problem by considering 3 nodes only (one at the interface and two at the boundaries).

5-31E A large plate lying on the ground is subjected to convection from its exposed surface. The finite difference formulation of this problem is to be obtained, and the top and bottom surface temperatures under steady conditions are to be determined.

Assumptions **1** Heat transfer through the plate is given to be steady and one-dimensional. **2** There is no heat generation in the plate and the soil. **3** The thermal contact resistance at the plate-soil interface is negligible. **4** Radiation heat transfer is negligible.

Properties The thermal conductivity of the plate and the soil are given to be k_{plate} = 7.2 Btu/h·ft·°F and k_{soil} = 0.49 Btu/h·ft·°F.

Analysis The nodal spacing is given to be Δx_1=1 in. in the plate, and be Δx_2=0.6 ft in the soil. Then the number of nodes becomes

$$M = \left(\frac{L}{\Delta x}\right)_{plate} + \left(\frac{L}{\Delta x}\right)_{soil} + 1 = \frac{5\text{ in}}{1\text{ in}} + \frac{3\text{ ft}}{0.6\text{ ft}} + 1 = 11$$

The temperature at node 10 (bottom of thee soil) is given to be T_{10} =50°F. Nodes 1, 2, 3, and 4 in the plate and 6, 7, 8, and 9 in the soil are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0 \rightarrow T_{m-1} - 2T_m + T_{m+1} = 0 \quad (\text{since } \dot{g} = 0)$$

The finite difference equation for node 0 on the left surface and node 5 at the interface are obtained by applying an energy balance on their respective volume elements and taking the direction of all heat transfers to be towards the node under consideration:

Node 0 (top surface): $h(T_\infty - T_0) + k_{plate}\dfrac{T_1 - T_0}{\Delta x_1} = 0$

Node 1 (interior): $T_0 - 2T_1 + T_2 = 0$

Node 2 (interior): $T_1 - 2T_2 + T_3 = 0$

Node 3 (interior): $T_2 - 2T_3 + T_4 = 0$

Node 4 (interior): $T3 - 2T_4 + T_5 = 0$

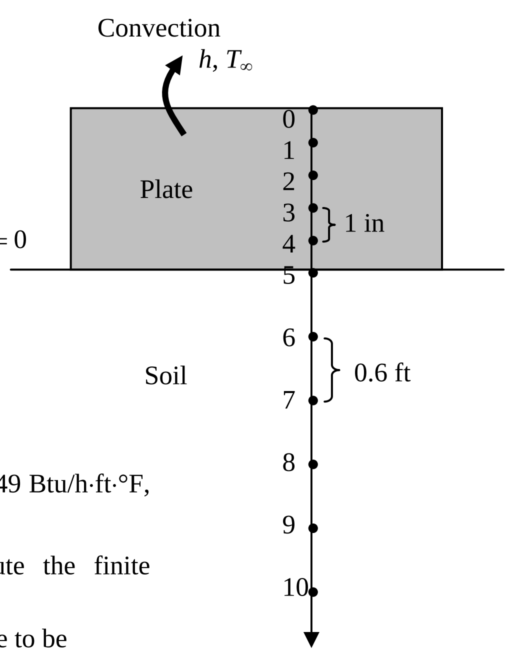
Node 5 (interface): $k_{plate}\dfrac{T_4 - T_5}{\Delta x_1} + k_{soil}\dfrac{T_6 - T_5}{\Delta x_2} = 0$

Node 6 (interior): $T_5 - 2T_6 + T_7 = 0$

Node 7 (interior): $T_6 - 2T_7 + T_8 = 0$

Node 8 (interior): $T_7 - 2T_8 + T_9 = 0$

Node 9 (interior): $T_8 - 2T_9 + T_{10} = 0$



where Δx_1=1/12 ft, Δx_2=0.6 ft, k_{plate} = 7.2 Btu/h·ft·°F, k_{soil} = 0.49 Btu/h·ft·°F, h = 3.5 Btu/h·ft^2·°F, $T_\infty = 80$°F, and T_{10} =50°F.

This system of 10 equations with 10 unknowns constitute the finite difference formulation of the problem.

(*b*) The temperatures are determined by solving equations above to be

$\mathbf{T_0}$ **=78.67°F**, T_1 =78.62°F, T_2 =78.57°F, T_3 =78.51°F, T_4 =78.46°F, $\mathbf{T_5}$ **=78.41°F**,

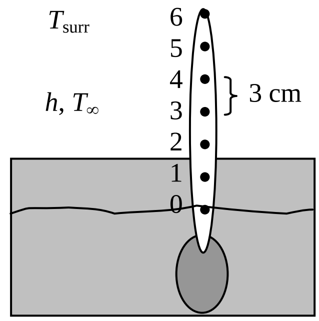
T_6 =72.7°F, T_7 =67.0°F, T_8 =61.4°F, T_9 =55.7°F

Discussion Note that the plate is essentially isothermal at about 78.6°F. Also, the temperature in each layer varies linearly and thus we could solve this problem by considering 3 nodes only (one at the interface and two at the boundaries).

5-32 The handle of a stainless steel spoon partially immersed in boiling water loses heat by convection and radiation. The finite difference formulation of the problem is to be obtained, and the tip temperature of the spoon as well as the rate of heat transfer from the exposed surfaces are to be determined.

Assumptions **1** Heat transfer through the handle of the spoon is given to be steady and one-dimensional. **2** Thermal conductivity and emissivity are constant. **3** Convection heat transfer coefficient is constant and uniform.

Properties The thermal conductivity and emissivity are given to be k = 15.1 W/m·°C and ε = 0.8.



Analysis The nodal spacing is given to be Δx=3 cm. Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{18\,\text{cm}}{3\,\text{cm}} + 1 = 7$$

The base temperature at node 0 is given to be T_0 = 95°C. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations to determine them uniquely. Nodes 1, 2, 3, 4, and 5 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$kA\frac{T_{m-1}-T_m}{\Delta x} + kA\frac{T_{m+1}-T_m}{\Delta x} + h(p\Delta x)(T_\infty - T_m) + \varepsilon\sigma(p\Delta x)[T_{\text{surr}}^4 - (T_m + 273)^4] = 0$$

or $T_{m-1} - 2T_m + T_{m+1} + h(p\Delta x^2 / kA)(T_\infty - T_m) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_m + 273)^4] = 0$, m = 1,2,3,4,5

The finite difference equation for node 6 at the fin tip is obtained by applying an energy balance on the half volume element about node 6. Then,

$$m\text{= 1: } T_0 - 2T_1 + T_2 + h(p\Delta x^2 / kA)(T_\infty - T_1) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_1 + 273)^4] = 0$$

$$m\text{= 2: } T_1 - 2T_2 + T_3 + h(p\Delta x^2 / kA)(T_\infty - T_2) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_2 + 273)^4] = 0$$

$$m\text{= 3: } T_2 - 2T_3 + T_4 + h(p\Delta x^2 / kA)(T_\infty - T_3) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_3 + 273)^4] = 0$$

$$m\text{= 4: } T_3 - 2T_4 + T_5 + h(p\Delta x^2 / kA)(T_\infty - T_4) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_4 + 273)^4] = 0$$

$$m\text{= 5: } T_4 - 2T_5 + T_6 + h(p\Delta x^2 / kA)(T_\infty - T_5) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_5 + 273)^4] = 0$$

$$\text{Node 6: } kA\frac{T_5 - T_6}{\Delta x} + h(p\Delta x/2 + A)(T_\infty - T_6) + \varepsilon\sigma(p\Delta x/2 + A)[T_{\text{surr}}^4 - (T_6 + 273)^4] = 0$$

where $\Delta x = 0.03\,\text{m},\ k = 15.1\,\text{W/m}\cdot°\text{C}, \varepsilon = 0.6,\ T_\infty = 25°\text{C}, T_0 = 95°\text{C},\ T_{surr} = 295\,\text{K},\ h = 13\,\text{W/m}^2\cdot°\text{C}$

and $A = (1\,\text{cm})(0.2\,\text{cm}) = 0.2\,\text{cm}^2 = 0.2\times10^{-4}\ \text{m}^2$ and $p = 2(1 + 0.2\,\text{cm}) = 2.4\,\text{cm} = 0.024\,\text{m}$

The system of 6 equations with 6 unknowns constitute the finite difference formulation of the problem.

(b) The nodal temperatures under steady conditions are determined by solving the 6 equations above simultaneously with an equation solver to be

T_1 =49.0°C, T_2 = 33.0°C, T_3 =27.4°C, T_4 =25.5°C, T_5 =24.8°C, and $\boldsymbol{T_6}$ **=24.6°C**,

(c) The total rate of heat transfer from the spoon handle is simply the sum of the heat transfer from each nodal element, and is determined from

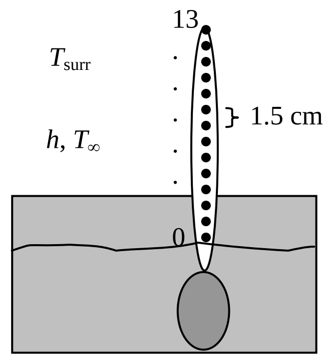
$$\dot{Q}_{\text{fin}} = \sum_{m=0}^{6}\dot{Q}_{\text{element},m} = \sum_{m=0}^{6} hA_{\text{surface},m}(T_m - T_\infty) + \sum_{m=0}^{6}\varepsilon\sigma A_{\text{surface},m}[(T_m + 273)^4 - T_{\text{surr}}^4] = \mathbf{0.92\ \ W}$$

where $A_{\text{surface, m}}$ =$p\Delta x$/2 for node 0, $A_{\text{surface, m}}$ =$p\Delta x$/2+A for node 6, and $A_{\text{surface, m}}$ =$p\Delta x$ for other nodes.

5-33 The handle of a stainless steel spoon partially immersed in boiling water loses heat by convection and radiation. The finite difference formulation of the problem for all nodes is to be obtained, and the temperature of the tip of the spoon as well as the rate of heat transfer from the exposed surfaces of the spoon are to be determined.

Assumptions **1** Heat transfer through the handle of the spoon is given to be steady and one-dimensional. **2** The thermal conductivity and emissivity are constant. **3** Heat transfer coefficient is constant and uniform.

Properties The thermal conductivity and emissivity are given to be k = 15.1 W/m·°C and ε = 0.8.



Analysis The nodal spacing is given to be Δx=1.5 cm. Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{18\,\text{cm}}{1.5\,\text{cm}} + 1 = 13$$

The base temperature at node 0 is given to be T_0 = 95°C. This problem involves 12 unknown nodal temperatures, and thus we need to have 6 equations to determine them uniquely. Nodes 1 through 12 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$kA\frac{T_{m-1} - T_m}{\Delta x} + kA\frac{T_{m+1} - T_m}{\Delta x} + h(p\Delta x)(T_\infty - T_m) + \varepsilon\sigma(p\Delta x)[T_{\text{surr}}^4 - (T_m + 273)^4] = 0$$

or $\quad T_{m-1} - 2T_m + T_{m+1} + h(p\Delta x^2 / kA)(T_\infty - T_m) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_m + 273)^4] = 0\,,\quad m = 1\text{-}12$

The finite difference equation for node 6 at the fin tip is obtained by applying an energy balance on the half volume element about node 13. Then,

$$m=1:\; T_0 - 2T_1 + T_2 + h(p\Delta x^2 / kA)(T_\infty - T_1) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_1 + 273)^4] = 0$$

$$m=2:\; T_1 - 2T_2 + T_3 + h(p\Delta x^2 / kA)(T_\infty - T_2) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_2 + 273)^4] = 0$$

$$m=3:\; T_2 - 2T_3 + T_4 + h(p\Delta x^2 / kA)(T_\infty - T_3) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_3 + 273)^4] = 0$$

$$m=4:\; T_3 - 2T_4 + T_5 + h(p\Delta x^2 / kA)(T_\infty - T_4) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_4 + 273)^4] = 0$$

$$m = 5:\quad T_4 - 2T_5 + T_6 + h(p\Delta x^2 / kA)(T_\infty - T_5) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_5 + 273)^4] = 0$$
$$m = 6:\quad T_5 - 2T_6 + T_7 + h(p\Delta x^2 / kA)(T_\infty - T_6) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_6 + 273)^4] = 0$$
$$m = 7:\quad T_6 - 2T_7 + T_8 + h(p\Delta x^2 / kA)(T_\infty - T_7) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_7 + 273)^4] = 0$$
$$m = 8:\quad T_7 - 2T_8 + T_9 + h(p\Delta x^2 / kA)(T_\infty - T_8) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_8 + 273)^4] = 0$$
$$m = 9:\quad T_8 - 2T_9 + T_{10} + h(p\Delta x^2 / kA)(T_\infty - T_9) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_9 + 273)^4] = 0$$
$$m = 10:\quad T_9 - 2T_{10} + T_{11} + h(p\Delta x^2 / kA)(T_\infty - T_{10}) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_{10} + 273)^4] = 0$$
$$m = 11:\quad T_{10} - 2T_{11} + T_{12} + h(p\Delta x^2 / kA)(T_\infty - T_{11}) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_{11} + 273)^4] = 0$$
$$m = 12:\quad T_{11} - 2T_{12} + T_{13} + h(p\Delta x^2 / kA)(T_\infty - T_{12}) + \varepsilon\sigma(p\Delta x^2 / kA)[T_{\text{surr}}^4 - (T_{12} + 273)^4] = 0$$

$$\text{Node 13:}\; kA\frac{T_{12} - T_{13}}{\Delta x} + h(p\Delta x/2 + A)(T_\infty - T_{13}) + \varepsilon\sigma(p\Delta x/2 + A)[T_{\text{surr}}^4 - (T_{13} + 273)^4] = 0$$

where $\quad \Delta x = 0.03\,\text{m},\; k = 15.1\,\text{W/m}\cdot{}^\circ\text{C},\; \varepsilon = 0.6,\; T_\infty = 25^\circ\text{C}, T_0 = 95^\circ\text{C},\; T_{surr} = 295\,\text{K},\; h = 13\,\text{W/m}^2\cdot{}^\circ\text{C}$

$$A = (1\,\text{cm})(0.2\,\text{cm}) = 0.2\,\text{cm}^2 = 0.2\times 10^{-4}\ \text{m}^2 \quad\text{and}\quad p = 2(1 + 0.2\,\text{cm}) = 2.4\,\text{cm} = 0.024\,\text{m}$$

(b) The nodal temperatures under steady conditions are determined by solving the equations above to be

T_1 =65.2°C, T_2 = 48.1°C, T_3 =38.2°C, T_4 =32.4°C, T_5 =29.1°C, T_6 =27.1°C, T_7 =26.0°C,

T_8 =25.3°C, T_9 = 24.9°C, T_{10} =24.7°C, T_{11} =24.6°C, T_{12} =24.5°C, and $\boldsymbol{T_{13}}$ **=24.5°C**, (*c*) The total rate of heat transfer from the spoon handle is the sum of the heat transfer from each element,

$$\dot{Q}_{\text{fin}} = \sum_{m=0}^{13} \dot{Q}_{\text{element},m} = \sum_{m=0}^{13} hA_{\text{surface},m}(T_m - T_\infty) + \sum_{m=0}^{13} \varepsilon\sigma A_{\text{surface},m}[(T_m + 273)^4 - T_{\text{surr}}^4] = \mathbf{0.83\ W}$$

where $A_{\text{surface, m}} = p\Delta x/2$ for node 0, $A_{\text{surface, m}} = p\Delta x/2 + \text{A}$ for node 13, and $A_{\text{surface, m}} = p\Delta x$ for other nodes.

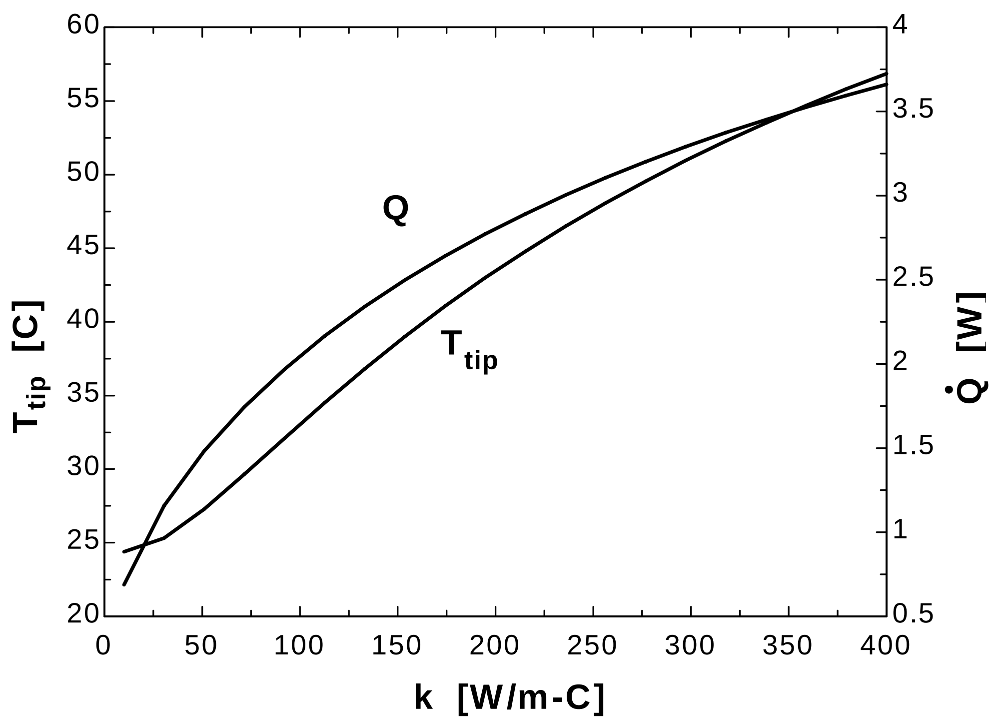
5-34 "!PROBLEM 5-34"

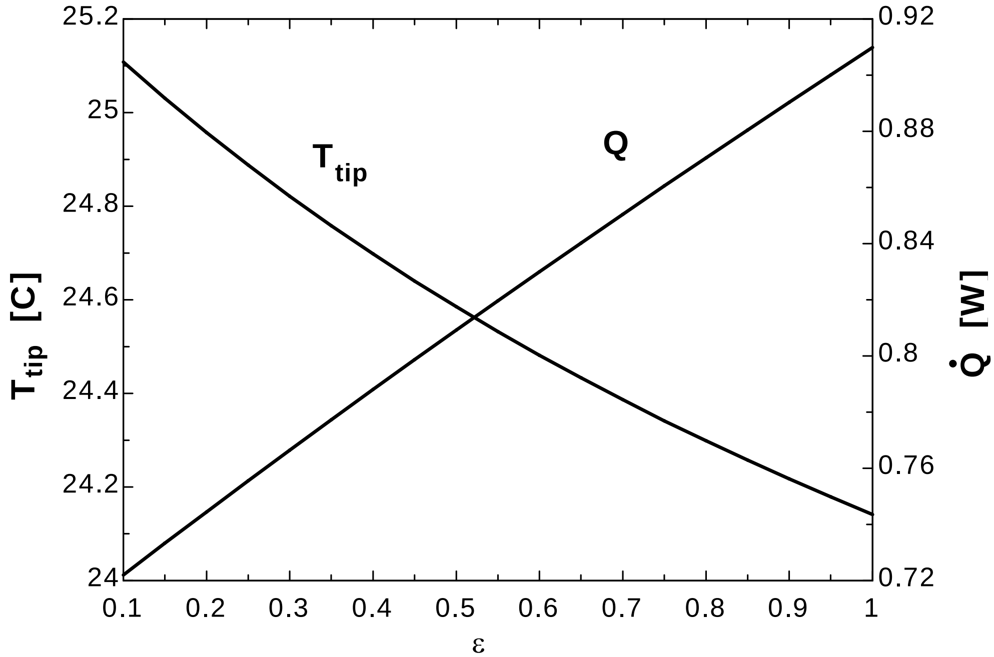
```
"GIVEN"
k=15.1 "[W/m-C], parameter to be varied"
"epsilon=0.6 parameter to be varied"
T_0=95 "[C]"
T_infinity=25 "[C]"
w=0.002 "[m]"
s=0.01 "[m]"
L=0.18 "[m]"
h=13 "[W/m^2-C]"
T_surr=295 "[K]"
DELTAx=0.015 "[m]"
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"

"ANALYSIS"
"(b)"
M=L/DELTAx+1 "Number of nodes"
A=w*s
p=2*(w+s)
"Using the finite difference method, the five equations for the unknown temperatures at 12 nodes
are determined to be"
T_0-2*T_1+T_2+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_1)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_1+273)^4)=0 "mode 1"
T_1-2*T_2+T_3+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_2)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_2+273)^4)=0 "mode 2"
T_2-2*T_3+T_4+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_3)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_3+273)^4)=0 "mode 3"
T_3-2*T_4+T_5+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_4)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_4+273)^4)=0 "mode 4"
T_4-2*T_5+T_6+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_5)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_5+273)^4)=0 "mode 5"
T_5-2*T_6+T_7+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_6)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_6+273)^4)=0 "mode 6"
T_6-2*T_7+T_8+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_7)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_7+273)^4)=0 "mode 7"
T_7-2*T_8+T_9+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_8)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_8+273)^4)=0 "mode 8"
T_8-2*T_9+T_10+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_9)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_9+273)^4)=0 "mode 9"
T_9-2*T_10+T_11+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_10)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_10+273)^4)=0 "mode 10"
T_10-2*T_11+T_12+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_11)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_11+273)^4)=0 "mode 11"
T_11-2*T_12+T_13+h*(p*DELTAx^2)/(k*A)*(T_infinity-
T_12)+epsilon*sigma*(p*DELTAx^2)/(k*A)*(T_surr^4-(T_12+273)^4)=0 "mode 12"

k*A*(T_12-T_13)/DELTAx+h*(p*DELTAx/2+A)*(T_infinity-
T_13)+epsilon*sigma*(p*DELTAx/2+A)*(T_surr^4-(T_13+273)^4)=0 "mode 13"

T_tip=T_13
"(c)"
A_s_0=p*DELTAx/2
A_s_13=p*DELTAx/2+A
A_s=p*DELTAx
Q_dot=Q_dot_0+Q_dot_1+Q_dot_2+Q_dot_3+Q_dot_4+Q_dot_5+Q_dot_6+Q_dot_7+Q_dot_8+
Q_dot_9+Q_dot_10+Q_dot_11+Q_dot_12+Q_dot_13 "where"
Q_dot_0=h*A_s_0*(T_0-T_infinity)+epsilon*sigma*A_s_0*((T_0+273)^4-T_surr^4)
```

```
Q_dot_1=h*A_s*(T_1-T_infinity)+epsilon*sigma*A_s*((T_1+273)^4-T_surr^4)
Q_dot_2=h*A_s*(T_2-T_infinity)+epsilon*sigma*A_s*((T_2+273)^4-T_surr^4)
Q_dot_3=h*A_s*(T_3-T_infinity)+epsilon*sigma*A_s*((T_3+273)^4-T_surr^4)
Q_dot_4=h*A_s*(T_4-T_infinity)+epsilon*sigma*A_s*((T_4+273)^4-T_surr^4)
Q_dot_5=h*A_s*(T_5-T_infinity)+epsilon*sigma*A_s*((T_5+273)^4-T_surr^4)
Q_dot_6=h*A_s*(T_6-T_infinity)+epsilon*sigma*A_s*((T_6+273)^4-T_surr^4)
Q_dot_7=h*A_s*(T_7-T_infinity)+epsilon*sigma*A_s*((T_7+273)^4-T_surr^4)
Q_dot_8=h*A_s*(T_8-T_infinity)+epsilon*sigma*A_s*((T_8+273)^4-T_surr^4)
Q_dot_9=h*A_s*(T_9-T_infinity)+epsilon*sigma*A_s*((T_9+273)^4-T_surr^4)
Q_dot_10=h*A_s*(T_10-T_infinity)+epsilon*sigma*A_s*((T_10+273)^4-T_surr^4)
Q_dot_11=h*A_s*(T_11-T_infinity)+epsilon*sigma*A_s*((T_11+273)^4-T_surr^4)
Q_dot_12=h*A_s*(T_12-T_infinity)+epsilon*sigma*A_s*((T_12+273)^4-T_surr^4)
Q_dot_13=h*A_s_13*(T_13-T_infinity)+epsilon*sigma*A_s_13*((T_13+273)^4-T_surr^4)
```

k [W/m.C]	T_{tip} [C]	Q [W]
10	24.38	0.6889
30.53	25.32	1.156
51.05	27.28	1.482
71.58	29.65	1.745
92.11	32.1	1.969
112.6	34.51	2.166
133.2	36.82	2.341
153.7	39	2.498
174.2	41.06	2.641
194.7	42.98	2.772
215.3	44.79	2.892
235.8	46.48	3.003
256.3	48.07	3.106
276.8	49.56	3.202
297.4	50.96	3.291
317.9	52.28	3.374
338.4	53.52	3.452
358.9	54.69	3.526
379.5	55.8	3.595
400	56.86	3.66

ε	T_{tip} [C]	Q [W]
0.1	25.11	0.722
0.15	25.03	0.7333
0.2	24.96	0.7445
0.25	24.89	0.7555
0.3	24.82	0.7665
0.35	24.76	0.7773
0.4	24.7	0.7881
0.45	24.64	0.7987
0.5	24.59	0.8092
0.55	24.53	0.8197
0.6	24.48	0.83
0.65	24.43	0.8403
0.7	24.39	0.8504
0.75	24.34	0.8605
0.8	24.3	0.8705
0.85	24.26	0.8805
0.9	24.22	0.8904
0.95	24.18	0.9001
1	24.14	0.9099





25.2
25
24.8
24.6
24.4
24.2
24
0.92
0.88
0.84
0.8
0.76
0.72
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
ε
T tip
Q
T tip [C]
Q̇ [W]

5-35 One side of a hot vertical plate is to be cooled by attaching aluminum fins of rectangular profile. The finite difference formulation of the problem for all nodes is to be obtained, and the nodal temperatures, the rate of heat transfer from a single fin and from the entire surface of the plate are to be determined.

Assumptions **1** Heat transfer along the fin is given to be steady and one-dimensional. **2** The thermal conductivity is constant. **3** Combined convection and radiation heat transfer coefficient is constant and uniform.

Properties The thermal conductivity is given to be k = 237 W/m·°C.

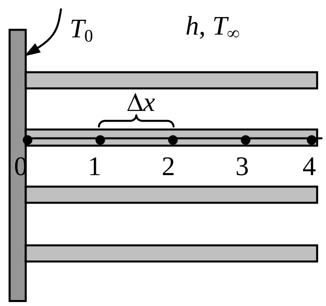
Analysis (a) The nodal spacing is given to be Δx=0.5 cm. Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{2\text{ cm}}{0.5\text{ cm}} + 1 = 5$$

The base temperature at node 0 is given to be T_0 = 130°C. This problem involves 4 unknown nodal temperatures, and thus we need to have 4 equations to determine them uniquely. Nodes 1, 2, and 3 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$kA\frac{T_{m-1}-T_m}{\Delta x} + kA\frac{T_{m+1}-T_m}{\Delta x} + h(p\Delta x)(T_\infty - T_m) = 0 \rightarrow T_{m-1} - 2T_m + T_{m+1} + h(p\Delta x^2 / kA)(T_\infty - T_m) = 0$$

The finite difference equation for node 4 at the fin tip is obtained by applying an energy balance on the half volume element about that node. Then,



$$m= 1:\ T_0 - 2T_1 + T_2 + h(p\Delta x^2 / kA)(T_\infty - T_1) = 0$$

$$m= 2:\ T_1 - 2T_2 + T_3 + h(p\Delta x^2 / kA)(T_\infty - T_2) = 0$$

$$m= 3:\ T_2 - 2T_3 + T_4 + h(p\Delta x^2 / kA)(T_\infty - T_3) = 0$$

$$\text{Node 4:}\quad kA\frac{T_3 - T_4}{\Delta x} + h(p\Delta x/2 + A)(T_\infty - T_4) = 0$$

where $\Delta x = 0.005\text{ m},\ k = 237\text{ W/m}\cdot°\text{C},\ T_\infty = 35°\text{C}, T_0 = 130°\text{C},\ h = 30\text{ W/m}^2\cdot°\text{C}$

and $A = (3\text{ m})(0.003\text{m}) = 0.009\text{ m}^2$ and $p = 2(3 + 0.003\text{ m}) = 6.006\text{ m}$.

This system of 4 equations with 4 unknowns constitute the finite difference formulation of the problem.

(b) The nodal temperatures under steady conditions are determined by solving the 4 equations above simultaneously with an equation solver to be

T_1 =129.2°C, T_2 =128.7°C, T_3 =128.3°C, T_4 =128.2°C

(c) The rate of heat transfer from a single fin is simply the sum of the heat transfer from each nodal element,

$$\dot{Q}_{\text{fin}} = \sum_{m=0}^{4}\dot{Q}_{\text{element},m} = \sum_{m=0}^{4} hA_{\text{surface},m}(T_m - T_\infty)$$
$$= hp(\Delta x/2)(T_0 - T_\infty) + hp\Delta x(T_1 + T_2 + T_3 - 3T_\infty) + h(p\Delta x/2 + A)(T_4 - T_\infty) = \mathbf{363\ W}$$

(d) The number of fins on the surface is

$$\text{No. of fins} = \frac{\text{Plate height}}{\text{Fin thickness} + \text{fin spacing}} = \frac{2\text{ m}}{(0.003 + 0.004)\text{ m}} = 286\text{ fins}$$

Then the rate of heat tranfer from the fins, the unfinned portion, and the entire finned surface become

$$\dot{Q}_{\text{fin, total}} = (\text{No. of fins})\dot{Q}_{\text{fin}} = 286(363\text{ W}) = 103{,}818\text{ W}$$

$$\dot{Q}_{\text{unfinned}} = hA_{\text{unfinned}}(T_0 - T_\infty) = (30\text{ W/m}^2\cdot°\text{C})(286 \times 3\text{ m} \times 0.004\text{ m})(130\text{ - }35)°\text{C} = 9781\text{ W}$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{fin, total}} + \dot{Q}_{\text{unfinned}} = 103{,}818 + 9781 = \mathbf{113{,}600\ W \cong 114\ kW}$$

5-36 One side of a hot vertical plate is to be cooled by attaching aluminum pin fins. The finite difference formulation of the problem for all nodes is to be obtained, and the nodal temperatures, the rate of heat transfer from a single fin and from the entire surface of the plate are to be determined.

Assumptions **1** Heat transfer along the fin is given to be steady and one-dimensional. **2** The thermal conductivity is constant. **3** Combined convection and radiation heat transfer coefficient is constant and uniform.

Properties The thermal conductivity is given to be k = 237 W/m·°C.

Analysis (*a*) The nodal spacing is given to be Δx=0.5 cm. Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{3\,\text{cm}}{0.5\,\text{cm}} + 1 = 7$$

The base temperature at node 0 is given to be T_0 = 100°C. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations to determine them uniquely. Nodes 1, 2, 3, 4, and 5 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$kA\frac{T_{m-1}-T_m}{\Delta x} + kA\frac{T_{m+1}-T_m}{\Delta x} + h(p\Delta x)(T_\infty - T_m) = 0 \quad\rightarrow\quad T_{m-1} - 2T_m + T_{m+1} + h(p\Delta x^2/kA)(T_\infty - T_m) = 0$$

The finite difference equation for node 6 at the fin tip is obtained by applying an energy balance on the half volume element about that node. Then,

$$m=1:\; T_0 - 2T_1 + T_2 + h(p\Delta x^2/kA)(T_\infty - T_1) = 0$$

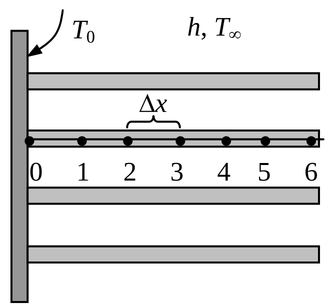
$$m=2:\; T_1 - 2T_2 + T_3 + h(p\Delta x^2/kA)(T_\infty - T_2) = 0$$

$$m=3:\; T_2 - 2T_3 + T_4 + h(p\Delta x^2/kA)(T_\infty - T_3) = 0$$

$$m=4:\; T_3 - 2T_4 + T_5 + h(p\Delta x^2/kA)(T_\infty - T_4) = 0$$

$$m=5:\; T_4 - 2T_5 + T_6 + h(p\Delta x^2/kA)(T_\infty - T_5) = 0$$

$$\text{Node 6:}\; kA\frac{T_5 - T_6}{\Delta x} + h(p\Delta x/2 + A)(T_\infty - T_6) = 0$$



where $\Delta x = 0.005\,\text{m},\; k = 237\,\text{W/m}\cdot°\text{C},\;\; T_\infty = 30°\text{C}, T_0 = 100°\text{C},\; h = 35\,\text{W/m}^2\cdot°\text{C}$

and
$$A = \pi D^2/4 = \pi(0.25\,\text{cm})^2/4 = 0.0491\,\text{cm}^2 = 0.0491\times10^{-4}\,\text{m}^2$$
$$p = \pi D = \pi(0.0025\,\text{m}) = 0.00785\,\text{m}$$

(*b*) The nodal temperatures under steady conditions are determined by solving the 6 equations above simultaneously with an equation solver to be

T_1 =97.9°C, T_2 =96.1°C, T_3 =94.7°C, T_4 =93.8°C, T_5 =93.1°C, T_6 =92.9°C

(*c*) The rate of heat transfer from a single fin is simply the sum of the heat transfer from the nodal elements,

$$\dot{Q}_{\text{fin}} = \sum_{m=0}^{6}\dot{Q}_{\text{element},m} = \sum_{m=0}^{6} hA_{\text{surface},m}(T_m - T_\infty)$$
$$= hp\Delta x/2(T_0 - T_\infty) + hp\Delta x(T_1 + T_2 + T_3 + T_4 + T_5 - 5T_\infty) + h(p\Delta x/2 + A)(T_6 - T_\infty) = \mathbf{0.5496\;\;W}$$

(*d*) The number of fins on the surface is $\quad \text{No. of fins} = \dfrac{1\,\text{m}^2}{(0.006\,\text{m})(0.006\,\text{m})} = 27{,}778\,\text{fins}$

Then the rate of heat tranfer from the fins, the unfinned portion, and the entire finned surface become

$$\dot{Q}_{\text{fin, total}} = (\text{No. of fins})\dot{Q}_{\text{fin}} = 27{,}778(0.5496\,\text{W}) = 15{,}267\,\text{W}$$

$$\dot{Q}_{\text{unfinned}} = hA_{\text{unfinned}}(T_0 - T_\infty) = (35\,\text{W/m}^2\cdot°\text{C})(1 - 27{,}778\times0.0491\times10^{-4}\,\text{m}^2)(100-30)°\text{C} = 2116\,\text{W}$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{fin, total}} + \dot{Q}_{\text{unfinned}} = 15{,}267 + 2116 = \mathbf{17{,}383\;W \cong 17.4\;\;kW}$$

5-37 One side of a hot vertical plate is to be cooled by attaching copper pin fins. The finite difference formulation of the problem for all nodes is to be obtained, and the nodal temperatures, the rate of heat transfer from a single fin and from the entire surface of the plate are to be determined.

Assumptions **1** Heat transfer along the fin is given to be steady and one-dimensional. **2** The thermal conductivity is constant. **3** Combined convection and radiation heat transfer coefficient is constant and uniform.

Properties The thermal conductivity is given to be k = 386 W/m·°C.

Analysis (*a*) The nodal spacing is given to be Δx=0.5 cm. Then the number of nodes M becomes

$$M = \frac{L}{\Delta x} + 1 = \frac{3\,\text{cm}}{0.5\,\text{cm}} + 1 = 7$$

The base temperature at node 0 is given to be T_0 = 100°C. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations to determine them uniquely. Nodes 1, 2, 3, 4, and 5 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$kA\frac{T_{m-1}-T_m}{\Delta x} + kA\frac{T_{m+1}-T_m}{\Delta x} + h(p\Delta x)(T_\infty - T_m) = 0 \;\rightarrow\; T_{m-1} - 2T_m + T_{m+1} + h(p\Delta x^2/kA)(T_\infty - T_m) = 0$$

The finite difference equation for node 6 at the fin tip is obtained by applying an energy balance on the half volume element about that node. Then,

$$m=1:\; T_0 - 2T_1 + T_2 + h(p\Delta x^2/kA)(T_\infty - T_1) = 0$$

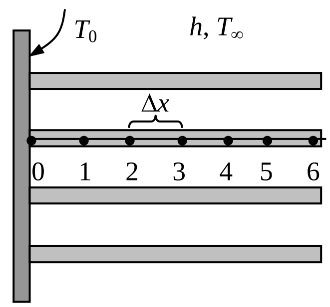
$$m=2:\; T_1 - 2T_2 + T_3 + h(p\Delta x^2/kA)(T_\infty - T_2) = 0$$

$$m=3:\; T_2 - 2T_3 + T_4 + h(p\Delta x^2/kA)(T_\infty - T_3) = 0$$

$$m=4:\; T_3 - 2T_4 + T_5 + h(p\Delta x^2/kA)(T_\infty - T_4) = 0$$

$$m=5:\; T_4 - 2T_5 + T_6 + h(p\Delta x^2/kA)(T_\infty - T_5) = 0$$

$$\text{Node 6:}\; kA\frac{T_5 - T_6}{\Delta x} + h(p\Delta x/2 + A)(T_\infty - T_6) = 0$$



where $\Delta x = 0.005\,\text{m},\; k = 386\,\text{W/m}\cdot°\text{C},\; T_\infty = 30°\text{C}, T_0 = 100°\text{C},\; h = 35\,\text{W/m}^2\cdot°\text{C}$

and
$$A = \pi D^2/4 = \pi(0.25\,\text{cm})^2/4 = 0.0491\,\text{cm}^2 = 0.0491\times10^{-4}\,\text{m}^2$$
$$p = \pi D = \pi(0.0025\,\text{m}) = 0.00785\,\text{m}$$

(*b*) The nodal temperatures under steady conditions are determined by solving the 6 equations above simultaneously with an equation solver to be

T_1 =98.6°C, T_2 =97.5°C, T_3 =96.7°C, T_4 =96.0°C, T_5 =95.7°C, T_6 =95.5°C

(*c*) The rate of heat transfer from a single fin is simply the sum of the heat transfer from the nodal elements,

$$\dot{Q}_{\text{fin}} = \sum_{m=0}^{6}\dot{Q}_{\text{element},m} = \sum_{m=0}^{6} hA_{\text{surface},m}(T_m - T_\infty)$$
$$= hp\Delta x/2(T_0 - T_\infty) + hp\Delta x(T_1 + T_2 + T_3 + T_4 + T_5 - 5T_\infty) + h(p\Delta x/2 + A)(T_6 - T_\infty) = \mathbf{0.5641\ W}$$

(*d*) The number of fins on the surface is $\quad \text{No. of fins} = \dfrac{1\,\text{m}^2}{(0.006\,\text{m})(0.006\,\text{m})} = 27{,}778\,\text{fins}$

Then the rate of heat tranfer from the fins, the unfinned portion, and the entire finned surface become

$$\dot{Q}_{\text{fin, total}} = (\text{No. of fins})\dot{Q}_{\text{fin}} = 27{,}778(0.5641\,\text{W}) = 15{,}670\,\text{W}$$

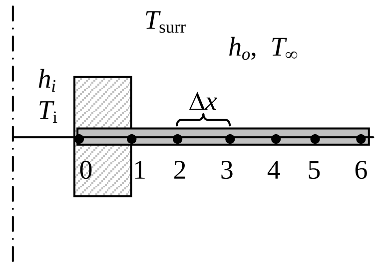
$$\dot{Q}_{\text{unfinned}} = hA_{\text{unfinned}}(T_0 - T_\infty) = (35\,\text{W/m}^2\cdot°\text{C})(1 - 27{,}778\times0.0491\times10^{-4}\,\text{m}^2)(100-30)°\text{C} = 2116\,\text{W}$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{fin, total}} + \dot{Q}_{\text{unfinned}} = 15{,}670 + 2116 = \mathbf{17{,}786\ W} \cong \mathbf{17.8\,kW}$$

5-38 Two cast iron steam pipes are connected to each other through two 1-cm thick flanges, and heat is lost from the flanges by convection and radiation. The finite difference formulation of the problem for all nodes is to be obtained, and the temperature of the tip of the flange as well as the rate of heat transfer from the exposed surfaces of the flange are to be determined.

Assumptions **1** Heat transfer through the flange is stated to be steady and one-dimensional. **2** The thermal conductivity and emissivity are constants. **3** Convection heat transfer coefficient is constant and uniform.

Properties The thermal conductivity and emissivity are given to be k = 52 W/m·°C and ε = 0.8.



Analysis (*a*) The distance between nodes 0 and 1 is the thickness of the pipe, Δx_1=0.4 cm=0.004 m. The nodal spacing along the flange is given to be Δx_2=1 cm = 0.01 m. Then the number of nodes *M* becomes

$$M = \frac{L}{\Delta x} + 2 = \frac{5\,\text{cm}}{1\,\text{cm}} + 2 = 7$$

This problem involves 7 unknown nodal temperatures, and thus we need to have 7 equations to determine them uniquely. Noting that the total thickness of the flange is t = 0.02 m, the heat conduction area at any location along the flange is $A_{\text{cond}} = 2\pi rt$ where the values of radii at the nodes and between the nodes (the mid points) are

r_0=0.046 m, r_1=0.05 m, r_2=0.06 m, r_3=0.07 m, r_4=0.08 m, r_5=0.09 m, r_6=0.10 m

r_{01}=0.048 m, r_{12}=0.055 m, r_{23}=0.065 m, r_{34}=0.075 m, r_{45}=0.085 m, r_{56}=0.095 m

Then the finite difference equations for each node are obtained from the energy balance to be as follows:

Node 0: $$h_i(2\pi t r_0)(T_i - T_0) + k(2\pi t r_{01})\frac{T_1 - T_0}{\Delta x_1} = 0$$

Node 1:

$$k(2\pi t r_{01})\frac{T_0 - T_1}{\Delta x_1} + k(2\pi t r_{12})\frac{T_2 - T_1}{\Delta x_2} + 2[2\pi t(r_1 + r_{12})/2)](\Delta x_2/2)\{h(T_\infty - T_1) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_1 + 273)^4]\} = 0$$

Node 2: $$k(2\pi t r_{12})\frac{T_1 - T_2}{\Delta x_2} + k(2\pi t r_{23})\frac{T_3 - T_2}{\Delta x_2} + 2(2\pi t r_2 \Delta x_2)\{h(T_\infty - T_2) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_2 + 273)^4]\} = 0$$

Node 3: $$k(2\pi t r_{23})\frac{T_2 - T_3}{\Delta x_2} + k(2\pi t r_{34})\frac{T_4 - T_3}{\Delta x_2} + 2(2\pi t r_3 \Delta x_2)\{h(T_\infty - T_3) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_3 + 273)^4]\} = 0$$

Node 4: $$k(2\pi t r_{34})\frac{T_3 - T_4}{\Delta x_2} + k(2\pi t r_{45})\frac{T_5 - T_4}{\Delta x_2} + 2(2\pi t r_4 \Delta x_2)\{h(T_\infty - T_4) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_4 + 273)^4]\} = 0$$

Node 5: $$k(2\pi t r_{45})\frac{T_4 - T_5}{\Delta x_2} + k(2\pi t r_{56})\frac{T_6 - T_5}{\Delta x_2} + 2(2\pi t r_5 \Delta x_2)\{h(T_\infty - T_5) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_5 + 273)^4]\} = 0$$

Node 6: $$k(2\pi t r_{56})\frac{T_5 - T_6}{\Delta x_2} + 2[2\pi t(\Delta x_2/2)(r_{56} + r_6)/2 + 2\pi r_6 t]\{h(T_\infty - T_6) + \varepsilon\sigma[T_{\text{surr}}^4 - (T_6 + 273)^4]\} = 0$$

where $\Delta x_1 = 0.004\text{ m}$, $\Delta x_2 = 0.01\text{ m}$, $k = 52\text{ W/m}\cdot°\text{C}$, $\varepsilon = 0.8$, $T_\infty = 8°\text{C}$, $T_{in} = 200°\text{C}$, $T_{surr} = 290\text{ K}$ and $h = 25\text{ W/m}^2\cdot°\text{C}$, $h_i = 180\text{ W/m}^2\cdot°\text{C}$, $\sigma = 5.67\times10^{-8}\text{ W/m}^2\cdot\text{K}^4$.

The system of 7 equations with 7 unknowns constitutes the finite difference formulation of the problem.

(*b*) The nodal temperatures under steady conditions are determined by solving the 7 equations above simultaneously with an equation solver to be

T_0 =119.7°C, T_1 =118.6°C, T_2 = 116.3°C, T_3 =114.3°C, T_4 =112.7°C, T_5 =111.2°C, and **T_6 = 109.9°C** (*c*) Knowing the inner surface temperature, the rate of heat transfer from the flange under steady conditions is simply the rate of heat transfer from the steam to the pipe at flange section

$$\dot{Q}_{\text{fin}} = \sum_{m=1}^{6} \dot{Q}_{\text{element},m} = \sum_{m=1}^{6} hA_{\text{surface},m}(T_m - T_\infty) + \sum_{m=1}^{6} \varepsilon\sigma A_{\text{surface},m}[(T_m + 273)^4 - T_{\text{surr}}^4] = \mathbf{83.6\ W}$$

where $A_{\text{surface, m}}$ are as given above for different nodes.

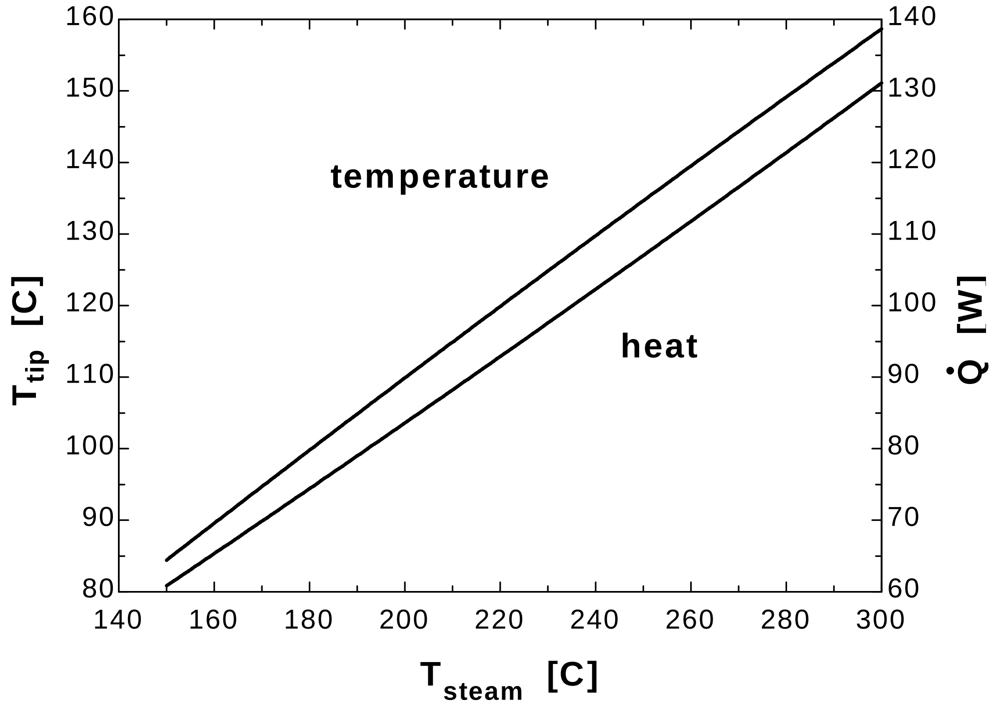
5-39 "!PROBLEM 5-39"

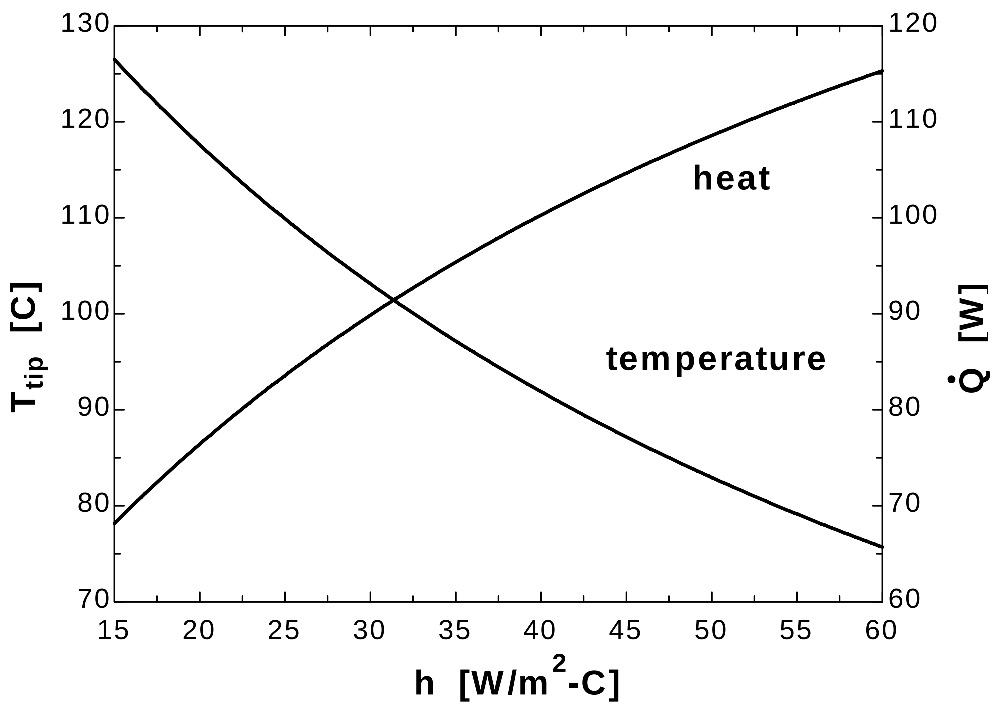
```
"GIVEN"
t_pipe=0.004 "[m]"
k=52 "[W/m-C]"
epsilon=0.8
D_o_pipe=0.10 "[m]"
t_flange=0.01 "[m]"
D_o_flange=0.20 "[m]"
T_steam=200 "[C], parameter to be varied"
h_i=180 "[W/m^2-C]"
T_infinity=8 "[C]"
"h=25 [W/m^2-C], parameter to be varied"
T_surr=290 "[K]"
DELTAx=0.01 "[m]"
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"

"ANALYSIS"
"(b)"
DELTAx_1=t_pipe "the distance between nodes 0 and 1"
DELTAx_2=t_flange "nodal spacing along the flange"
L=(D_o_flange-D_o_pipe)/2
M=L/DELTAx_2+2 "Number of nodes"
t=2*t_flange "total thixkness of the flange"
"The values of radii at the nodes and between the nodes /-(the midpoints) are"
r_0=0.046 "[m]"
r_1=0.05 "[m]"
r_2=0.06 "[m]"
r_3=0.07 "[m]"
r_4=0.08 "[m]"
r_5=0.09 "[m]"
r_6=0.10 "[m]"
r_01=0.048 "[m]"
r_12=0.055 "[m]"
r_23=0.065 "[m]"
r_34=0.075 "[m]"
r_45=0.085 "[m]"
r_56=0.095 "[m]"
"Using the finite difference method, the five equations for the unknown temperatures at 7 nodes
are determined to be"
h_i*(2*pi*t*r_0)*(T_steam-T_0)+k*(2*pi*t*r_01)*(T_1-T_0)/DELTAx_1=0 "node 0"
k*(2*pi*t*r_01)*(T_0-T_1)/DELTAx_1+k*(2*pi*t*r_12)*(T_2-
T_1)/DELTAx_2+2*2*pi*t*(r_1+r_12)/2*(DELTAx_2/2)*(h*(T_infinity-
T_1)+epsilon*sigma*(T_surr^4-(T_1+273)^4))=0 "node 1"
k*(2*pi*t*r_12)*(T_1-T_2)/DELTAx_2+k*(2*pi*t*r_23)*(T_3-
T_2)/DELTAx_2+2*2*pi*t*r_2*DELTAx_2*(h*(T_infinity-T_2)+epsilon*sigma*(T_surr^4-
(T_2+273)^4))=0 "node 2"
k*(2*pi*t*r_23)*(T_2-T_3)/DELTAx_2+k*(2*pi*t*r_34)*(T_4-
T_3)/DELTAx_2+2*2*pi*t*r_3*DELTAx_2*(h*(T_infinity-T_3)+epsilon*sigma*(T_surr^4-
(T_3+273)^4))=0 "node 3"
k*(2*pi*t*r_34)*(T_3-T_4)/DELTAx_2+k*(2*pi*t*r_45)*(T_5-
T_4)/DELTAx_2+2*2*pi*t*r_4*DELTAx_2*(h*(T_infinity-T_4)+epsilon*sigma*(T_surr^4-
(T_4+273)^4))=0 "node 4"
k*(2*pi*t*r_45)*(T_4-T_5)/DELTAx_2+k*(2*pi*t*r_56)*(T_6-
T_5)/DELTAx_2+2*2*pi*t*r_5*DELTAx_2*(h*(T_infinity-T_5)+epsilon*sigma*(T_surr^4-
(T_5+273)^4))=0 "node 5"
```

k*(2*pi*t*r_56)*(T_5-
T_6)/DELTAx_2+2*(2*pi*t*(r_56+r_6)/2*(DELTAx_2/2)+2*pi*t*r_6)*(h*(T_infinity-
T_6)+epsilon*sigma*(T_surr^4-(T_6+273)^4))=0 "node 6"
T_tip=T_6
"(c)"
Q_dot=Q_dot_1+Q_dot_2+Q_dot_3+Q_dot_4+Q_dot_5+Q_dot_6 "where"
Q_dot_1=h*2*2*pi*t*(r_1+r_12)/2*DELTAx_2/2*(T_1-
T_infinity)+epsilon*sigma*2*2*pi*t*(r_1+r_12)/2*DELTAx_2/2*((T_1+273)^4-T_surr^4)
Q_dot_2=h*2*2*pi*t*r_2*DELTAx_2*(T_2-
T_infinity)+epsilon*sigma*2*2*pi*t*r_2*DELTAx_2*((T_2+273)^4-T_surr^4)
Q_dot_3=h*2*2*pi*t*r_3*DELTAx_2*(T_3-
T_infinity)+epsilon*sigma*2*2*pi*t*r_3*DELTAx_2*((T_3+273)^4-T_surr^4)
Q_dot_4=h*2*2*pi*t*r_4*DELTAx_2*(T_4-
T_infinity)+epsilon*sigma*2*2*pi*t*r_4*DELTAx_2*((T_4+273)^4-T_surr^4)
Q_dot_5=h*2*2*pi*t*r_5*DELTAx_2*(T_5-
T_infinity)+epsilon*sigma*2*2*pi*t*r_5*DELTAx_2*((T_5+273)^4-T_surr^4)
Q_dot_6=h*2*(2*pi*t*(r_56+r_6)/2*(DELTAx_2/2)+2*pi*t*r_6)*(T_6-
T_infinity)+epsilon*sigma*2*(2*pi*t*(r_56+r_6)/2*(DELTAx_2/2)+2*pi*t*r_6)*((T_6+273)^4-
T_surr^4)

T_{steam} [C]	T_{tip} [C]	Q [W]
150	84.42	60.83
160	89.57	65.33
170	94.69	69.85
180	99.78	74.4
190	104.8	78.98
200	109.9	83.58
210	114.9	88.21
220	119.9	92.87
230	124.8	97.55
240	129.7	102.3
250	134.6	107
260	139.5	111.8
270	144.3	116.6
280	149.1	121.4
290	153.9	126.2
300	158.7	131.1

h [W/m^2.C]	T_{tip} [C]	Q [W]
15	126.5	68.18
20	117.6	76.42
25	109.9	83.58
30	103.1	89.85
35	97.17	95.38
40	91.89	100.3
45	87.17	104.7
50	82.95	108.6
55	79.14	112.1
60	75.69	115.3


temperature
heat
T_{tip} [C]
$\dot{Q}$ [W]
T_{steam} [C]
160
150
140
130
120
110
100
90
80
140
130
120
110
100
90
80
70
60
140
160
180
200
220
240
260
280
300


heat
temperature
T_{tip} [C]
$\dot{Q}$ [W]
h [W/m^2-C]
130
120
110
100
90
80
70
120
110
100
90
80
70
60
15
20
25
30
35
40
45
50
55
60

5-40 Using an equation solver or an iteration method, the solutions of the following systems of algebraic equations are determined to be as follows:

(*a*)

$$3x_1 - x_2 + 3x_3 = 0$$
$$-x_1 + 2x_2 + x_3 = 3$$
$$2x_1 - x_2 - x_3 = 2$$

Solution: x_1=2, x_2=3, x_3=1

(*b*)

$$4x_1 - 2x_2^2 + 0.5x_3 = -2$$
$$x_1^3 - x_2 + x_3 = 11.964$$
$$x_1 + x_2 + x_3 = 3$$

Solution: x_1=2.532, x_2=2.364, x_3=-1.896

```
"ANALYSIS"
"(a)"
3*x_1a-x_2a+3*x_3a=0
-x_1a+2*x_2a+x_3a=3
2*x_1a-x_2a-x_3a=2
"(b)"
4*x_1b-2*x_2b^2+0.5*x_3b=-2
x_1b^3-x_2b+-x_3b=11.964
x_1b+x_2b+x_3b=3
```

5-41 Using an equation solver or an iteration method, the solutions of the following systems of algebraic equations are determined to be as follows:

(*a*)

$$3x_1 - 2x_2 - x_3 + x_4 = 6$$
$$x_1 + 2x_2 - x_4 = -3$$
$$-2x_1 + x_2 + 3x_3 + x_4 = 2$$
$$3x_2 + x_3 - 4x_4 = -6$$

Solution: x_1=13, x_2=-9, x_3=13, x_4= -2

(*b*)

$$3x_1 + x_2^2 + 2x_3 = 8$$
$$-x_1^2 + 3x_2 + 2x_3 = -6.293$$
$$2x_1 - x_2^4 + 4x_3 = -12$$

Solution: x_1=2.825, x_2=1.791, x_3=-1.841

```
"ANALYSIS"
"(a)"
3*x_1a+2*x_2a-x_3a+x_4a=6
x_1a+2*x_2a-x_4a=-3
-2*x_1a+x_2a+3*x_3a+x_4a=2
3*x_2a+x_3a-4*x_4a=-6
"(b)"
3*x_1b+x_2b^2+2*x_3b=8
-x_1b^2+3*x_2b+2*x_3b=-6.293
2*x_1b-x_2b^4+4*x_3b=-12
```

5-42 Using an equation solver or an iteration method, the solutions of the following systems of algebraic equations are determined to be as follows:

(*a*)

$$\begin{aligned} 4x_1 - x_2 + 2x_3 + x_4 &= -6 \\ x_1 + 3x_2 - x_3 + 4x_4 &= -1 \\ -x_1 + 2x_2 + 5x_4 &= 5 \\ 2x_2 - 4x_3 - 3x_4 &= 2 \end{aligned}$$

Solution: x_1=-0.744, x_2=-8, x_3=-7.54, x_4= 4.05

(*b*)

$$\begin{aligned} 2x_1 + x_2^4 - 2x_3 + x_4 &= 1 \\ x_1^2 + 4x_2 + 2x_3^2 - 2x_4 &= -3 \\ -x_1 + x_2^4 + 5x_3 &= 10 \\ 3x_1 - x_3^2 + 8x_4 &= 15 \end{aligned}$$

Solution: x_1=0.263, x_2=-1.15, x_3=1.70, x_4=2.14

```
"ANALYSIS"
"(a)"
4*x_1a-x_2a+2*x_3a+x_4a=-6
x_1a+3*x_2a-x_3a+4*x_4a=-1
-x_1a+2*x_2a+5*x_4a=5
2*x_2a-4*x_3a-3*x_4a=2
"(b)"
2*x_1b+x_2b^4-2*x_3b+x_4b=1
x_1b^2+4*x_2b+2*x_3b^2-2*x_4b=-3
-x_1b+x_2b^4+5*x_3b=10
3*x_1b-x_3b^2+8*x_4b=15
```

Two-Dimensional Steady Heat Conduction

5-43C For a medium in which the finite difference formulation of a general interior node is given in its simplest form as $T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \dfrac{\dot{g}_{\text{node}} l^2}{k} = 0$:

(*a*) Heat transfer is steady, (*b*) heat transfer is two-dimensional, (*c*) there is heat generation in the medium, (*d*) the nodal spacing is constant, and (*e*) the thermal conductivity of the medium is constant.

5-44C For a medium in which the finite difference formulation of a general interior node is given in its simplest form as $T_{\text{node}} = (T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}})/4$:

(*a*) Heat transfer is steady, (*b*) heat transfer is two-dimensional, (*c*) there is no heat generation in the medium, (*d*) the nodal spacing is constant, and (*e*) the thermal conductivity of the medium is constant.

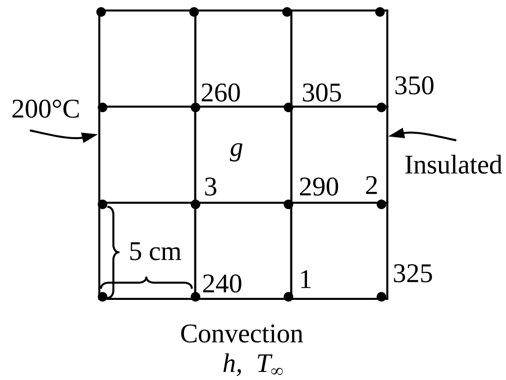
5-45C A region that cannot be filled with simple volume elements such as strips for a plane wall, and rectangular elements for two-dimensional conduction is said to have *irregular boundaries*. A practical way of dealing with such geometries in the finite difference method is to replace the elements bordering the irregular geometry by a series of simple volume elements.

5-46 A long solid body is subjected to steady two-dimensional heat transfer. The unknown nodal temperatures and the rate of heat loss from the bottom surface through a 1-m long section are to be determined.

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** Heat is generated uniformly in the body. **3** Radiation heat transfer is negligible.

Properties The thermal conductivity is given to be k = 45 W/m·°C.

Analysis The nodal spacing is given to be Δx=Δx=l=0.05 m, and the general finite difference form of an interior node for steady two-dimensional heat conduction is expressed as



$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0$$

where

$$\frac{\dot{g}_{\text{node}} l^2}{k} = \frac{\dot{g}_0 l^2}{k} = \frac{(8\times 10^6 \text{ W/m}^3)(0.05\text{ m})^2}{214\text{ W/m}\cdot{}^\circ\text{C}} = 93.5^\circ\text{C}$$

The finite difference equations for boundary nodes are obtained by applying an energy balance on the volume elements and taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (convection)}: \quad k\frac{l}{2}\frac{240-T_1}{l} + kl\frac{290-T_1}{l} + k\frac{l}{2}\frac{325-T_1}{l} + hl(T_\infty - T_1) + \frac{\dot{g}_0 l^2}{2k} = 0$$

$$\text{Node 2 (interior)}: \quad 350 + 290 + 325 + 290 - 4T_2 + \frac{\dot{g}_0 l^2}{k} = 0$$

$$\text{Node 3 (interior)}: \quad 260 + 290 + 240 + 200 - 4T_3 + \frac{\dot{g}_0 l^2}{k} = 0$$

where $\quad k = 45\text{ W/m.°C}, \ \ h = 50\text{ W/m}^2\text{.°C}, \ \ \dot{g} = 8\times 10^6 \text{ W/m}^3, \ \ T_\infty = 20^\circ\text{C}$

Substituting, $\quad T_1$ **= 280.9°C**, T_2 **= 397.1°C**, T_3 **= 330.8°C**,

(*b*) The rate of heat loss from the bottom surface through a 1-m long section is

$$\begin{aligned}\dot{Q} &= \sum_m \dot{Q}_{\text{element},m} = \sum_m hA_{\text{surface},m}(T_m - T_\infty) \\ &= h(l/2)(200 - T_\infty) + hl(240 - T_\infty) + hl(T_1 - T_\infty) + h(l/2)(325 - T_\infty) \\ &= (50\text{ W/m}^2\cdot{}^\circ\text{C})(0.05\text{ m}\times 1\text{ m})[(200-20)/2 + (240-20) + (280.9-20) + (325-20)/2]^\circ\text{C} = \mathbf{1808\ \ W}\end{aligned}$$

5-47 A long solid body is subjected to steady two-dimensional heat transfer. The unknown nodal temperatures are to be determined.

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** There is no heat generation in the body.

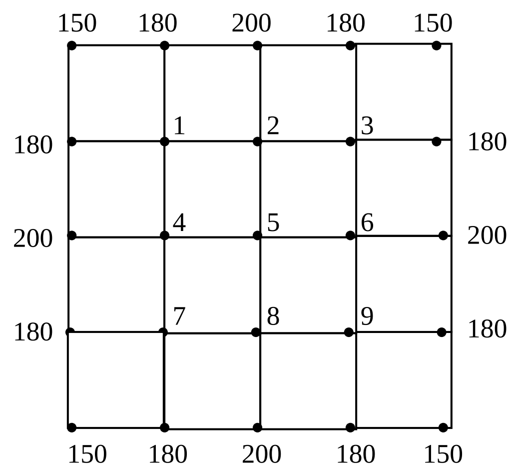
Properties The thermal conductivity is given to be k = 45 W/m·°C.

Analysis The nodal spacing is given to be Δx=Δx=l=0.01 m, and the general finite difference form of an interior node for steady two-dimensional heat conduction for the case of no heat generation is expressed as

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0 \;\rightarrow\; T_{\text{node}} = (T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}})/4$$

There is symmetry about the horizontal, vertical, and diagonal lines passing through the midpoint, and thus we need to consider only 1/8th of the region. Then,

$$T_1 = T_3 = T_7 = T_9$$
$$T_2 = T_4 = T_6 = T_8$$



Therefore, there are there are only 3 unknown nodal temperatures, $T_1, T_3,$ and T_5, and thus we need only 3 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept when writing the finite difference equations for the interior nodes.

Node 1 (interior): $T_1 = (180 + 180 + 2T_2)/4$

Node 2 (interior): $T_2 = (200 + T_5 + 2T_1)/4$

Node 3 (interior): $T_5 = 4T_2/4 = T_2$

Solving the equations above simultaneously gives

$$T_1 = T_3 = T_7 = T_9 = \mathbf{185^\circ C}$$
$$T_2 = T_4 = T_5 = T_6 = T_8 = \mathbf{190^\circ C}$$

Discussion Note that taking advantage of symmetry simplified the problem greatly.

5-48 A long solid body is subjected to steady two-dimensional heat transfer. The unknown nodal temperatures are to be determined.

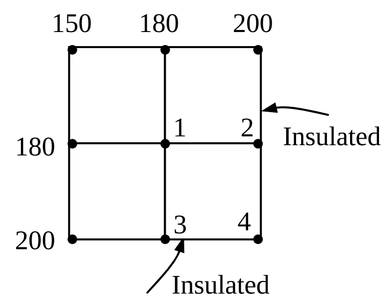
Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** There is no heat generation in the body.

Properties The thermal conductivity is given to be k = 20 W/m·°C.

Analysis The nodal spacing is given to be Δx=Δx=l=0.02 m, and the general finite difference form of an interior node for steady two-dimensional heat conduction for the case of no heat generation is expressed as

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0 \;\rightarrow\; T_{\text{node}} = (T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}})/4$$

(*a*) There is symmetry about the insulated surfaces as well as about the diagonal line. Therefore $T_3 = T_2$, and T_1, T_2, and T_4 are the only 3 unknown nodal temperatures. Thus we need only 3 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept when writing the finite difference equations for the interior nodes.



Node 1 (interior): $T_1 = (180 + 180 + T_2 + T_3)/4$

Node 2 (interior): $T_2 = (200 + T_4 + 2T_1)/4$

Node 4 (interior): $T_4 = (2T_2 + 2T_3)/4$

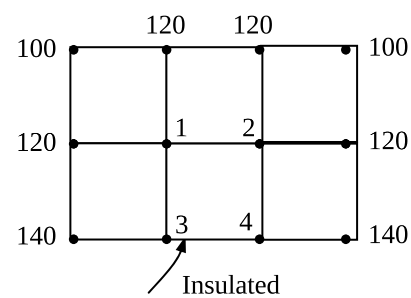
Also, $T_3 = T_2$

Solving the equations above simultaneously gives

$T_2 = T_3 = T_4 = \mathbf{190°C}$

$T_1 = \mathbf{185°C}$

(*b*) There is symmetry about the insulated surface as well as the diagonal line. Replacing the symmetry lines by insulation, and utilizing the mirror-image concept, the finite difference equations for the interior nodes can be written as



Node 1 (interior): $T_1 = (120 + 120 + T_2 + T_3)/4$

Node 2 (interior): $T_2 = (120 + 120 + T_4 + T_1)/4$

Node 3 (interior): $T_3 = (140 + 2T1 + T_4)/4 = T_2$

Node 4 (interior): $T_4 = (2T_2 + 140 + 2T_3)/4$

Solving the equations above simultaneously gives

$T_1 = T_2 = \mathbf{122.9°C}$

$T_3 = T_4 = \mathbf{128.6°C}$

Discussion Note that taking advantage of symmetry simplified the problem greatly.

5-49 Starting with an energy balance on a volume element, the steady two-dimensional finite difference equation for a general interior node in rectangular coordinates for $T(x, y)$ for the case of variable thermal conductivity and uniform heat generation is to be obtained.

Analysis We consider a *volume element* of size $\Delta x \times \Delta y \times 1$ centered about a general interior node (m, n) in a region in which heat is generated at a constant rate of $\dot{g}$ and the thermal conductivity k is variable (see Fig. 5-24 in the text). Assuming the direction of heat conduction to be *towards* the node under consideration at all surfaces, the energy balance on the volume element can be expressed as

$$\dot{Q}_{\text{cond, left}} + \dot{Q}_{\text{cond, top}} + \dot{Q}_{\text{cond, right}} + \dot{Q}_{\text{cond, bottom}} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t} = 0$$

for the *steady* case. Again assuming the temperatures between the adjacent nodes to vary linearly and noting that the heat transfer area is $\Delta y \times 1$ in the x direction and $\Delta x \times 1$ in the y direction, the energy balance relation above becomes

$$k_{m,n}(\Delta y \times 1)\frac{T_{m-1,n} - T_{m,n}}{\Delta x} + k_{m,n}(\Delta x \times 1)\frac{T_{m,n+1} - T_{m,n}}{\Delta y} + k_{m,n}(\Delta y \times 1)\frac{T_{m+1,n} - T_{m,n}}{\Delta x}$$

$$+ k_{m,n}(\Delta x \times 1)\frac{T_{m,n-1} - T_{m,n}}{\Delta y} + \dot{g}_0(\Delta x \times \Delta y \times 1) = 0$$

Dividing each term by $\Delta x \times \Delta y \times 1$ and simplifying gives

$$\frac{T_{m-1,n} - 2T_{m,n} + T_{m+1,n}}{\Delta x^2} + \frac{T_{m,n-1} - 2T_{m,n} + T_{m,n+1}}{\Delta y^2} + \frac{\dot{g}_0}{k_{m,n}} = 0$$

For a square mesh with $\Delta x = \Delta y = l$, and the relation above simplifies to

$$T_{m-1,n} + T_{m+1,n} + T_{m,n-1} + T_{m,n-1} - 4T_{m,n} + \frac{\dot{g}_0 l^2}{k_{m,n}} = 0$$

It can also be expressed in the following easy-to-remember form:

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_0 l^2}{k_{\text{node}}} = 0$$

5-50 A long solid body is subjected to steady two-dimensional heat transfer. The unknown nodal temperatures and the rate of heat loss from the top surface are to be determined.

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** Heat is generated uniformly in the body.

Properties The thermal conductivity is given to be k = 180 W/m·°C.

Analysis (*a*) The nodal spacing is given to be Δx=Δx=l=0.1 m, and the general finite difference form of an interior node equation for steady two-dimensional heat conduction for the case of constant heat generation is expressed as

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0$$

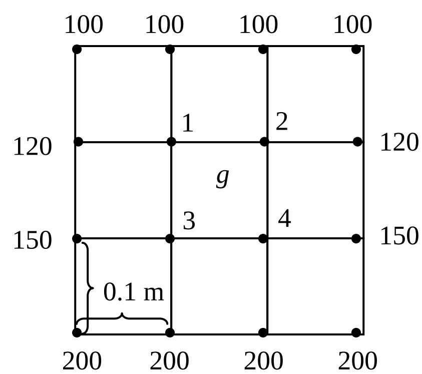
There is symmetry about a vertical line passing through the middle of the region, and thus we need to consider only half of the region. Then,

$$T_1 = T_2 \text{ and } T_3 = T_4$$

Therefore, there are there are only 2 unknown nodal temperatures, T_1 and T_3, and thus we need only 2 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept when writing the finite difference equations for the interior nodes.

Node 1 (interior): $$100 + 120 + T_2 + T_3 - 4T_1 + \frac{\dot{g}l^2}{k} = 0$$

Node 3 (interior): $$150 + 200 + T_1 + T_4 - 4T_3 + \frac{\dot{g}l^2}{k} = 0$$



Noting that $T_1 = T_2$ and $T_3 = T_4$ and substituting,

$$220 + T_3 - 3T_1 + \frac{(10^7 \text{ W/m}^3)(0.1\text{ m})^2}{180 \text{ W/m}\cdot°\text{C}} = 0$$

$$350 + T_1 - 3T_3 + \frac{(10^7 \text{ W/m}^3)(0.1\text{ m})^2}{180 \text{ W/m}\cdot°\text{C}} = 0$$

The solution of the above system is

$$T_1 = T_2 = \mathbf{411.5°C}$$
$$T_3 = T_4 = \mathbf{439.0°C}$$

(*b*) The total rate of heat transfer from the top surface $\dot{Q}_{\text{top}}$ can be determined from an energy balance on a volume element at the top surface whose height is l/2, length 0.3 m, and depth 1 m:

$$\dot{Q}_{\text{top}} + \dot{g}_0 (0.3\times 1\times l/2) + \left(2k\frac{l\times 1}{2}\frac{120-100}{l} + 2kl\times 1\frac{T_1 - 100}{l}\right) = 0$$

$$\dot{Q}_{\text{top}} = -(10^7 \text{ W/m}^3)(0.3\times 0.1/2)\text{m}^3 - 2(180 \text{ W/m}\cdot°\text{C})\left(\frac{1\text{ m}}{2}(120-100)°\text{C} + (1\text{ m})(411.5\text{-}100)°\text{C}\right)$$

$$= \mathbf{265{,}750\ W} \quad \text{(per } m \text{ depth)}$$

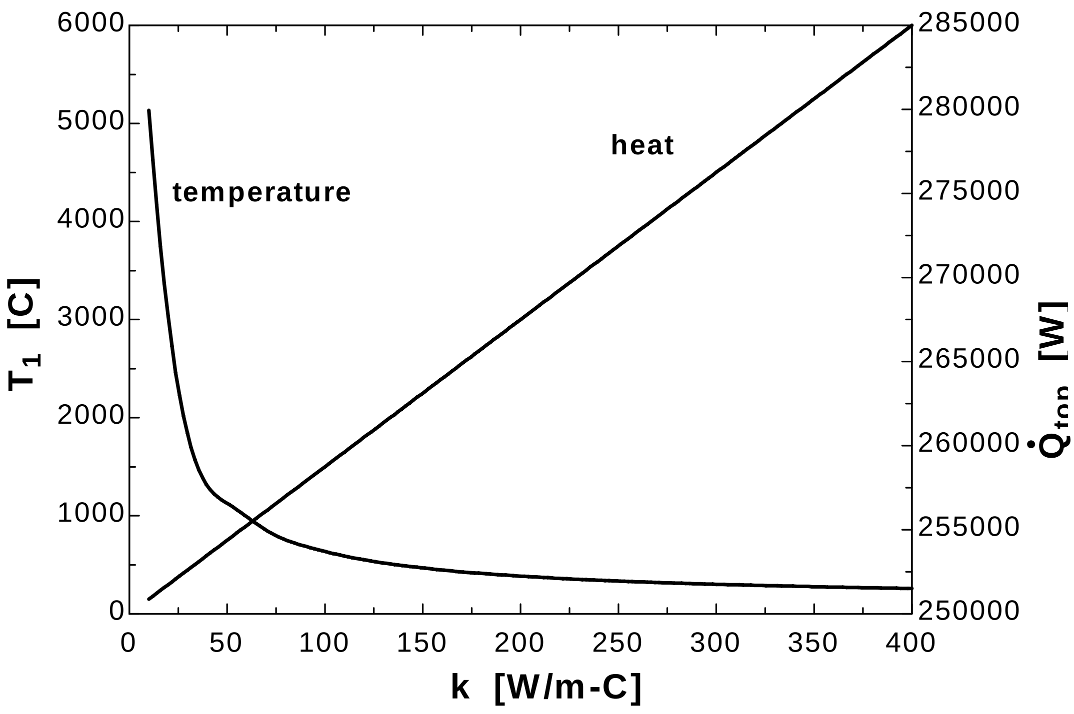
5-51 "!PROBLEM 5-51"

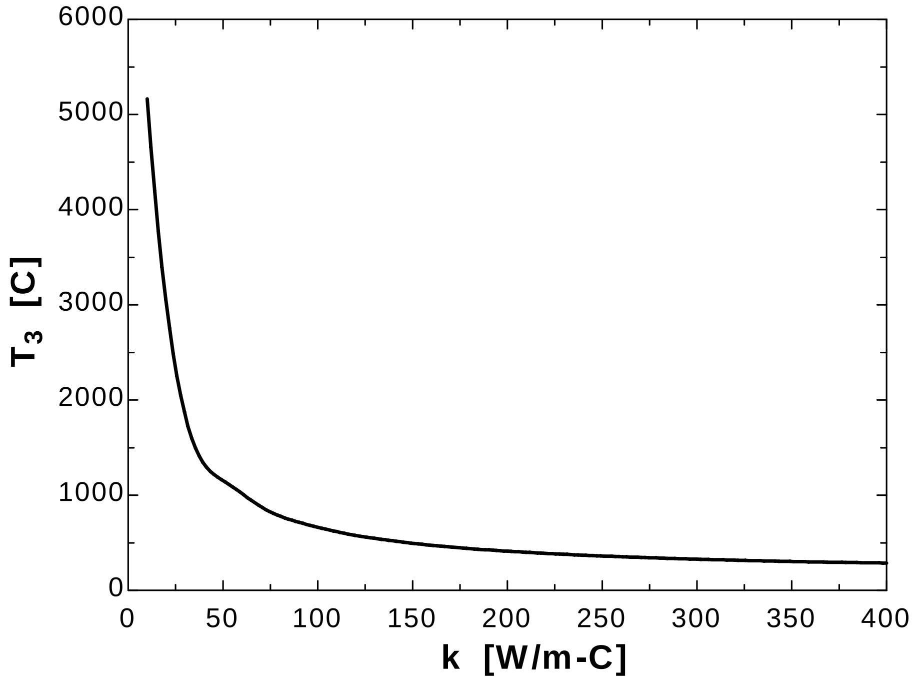
"GIVEN"
k=180 "[W/m-C], parameter to be varied"
g_dot=1E7 "[W/m^3], parameter to be varied"
DELTAx=0.10 "[m]"
DELTAy=0.10 "[m]"
d=1 "[m], depth"
"Temperatures at the selected nodes are also given in the figure"

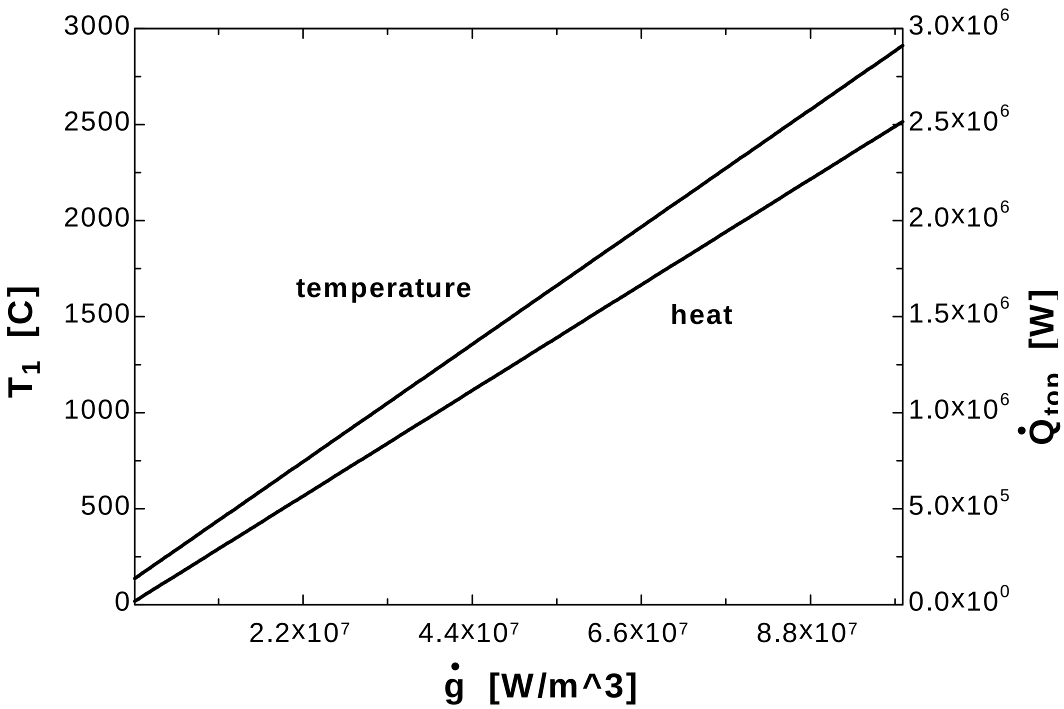
"ANALYSIS"
"(a)"
l=DELTAx
T_1=T_2 "due to symmetry"
T_3=T_4 "due to symmetry"
"Using the finite difference method, the two equations for the two unknown temperatures are determined to be"
120+120+T_2+T_3-4*T_1+(g_dot*l^2)/k=0
150+200+T_1+T_4-4*T_3+(g_dot*l^2)/k=0
"(b)"
"The rate of heat loss from the top surface can be determined from an energy balance on a volume element whose height is l/2, length 3*l, and depth d=1 m"
-Q_dot_top+g_dot*(3*l*d*l/2)+2*(k*(l*d)/2*(120-100)/l+k*l*d*(T_1-100)/l)=0

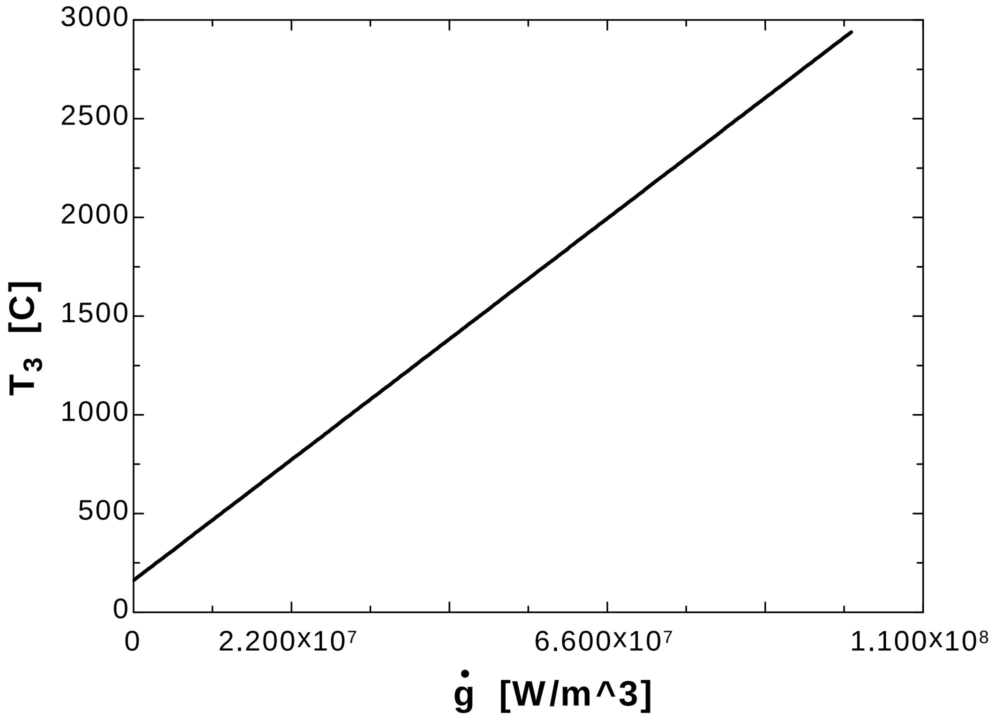
k [W/m.C]	T_1 [C]	T_3 [C]	Q_{top} [W]
10	5134	5161	250875
30.53	1772	1799	252671
51.05	1113	1141	254467
71.58	832.3	859.8	256263
92.11	676.6	704.1	258059
112.6	577.7	605.2	259855
133.2	509.2	536.7	261651
153.7	459.1	486.6	263447
174.2	420.8	448.3	265243
194.7	390.5	418	267039
215.3	366	393.5	268836
235.8	345.8	373.3	270632
256.3	328.8	356.3	272428
276.8	314.4	341.9	274224
297.4	301.9	329.4	276020
317.9	291	318.5	277816
338.4	281.5	309	279612
358.9	273	300.5	281408
379.5	265.5	293	283204
400	258.8	286.3	285000

g [W/m³]	T_1 [C]	T_3 [C]	Q_{top} [W]
100000	136.5	164	18250
5.358E+06	282.6	310.1	149697
1.061E+07	428.6	456.1	281145
1.587E+07	574.7	602.2	412592
2.113E+07	720.7	748.2	544039
2.639E+07	866.8	894.3	675487
3.165E+07	1013	1040	806934
3.691E+07	1159	1186	938382
4.216E+07	1305	1332	1.070E+06
4.742E+07	1451	1479	1.201E+06
5.268E+07	1597	1625	1.333E+06
5.794E+07	1743	1771	1.464E+06
6.319E+07	1889	1917	1.596E+06
6.845E+07	2035	2063	1.727E+06
7.371E+07	2181	2209	1.859E+06
7.897E+07	2327	2355	1.990E+06
8.423E+07	2473	2501	2.121E+06
8.948E+07	2619	2647	2.253E+06
9.474E+07	2765	2793	2.384E+06
1.000E+08	2912	2939	2.516E+06




6000
5000
4000
3000
2000
1000
0
T3 [C]
0
50
100
150
200
250
300
350
400
k [W/m-C]


3000
2500
2000
1500
1000
500
0
T1 [C]
3.0X10^6
2.5X10^6
2.0X10^6
1.5X10^6
1.0X10^6
5.0X10^5
0.0X10^0
Qtop [W]
temperature
heat
2.2X10^7
4.4X10^7
6.6X10^7
8.8X10^7
g [W/m^3]


3000
2500
2000
1500
1000
500
0
T3 [C]
0
2.200x10^7
6.600x10^7
1.100x10^8
g [W/m^3]

5-52 A long solid body is subjected to steady two-dimensional heat transfer. The unknown nodal temperatures are to be determined.

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** There is no heat generation in the body.

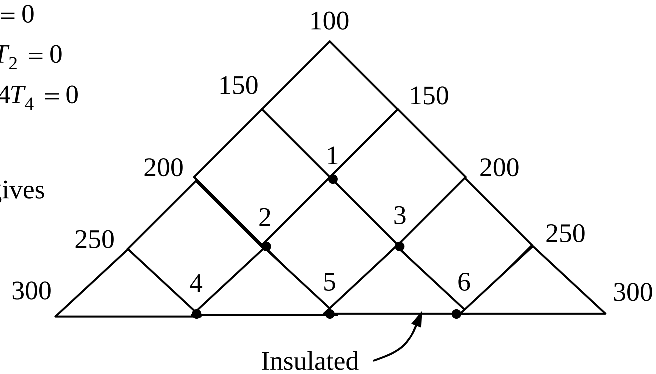
Properties The thermal conductivity is given to be k = 20 W/m·°C.

Analysis The nodal spacing is given to be Δx=Δx=l=0.01m, and the general finite difference form of an interior node for steady two-dimensional heat conduction for the case of no heat generation is expressed as

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0 \;\rightarrow\; T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} = 0$$

(a) There is symmetry about a vertical line passing through the nodes 1 and 3. Therefore, $T_3 = T_2$, $T_6 = T_4$, and $T_1, T_2, T_4,$ and T_5 are the only 4 unknown nodal temperatures, and thus we need only 4 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept when writing the finite difference equations for the interior nodes.

$$\text{Node 1 (interior)}: \quad 150 + 150 + 2T_2 - 4T_1 = 0$$
$$\text{Node 2 (interior)}: \quad 200 + T_1 + T_5 + T_4 - 4T_2 = 0$$
$$\text{Node 4 (interior)}: \quad 250 + 250 + T_2 + T_5 - 4T_4 = 0$$
$$\text{Node 5 (interior)}: \quad 4T_2 - 4T_5 = 0$$

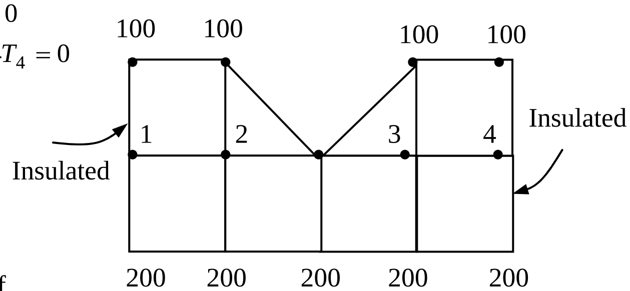


Solving the 4 equations above simultaneously gives

$T_1 = \mathbf{175°C}$
$T_2 = T_3 = \mathbf{200°C}$
$T_4 = T_6 = \mathbf{225°C}$
$T_5 = \mathbf{200°C}$

(b) There is symmetry about a vertical line passing through the middle. Therefore, $T_3 = T_2$ and $T_4 = T_1$. Replacing the symmetry lines by insulation and utilizing the mirror-image concept, the finite difference equations for the interior nodes 1 and 2 are determined to be

$$\text{Node 1 (interior)}: \quad 100 + 200 + 2T_2 - 4T_1 = 0$$
$$\text{Node 2 (interior)}: \quad 100 + 100 + 200 + T_1 - 4T_4 = 0$$



Solving the 2 equations above simultaneously gives

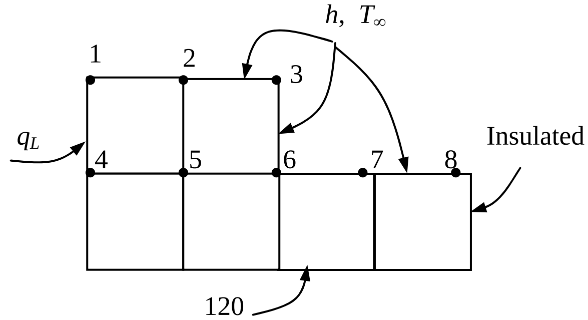
$T_1 = T_4 = \mathbf{143°C}$
$T_2 = T_3 = \mathbf{136°C}$

Discussion Note that taking advantage of symmetry simplified the problem greatly.

5-53 Heat conduction through a long L-shaped solid bar with specified boundary conditions is considered. The unknown nodal temperatures are to be determined with the finite difference method. √

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** Thermal conductivity is constant. **3** Heat generation is uniform.

Properties The thermal conductivity is given to be k = 45 W/m·°C.



Analysis (*a*) The nodal spacing is given to be $\Delta x=\Delta x=l$=0.015 m, and the general finite difference form of an interior node for steady two-dimensional heat conduction for the case of constant heat generation is expressed as

$$T_{\text{left}}+T_{\text{top}}+T_{\text{right}}+T_{\text{bottom}}-4T_{\text{node}}+\frac{\dot{g}_0 l^2}{k}=0$$

We observe that all nodes are boundary nodes except node 5 that is an interior node. Therefore, we will have to rely on energy balances to obtain the finite difference equations. Using energy balances, the finite difference equations for each of the 8 nodes are obtained as follows:

$$\text{Node 1: } \dot{q}_L\frac{l}{2}+h\frac{l}{2}(T_\infty-T_1)+k\frac{l}{2}\frac{T_2-T_1}{l}+k\frac{l}{2}\frac{T_4-T_1}{l}+\dot{g}_0\frac{l^2}{4}=0$$

$$\text{Node 2: } hl(T_\infty-T_2)+k\frac{l}{2}\frac{T_1-T_2}{l}+k\frac{l}{2}\frac{T_3-T_2}{l}+kl\frac{T_5-T_2}{l}+\dot{g}_0\frac{l^2}{2}=0$$

$$\text{Node 3: } hl(T_\infty-T_3)+k\frac{l}{2}\frac{T_2-T_3}{l}+k\frac{l}{2}\frac{T_6-T_3}{l}+\dot{g}_0\frac{l^2}{4}=0$$

$$\text{Node 4: } \dot{q}_L l+k\frac{l}{2}\frac{T_1-T_4}{l}+k\frac{l}{2}\frac{120-T_4}{l}+kl\frac{T_5-T_4}{l}+\dot{g}_0\frac{l^2}{2}=0$$

$$\text{Node 5: } T_4+T_2+T_6+120-4T_5+\frac{\dot{g}_0 l^2}{k}=0$$

$$\text{Node 6: } hl(T_\infty-T_6)+k\frac{l}{2}\frac{T_3-T_6}{l}+kl\frac{T_5-T_6}{l}+kl\frac{120-T_6}{l}+k\frac{l}{2}\frac{T_7-T_6}{l}+\dot{g}_0\frac{3l^2}{4}=0$$

$$\text{Node 7: } hl(T_\infty-T_7)+k\frac{l}{2}\frac{T_6-T_7}{l}+k\frac{l}{2}\frac{T_8-T_7}{l}+kl\frac{120-T_7}{l}+\dot{g}_0\frac{l^2}{2}=0$$

$$\text{Node 8: } h\frac{l}{2}(T_\infty-T_8)+k\frac{l}{2}\frac{T_7-T_8}{l}+k\frac{l}{2}\frac{120-T_8}{l}+\dot{g}_0\frac{l^2}{4}=0$$

where $\dot{g}_0=5\times10^6$ W/m^3, $\dot{q}_L=8000$ W/m^2, l = 0.015 m, k = 45 W/m·°C, h = 55 W/m^2·°C, and T_∞ =30°C. This system of 8 equations with 8 unknowns is the finite difference formulation of the problem.

(*b*) The 8 nodal temperatures under steady conditions are determined by solving the 8 equations above simultaneously with an equation solver to be

T_1 =163.6°C, T_2 =160.5°C, T_3 =156.4°C, T_4 =154.0°C, T_5 =151.0°C, T_6 =144.4°C, T_7 =134.5°C, T_8 =132.6°C

Discussion The accuracy of the solution can be improved by using more nodal points.

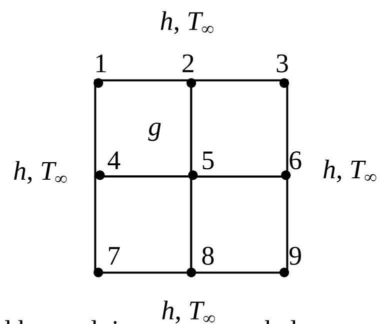
5-54E A long solid bar is subjected to steady two-dimensional heat transfer. The unknown nodal temperatures and the rate of heat loss from the bar through a 1-ft long section are to be determined.

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** Heat is generated uniformly in the body. **3** The heat transfer coefficient also includes the radiation effects.

Properties The thermal conductivity is given to be k = 16 Btu/h.ft·°C.

Analysis The nodal spacing is given to be Δx=Δx=l=0.2 ft, and the general finite difference form of an interior node for steady two-dimensional heat conduction is expressed as

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0$$



(a) There is symmetry about the vertical, horizontal, and diagonal lines passing through the center. Therefore, $T_1 = T_3 = T_7 = T_9$ and $T_2 = T_4 = T_6 = T_8$, and T_1, T_2, and T_5 are the only 3 unknown nodal temperatures, and thus we need only 3 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept for the interior nodes.

The finite difference equations for boundary nodes are obtained by applying an energy balance on the volume elements and taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (convection)}: \quad 2k\frac{l}{2}\frac{T_2 - T_1}{l} + 2h\frac{l}{2}(T_\infty - T_1) + \frac{\dot{g}_0 l^2}{4} = 0$$

$$\text{Node 2 (convection)}: \quad 2k\frac{l}{2}\frac{T_1 - T_2}{l} + kl\frac{T_5 - T_2}{l} + hl(T_\infty - T_2) + \frac{\dot{g}_0 l^2}{2} = 0$$

$$\text{Node 5 (interior)}: \quad 4T_2 - 4T_5 + \frac{\dot{g}_0 l^2}{k} = 0$$

where $\dot{g}_0 = 0.19\times10^5\ \text{Btu/h}\cdot\text{ft}^3$, l = 0.2 ft, k = 16 Btu/h.ft·°F, h =7.9 Btu/h.ft^2·°F, and T_∞ =70°F. The 3 nodal temperatures under steady conditions are determined by solving the 3 equations above simultaneously with an equation solver to be

$T_1 = T_3 = T_7 = T_9$ **=304.85°F,**

$T_2 = T_4 = T_6 = T_8$ **=316.16°F,** T_5 **=328.04°F**

(b) The rate of heat loss from the bar through a 1-ft long section is determined from an energy balance on one-eight section of the bar, and multiplying the result by 8:

$$\dot{Q} = 8\times\dot{Q}_{\text{one-eight section, conv}} = 8\times\left[h\frac{l}{2}(T_1 - T_\infty) + h\frac{l}{2}(T_2 - T_\infty)\right](1\,\text{ft}) = 8\times h\frac{l}{2}\left[T_1 + T_2 - 2T_\infty\right](1\,\text{ft})$$

$$= 8(7.9\ \text{Btu/h}\cdot\text{ft}^2\cdot°\text{F})(0.2/2\ \text{ft})(1\,\text{ft})[304.85 + 316.16 - 2\times70]°\text{F}$$

$$= \mathbf{3040\ Btu/h} \quad \text{(per ft flength)}$$

Discussion Under steady conditions, the rate of heat loss from the bar is equal to the rate of heat generation within the bar per unit length, and is determined to be

$$\dot{Q} = \dot{E}_{\text{gen}} = \dot{g}_0 V = (0.19\times10^5\ \text{Btu/h.ft}^3)(0.4\ \text{ft}\times0.4\ \text{ft}\times1\,\text{ft}) = 3040\ \text{Btu/h} \ \ \text{(per ft length)}$$

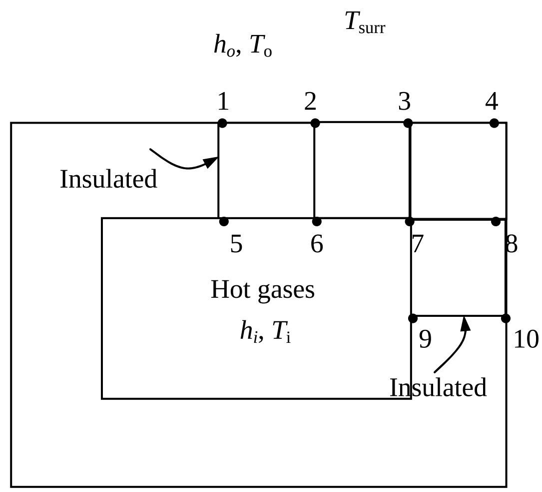
which confirms the results obtained by the finite difference method.

5-55 Heat transfer through a square chimney is considered. The nodal temperatures and the rate of heat loss per unit length are to be determined with the finite difference method.

Assumptions **1** Heat transfer is given to be steady and two-dimensional since the height of the chimney is large relative to its cross-section, and thus heat conduction through the chimney in the axial direction is negligible. It is tempting to simplify the problem further by considering heat transfer in each wall to be one dimensional which would be the case if the walls were thin and thus the corner effects were negligible. This assumption cannot be justified in this case since the walls are very thick and the corner sections constitute a considerable portion of the chimney structure. **2** There is no heat generation in the chimney. **3** Thermal conductivity is constant.

Properties The thermal conductivity and emissivity are given to be k = 1.4 W/m·°C and ε = 0.9.

Analysis (*a*) The most striking aspect of this problem is the apparent symmetry about the horizontal and vertical lines passing through the midpoint of the chimney. Therefore, we need to consider only one-fourth of the geometry in the solution whose nodal network consists of 10 equally spaced nodes. No heat can cross a symmetry line, and thus symmetry lines can be treated as insulated surfaces and thus "mirrors" in the finite-difference formulation. Considering a unit depth and using the energy balance approach for the boundary nodes (again assuming all heat transfer to be into the volume element for convenience), the finite difference formulation is obtained to be



$$\text{Node 1: } h_0\frac{l}{2}(T_0-T_1)+k\frac{l}{2}\frac{T_2-T_1}{l}+k\frac{l}{2}\frac{T_5-T_1}{l}+\varepsilon\sigma\frac{l}{2}[T_{surr}^4-(T_1+273)^4]=0$$

$$\text{Node 2: } h_0l(T_0-T_2)+k\frac{l}{2}\frac{T_1-T_2}{l}+k\frac{l}{2}\frac{T_3-T_2}{l}+kl\frac{T_6-T_2}{l}+\varepsilon\sigma l[T_{surr}^4-(T_2+273)^4]=0$$

$$\text{Node 3: } h_0l(T_0-T_3)+k\frac{l}{2}\frac{T_2-T_3}{l}+k\frac{l}{2}\frac{T_4-T_3}{l}+kl\frac{T_7-T_3}{l}+\varepsilon\sigma l[T_{surr}^4-(T_3+273)^4]=0$$

$$\text{Node 4: } h_0l(T_0-T_4)+k\frac{l}{2}\frac{T_3-T_4}{l}+k\frac{l}{2}\frac{T_8-T_4}{l}+\varepsilon\sigma l[T_{surr}^4-(T_4+273)^4]=0$$

$$\text{Node 5: } h_i\frac{l}{2}(T_i-T_5)+k\frac{l}{2}\frac{T_6-T_5}{l}+k\frac{l}{2}\frac{T_1-T_5}{l}=0$$

$$\text{Node 6: } h_il(T_i-T_6)+k\frac{l}{2}\frac{T_5-T_6}{l}+k\frac{l}{2}\frac{T_7-T_6}{l}+kl\frac{T_2-T_6}{l}=0$$

$$\text{Node 7: } h_il(T_i-T_7)+k\frac{l}{2}\frac{T_6-T_7}{l}+k\frac{l}{2}\frac{T_9-T_7}{l}+kl\frac{T_3-T_7}{l}+kl\frac{T_8-T_7}{l}=0$$

$$\text{Node 8: } h_0l(T_0-T_8)+k\frac{l}{2}\frac{T_4-T_8}{l}+k\frac{l}{2}\frac{T_{10}-T_8}{l}+kl\frac{T_7-T_8}{l}+\varepsilon\sigma l[T_{surr}^4-(T_8+273)^4]=0$$

$$\text{Node 9: } h_i\frac{l}{2}(T_i-T_9)+k\frac{l}{2}\frac{T_7-T_9}{l}+k\frac{l}{2}\frac{T_{10}-T_9}{l}=0$$

$$\text{Node 10: } h_0\frac{l}{2}(T_0-T_{10})+k\frac{l}{2}\frac{T_8-T_{10}}{l}+k\frac{l}{2}\frac{T_9-T_{10}}{l}+\varepsilon\sigma\frac{l}{2}[T_{surr}^4-(T_{10}+273)^4]=0$$

where $l = 0.1$ m, $k = 1.4$ W/m·°C, $h_i = 75$ W/m^2·°C, T_i =280°C, $h_o = 18$ W/m^2·°C, T_0 =15°C, T_{surr} =250 K, $\varepsilon = 0.9$, and $\sigma = 5.67\times10^{-8}$ W/m^2.K^4. This system of 10 equations with 10 unknowns constitutes the finite difference formulation of the problem.

(*b*) The 10 nodal temperatures under steady conditions are determined by solving the 10 equations above simultaneously with an equation solver to be

$\boldsymbol{T_1}$ **=94.5°C,** $\boldsymbol{T_2}$ **=93.0°C,** $\boldsymbol{T_3}$ **=82.1°C,** $\boldsymbol{T_4}$ **=36.1°C,** $\boldsymbol{T_5}$ **=250.6°C,**

$\boldsymbol{T_6}$ **=249.2°C,** $\boldsymbol{T_7}$ **=229.7°C,** $\boldsymbol{T_8}$ **=82.3°C,** $\boldsymbol{T_9}$ **=261.5°C,** $\boldsymbol{T_{10}}$ **=94.6°C**

(*c*) The rate of heat loss through a 1-m long section of the chimney is determined from

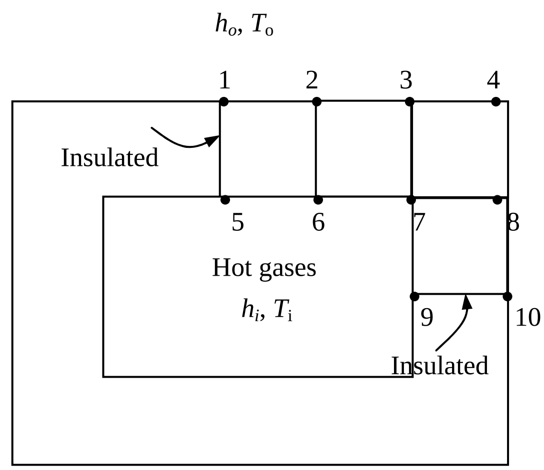
$$
\begin{aligned}
\dot{Q} &= 4\sum \dot{Q}_{\text{one-fourth of chimney}} = 4\sum \dot{Q}_{\text{element, inner surface}} = 4\sum_m h_i A_{\text{surface},m}(T_i - T_m) \\
&= 4[h_i(l/2)(T_i - T_5) + h_i l(T_i - T_6) + h_i l(T_i - T_7) + h_i(l/2)(T_i - T_9)] \\
&= 4(75 \text{ W/m}^2\cdot°\text{C})(0.1\text{ m}\times 1\text{ m})[(280-250.6)/2 + (280-249.2) + (280-229.7) + (280-261.5)/2]°\text{C} \\
&= \mathbf{3153\ \ W}
\end{aligned}
$$

Discussion The rate of heat transfer can also be determined by calculating the heat loss from the outer surface by convection and radiation.

5-56 Heat transfer through a square chimney is considered. The nodal temperatures and the rate of heat loss per unit length are to be determined with the finite difference method.

Assumptions **1** Heat transfer is given to be steady and two-dimensional since the height of the chimney is large relative to its cross-section, and thus heat conduction through the chimney in the axial direction is negligible. It is tempting to simplify the problem further by considering heat transfer in each wall to be one dimensional which would be the case if the walls were thin and thus the corner effects were negligible. This assumption cannot be justified in this case since the walls are very thick and the corner sections constitute a considerable portion of the chimney structure. **2** There is no heat generation in the chimney. **3** Thermal conductivity is constant. **4** Radiation heat transfer is negligible.

Properties The thermal conductivity of chimney is given to be k = 1.4 W/m·°C.



Analysis (*a*) The most striking aspect of this problem is the apparent symmetry about the horizontal and vertical lines passing through the midpoint of the chimney. Therefore, we need to consider only one-fourth of the geometry in the solution whose nodal network consists of 10 equally spaced nodes. No heat can cross a symmetry line, and thus symmetry lines can be treated as insulated surfaces and thus "mirrors" in the finite-difference formulation. Considering a unit depth and using the energy balance approach for the boundary nodes (again assuming all heat transfer to be into the volume element for convenience), the finite difference formulation is obtained to be

Node 1: $h_0 \frac{l}{2}(T_0 - T_1) + k\frac{l}{2}\frac{T_2 - T_1}{l} + k\frac{l}{2}\frac{T_5 - T_1}{l} = 0$

Node 2: $h_0 l(T_0 - T_2) + k\frac{l}{2}\frac{T_1 - T_2}{l} + k\frac{l}{2}\frac{T_3 - T_2}{l} + kl\frac{T_6 - T_2}{l} = 0$

Node 3: $h_0 l(T_0 - T_3) + k\frac{l}{2}\frac{T_2 - T_3}{l} + k\frac{l}{2}\frac{T_4 - T_3}{l} + kl\frac{T_7 - T_3}{l} = 0$

Node 4: $h_0 l(T_0 - T_4) + k\frac{l}{2}\frac{T_3 - T_4}{l} + k\frac{l}{2}\frac{T_8 - T_4}{l} = 0$

Node 5: $h_i \frac{l}{2}(T_i - T_5) + k\frac{l}{2}\frac{T_6 - T_5}{l} + k\frac{l}{2}\frac{T_1 - T_5}{l} = 0$

Node 6: $h_i l(T_i - T_6) + k\frac{l}{2}\frac{T_5 - T_6}{l} + k\frac{l}{2}\frac{T_7 - T_6}{l} + kl\frac{T_2 - T_6}{l} = 0$

Node 7: $h_i l(T_i - T_7) + k\frac{l}{2}\frac{T_6 - T_7}{l} + k\frac{l}{2}\frac{T_9 - T_7}{l} + kl\frac{T_3 - T_7}{l} + kl\frac{T_8 - T_7}{l} = 0$

Node 8: $h_0 l(T_0 - T_8) + k\frac{l}{2}\frac{T_4 - T_8}{l} + k\frac{l}{2}\frac{T_{10} - T_8}{l} + kl\frac{T_7 - T_8}{l} = 0$

Node 9: $h_i \frac{l}{2}(T_i - T_9) + k\frac{l}{2}\frac{T_7 - T_9}{l} + k\frac{l}{2}\frac{T_{10} - T_9}{l} = 0$

Node 10: $h_0 \frac{l}{2}(T_0 - T_{10}) + k\frac{l}{2}\frac{T_8 - T_{10}}{l} + k\frac{l}{2}\frac{T_9 - T_{10}}{l} = 0$

where l = 0.1 m, k = 1.4 W/m·°C, h_i = 75 W/m^2·°C, T_i =280°C, h_o = 18 W/m^2·°C, T_0 =15°C, and σ = 5.67×10^{-8} W/m^2.K^4. This system of 10 equations with 10 unknowns constitutes the finite difference formulation of the problem.

(*b*) The 10 nodal temperatures under steady conditions are determined by solving the 10 equations above simultaneously with an equation solver to be

$$\boldsymbol{T_1 = 118.8°C,\quad T_2 = 116.7°C,\quad T_3 = 103.4°C,\quad T_4 = 53.7°C,\quad T_5 = 254.4°C,}$$

$$\boldsymbol{T_6 = 253.0°C,\quad T_7 = 235.2°C,\quad T_8 = 103.5°C,\quad T_9 = 263.7°C,\quad T_{10} = 117.6°C}$$

(*c*) The rate of heat loss through a 1-m long section of the chimney is determined from

$$\begin{aligned}
\dot{Q} &= 4\sum \dot{Q}_{\text{one-fourth of chimney}} = 4\sum \dot{Q}_{\text{element, inner surface}} = 4\sum_m h_i A_{\text{surface},m}(T_i - T_m) \\
&= 4[h_i(l/2)(T_i - T_5) + h_i l(T_i - T_6) + h_i l(T_i - T_7) + h_i(l/2)(T_i - T_9)] \\
&= 4(75\ \text{W/m}^2\cdot°\text{C})(0.1\ \text{m}\times 1\ \text{m})[(280-254.4)/2 + (280-253.0) + (280-235.2) + (280-263.7)/2]°\text{C} \\
&= \mathbf{2783\ \ W}
\end{aligned}$$

Discussion The rate of heat transfer can also be determined by calculating the heat loss from the outer surface by convection.

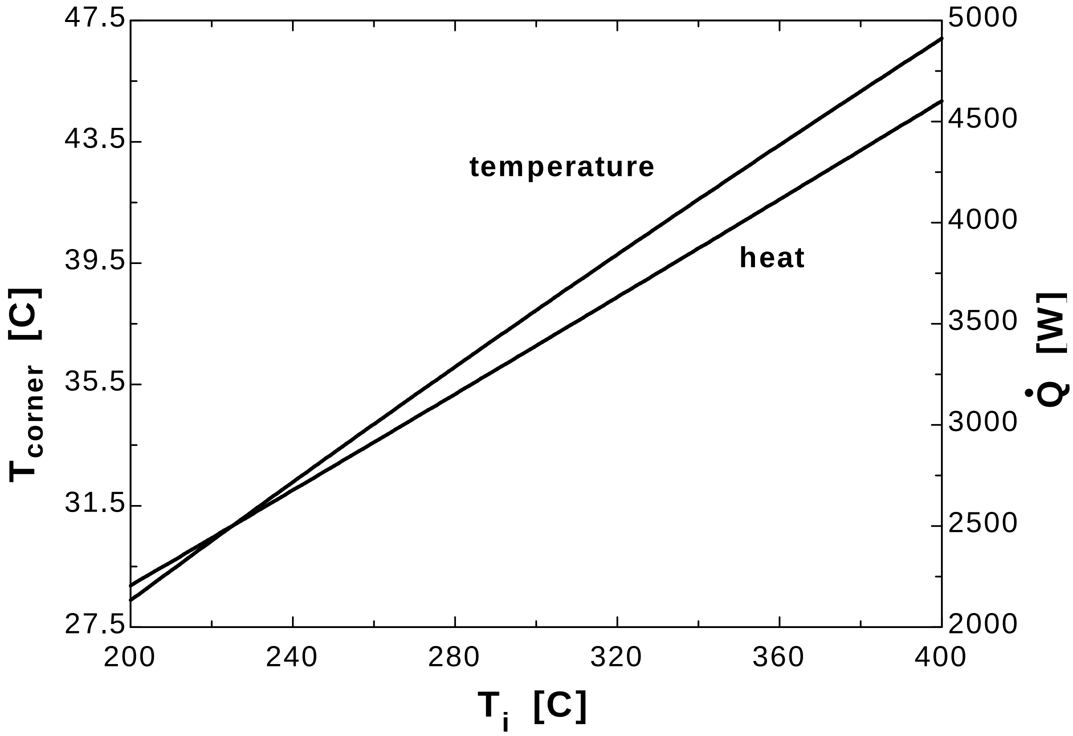
5-57 "!PROBLEM 5-57"

```
"GIVEN"
k=1.4 "[W/m-C]"
A_flow=0.20*0.40 "[m^2]"
t=0.10 "[m]"
T_i=280 "[C], parameter to ve varied"
h_i=75 "[W/m^2-C]"
T_o=15 "[C]"
h_o=18 "[W/m^2-C]"
epsilon=0.9 "parameter to ve varied"
T_sky=250 "[K]"
DELTAx=0.10 "[m]"
DELTAy=0.10 "[m]"
d=1 "[m], unit depth is considered"
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"

"ANALYSIS"
"(b)"
l=DELTAx
"We consider only one-fourth of the geometry whose nodal network consists of 10 nodes. Using
the finite difference method, 10 equations for 10 unknown temperatures are determined to be"
h_o*l/2*(T_o-T_1)+k*l/2*(T_2-T_1)/l+k*l/2*(T_5-T_1)/l+epsilon*sigma*l/2*(T_sky^4-
(T_1+273)^4)=0 "Node 1"
h_o*l*(T_o-T_2)+k*l/2*(T_1-T_2)/l+k*l/2*(T_3-T_2)/l+k*l*(T_6-T_2)/l+epsilon*sigma*l*(T_sky^4-
(T_2+273)^4)=0 "Node 2"
h_o*l*(T_o-T_3)+k*l/2*(T_2-T_3)/l+k*l/2*(T_4-T_3)/l+k*l*(T_7-T_3)/l+epsilon*sigma*l*(T_sky^4-
(T_3+273)^4)=0 "Node 3"
h_o*l*(T_o-T_4)+k*l/2*(T_3-T_4)/l+k*l/2*(T_8-T_4)/l+epsilon*sigma*l*(T_sky^4-(T_4+273)^4)=0
"Node 4"
h_i*l/2*(T_i-T_5)+k*l/2*(T_6-T_5)/l+k*l/2*(T_1-T_5)/l=0 "Node 5"
h_i*l*(T_i-T_6)+k*l/2*(T_5-T_6)/l+k*l/2*(T_7-T_6)/l+k*l*(T_2-T_6)/l=0 "Node 6"
h_i*l*(T_i-T_7)+k*l/2*(T_6-T_7)/l+k*l/2*(T_9-T_7)/l+k*l*(T_3-T_7)/l+k*l*(T_8-T_7)/l=0 "Node 7"
h_o*l*(T_o-T_8)+k*l/2*(T_4-T_8)/l+k*l/2*(T_10-T_8)/l+k*l*(T_7-T_8)/l+epsilon*sigma*l*(T_sky^4-
(T_8+273)^4)=0 "Node 8"
h_i*l*(T_i-T_9)+k*l/2*(T_7-T_9)/l+k*l/2*(T_10-T_9)/l=0 "Node 9"
h_o*l/2*(T_o-T_10)+k*l/2*(T_8-T_10)/l+k*l/2*(T_9-T_10)/l+epsilon*sigma*l/2*(T_sky^4-
(T_10+273)^4)=0 "Node 10"
"Right top corner is considered. The locations of nodes are as follows:"
"Node 1: Middle of top surface
Node 2: At the right side of node 1
Node 3: At the right side of node 2
Node 4: Corner node
Node 5: The node below node 1, at the middle of inner top surface
Node 6: The node below node 2
Node 7: The node below node 3, at the inner corner
Node 8: The node below node 4
Node 9: The node below node 7,at the middle of inner right surface
Node 10: The node below node 8, at the middle of outer right surface"
T_corner=T_4
T_inner_middle=T_9
"(c)"
"The rate of heat loss through a unit depth d=1 m of the chimney is"
Q_dot=4*(h_i*l/2*d*(T_i-T_5)+h_i*l*d*(T_i-T_6)+h_i*l*d*(T_i-T_7)+h_i*l/2*d*(T_i-T_9))
```

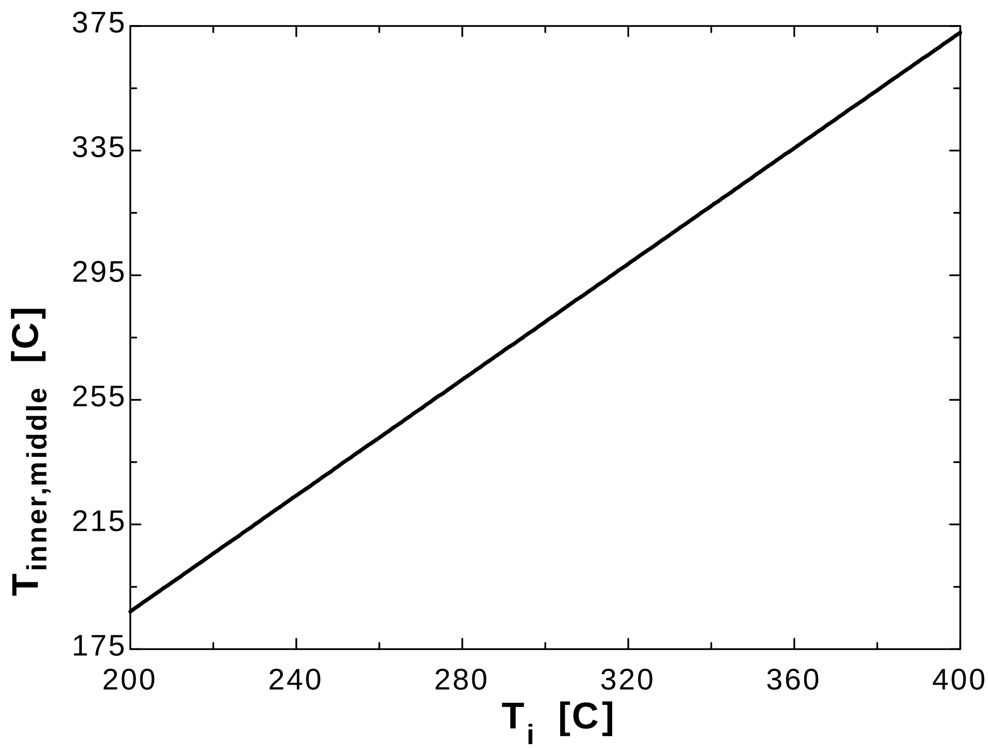
T_i [C]	T_{corner} [C]	$T_{inner, middle}$ [C]	Q [W]
200	28.38	187	2206
210	29.37	196.3	2323
220	30.35	205.7	2441

230	31.32	215	2559
240	32.28	224.3	2677
250	33.24	233.6	2796
260	34.2	242.9	2914
270	35.14	252.2	3033
280	36.08	261.5	3153
290	37.02	270.8	3272
300	37.95	280.1	3392
310	38.87	289.3	3512
320	39.79	298.6	3632
330	40.7	307.9	3752
340	41.6	317.2	3873
350	42.5	326.5	3994
360	43.39	335.8	4115
370	44.28	345.1	4237
380	45.16	354.4	4358
390	46.04	363.6	4480
400	46.91	372.9	4602

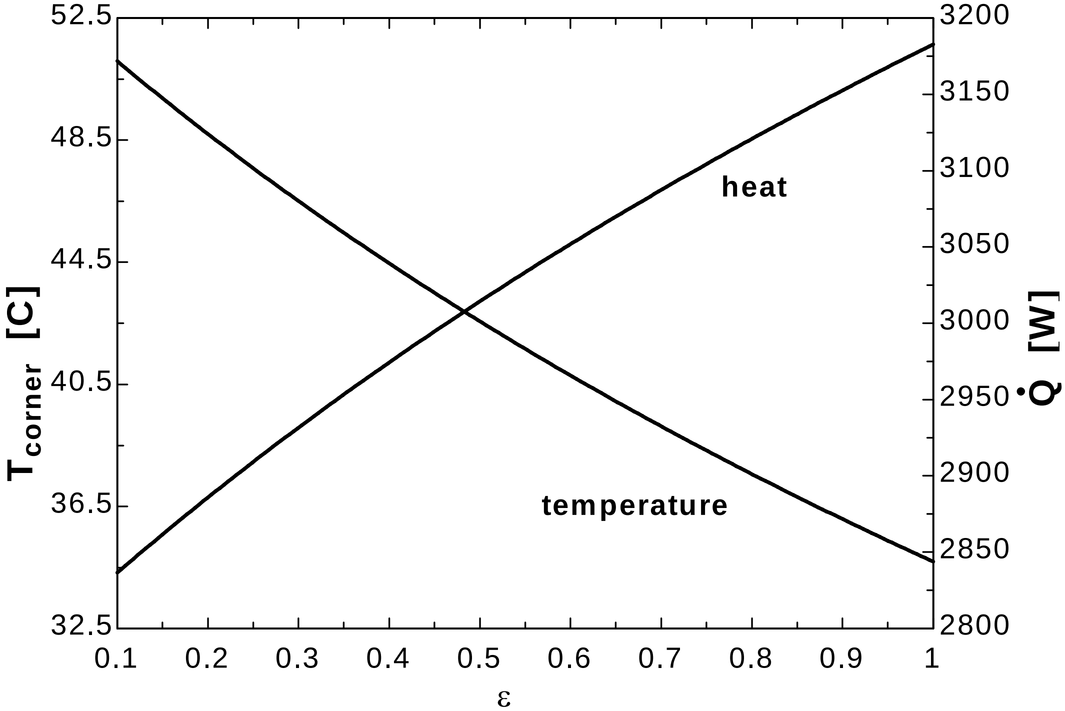
ε	**T_{corner} [C]**	**$T_{inner,\ middle}$ [C]**	**Q [W]**
0.1	51.09	263.4	2836
0.15	49.87	263.2	2862
0.2	48.7	263.1	2886
0.25	47.58	262.9	2909
0.3	46.5	262.8	2932
0.35	45.46	262.7	2953
0.4	44.46	262.5	2974
0.45	43.5	262.4	2995
0.5	42.56	262.3	3014
0.55	41.66	262.2	3033
0.6	40.79	262.1	3052
0.65	39.94	262	3070
0.7	39.12	261.9	3087
0.75	38.33	261.8	3104
0.8	37.56	261.7	3121
0.85	36.81	261.6	3137
0.9	36.08	261.5	3153
0.95	35.38	261.4	3168
1	34.69	261.3	3183



47.5
43.5
39.5
35.5
31.5
27.5
5000
4500
4000
3500
3000
2500
2000
200
240
280
320
360
400
temperature
heat
T corner [C]
Q [W]
T i [C]



375
335
295
255
215
175
200
240
280
320
360
400
T inner,middle [C]
T i [C]



52.5
48.5
44.5
40.5
36.5
32.5
T_{corner} [C]
3200
3150
3100
3050
3000
2950
2900
2850
2800
$\dot{Q}$ [W]
heat
temperature
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
ε

5-58 The exposed surface of a long concrete damn of triangular cross-section is subjected to solar heat flux and convection and radiation heat transfer. The vertical section of the damn is subjected to convection with water. The temperatures at the top, middle, and bottom of the exposed surface of the damn are to be determined.

Assumptions **1** Heat transfer through the damn is given to be steady and two-dimensional. **2** There is no heat generation within the damn. **3** Heat transfer through the base is negligible. **4** Thermal properties and heat transfer coefficients are constant.

Properties The thermal conductivity and solar absorptivity are given to be k = 0.6 W/m·°C and α_s = 0.7.

Analysis The nodal spacing is given to be $\Delta x=\Delta x=l=1$ m, and all nodes are boundary nodes. Node 5 on the insulated boundary can be treated as an interior node for which $T_{\text{left}}+T_{\text{top}}+T_{\text{right}}+T_{\text{bottom}}-4T_{\text{node}}=0$. Using the energy balance approach and taking the direction of all heat transfer to be towards the node, the finite difference equations for the nodes are obtained to be as follows:

Node 1: $h_i\dfrac{l}{2}(T_i-T_1)+k\dfrac{l}{2}\dfrac{T_2-T_1}{l}+\dfrac{l/2}{\sin 45}\left[\alpha_s\dot{q}_s+h_0(T_0-T_1)\right]=0$

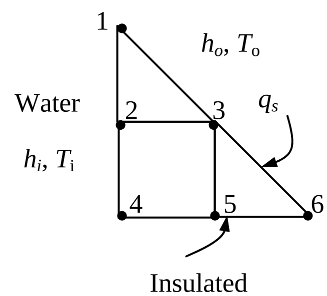
Node 2: $h_i l(T_i-T_1)+k\dfrac{l}{2}\dfrac{T_1-T_2}{l}+k\dfrac{l}{2}\dfrac{T_4-T_2}{l}+kl\dfrac{T_3-T_2}{l}=0$

Node 3: $kl\dfrac{T_2-T_3}{l}+kl\dfrac{T_5-T_3}{l}+\dfrac{l}{\sin 45}\left[\alpha_s\dot{q}_s+h_0(T_0-T_3)\right]=0$

Node 4: $h_i\dfrac{l}{2}(T_i-T_4)+k\dfrac{l}{2}\dfrac{T_2-T_4}{l}+k\dfrac{l}{2}\dfrac{T_5-T_4}{l}=0$

Node 5: $T_4+2T_3+T_6-4T_5=0$

Node 6: $k\dfrac{l}{2}\dfrac{T_5-T_6}{l}+\dfrac{l/2}{\sin 45}\left[\alpha_s\dot{q}_s+h_0(T_0-T_6)\right]=0$



where l = 1 m, k = 0.6 W/m·°C, h_i =150 W/m^2·°C, T_i =15°C, h_o = 30 W/m^2·°C, T_0 =25°C, α_s = 0.7, and $\dot{q}_s = 800\ \text{W/m}^2$. The system of 6 equations with 6 unknowns constitutes the finite difference formulation of the problem. The 6 nodal temperatures under steady conditions are determined by solving the 6 equations above simultaneously with an equation solver to be

$T_1 = T_{\text{top}}$ =**21.3**°C, T_2 =15.1°C, $T_3 = T_{\text{middle}}$ =**43.2**°C, T_4 =15.1°C, T_5 =36.3°C, $T_6 = T_{\text{bottom}}$ =**43.6**°C

Discussion Note that the highest temperature occurs at a location furthest away from the water, as expected.

5-59E The top and bottom surfaces of a V-grooved long solid bar are maintained at specified temperatures while the left and right surfaces are insulated. The temperature at the middle of the insulated surface is to be determined.

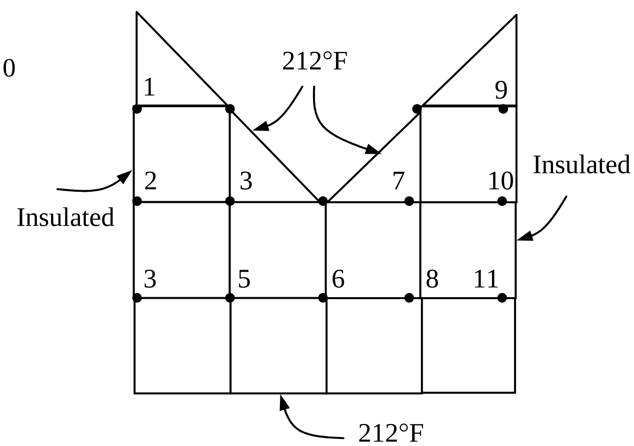
Assumptions 1 Heat transfer through the bar is given to be steady and two-dimensional. **2** There is no heat generation within the bar. **3** Thermal properties are constant.

Analysis The nodal spacing is given to be $\Delta x=\Delta y=l=1$ ft, and the general finite difference form of an interior node for steady two-dimensional heat conduction with no heat generation is expressed as

$$T_{\text{left}}+T_{\text{top}}+T_{\text{right}}+T_{\text{bottom}}-4T_{\text{node}}+\frac{\dot{g}_{\text{node}}l^2}{k}=0 \;\rightarrow\; T_{\text{left}}+T_{\text{top}}+T_{\text{right}}+T_{\text{bottom}}-4T_{\text{node}}=0$$

There is symmetry about the vertical plane passing through the center. Therefore, $T_1 = T_9$, $T_2 = T_{10}$, $T_3 = T_{11}$, $T_4 = T_7$, and $T_5 = T_8$. Therefore, there are only 6 unknown nodal temperatures, and thus we need only 6 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept when writing the finite difference equations for the interior nodes.

The finite difference equations for boundary nodes are obtained by applying an energy balance on the volume elements and taking the direction of all heat transfers to be towards the node under consideration:



Node 1: $k\frac{l}{2}\frac{32-T_1}{l}+kl\frac{32-T_1}{l}+k\frac{l}{2}\frac{T_2-T_1}{l}=0$

(Note that k and l cancel out)

Node 2: $T_1+2T_4+T_3-4T_2=0$

Node 3: $T_2+212+2T_5-4T_3=0$

Node 4: $2\times 32+T_2+T_5-4T_4=0$

Node 5: $T_3+212+T_4+T_6-4T_5=0$

Node 6: $32+212+2T_5-4T_6=0$

The 6 nodal temperatures under steady conditions are determined by solving the 6 equations above simultaneously with an equation solver to be

T_1 = 44.7°F, T_2 =82.8°F, T_3 =143.4°F, T_4 = 71.6°F, T_5 =139.4°F, T_6 =130.7°F

Therefore, the temperature at the middle of the insulated surface will be $\boldsymbol{T_2}$ **=82.8°F.**

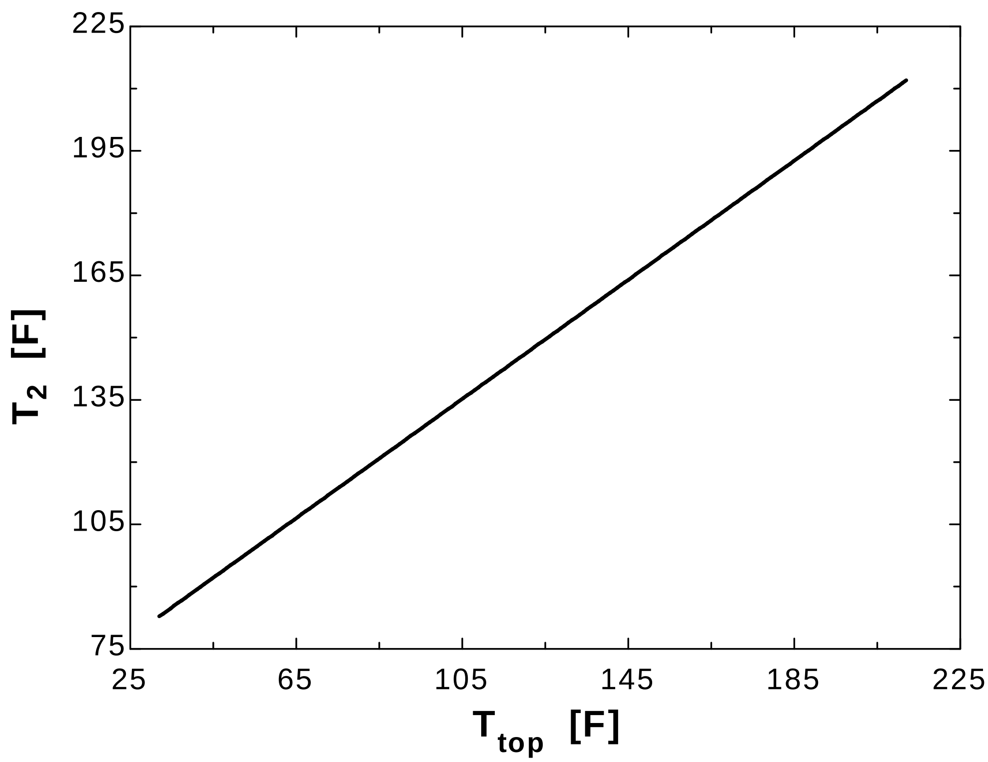
5-60 "!PROBLEM 5-60E"

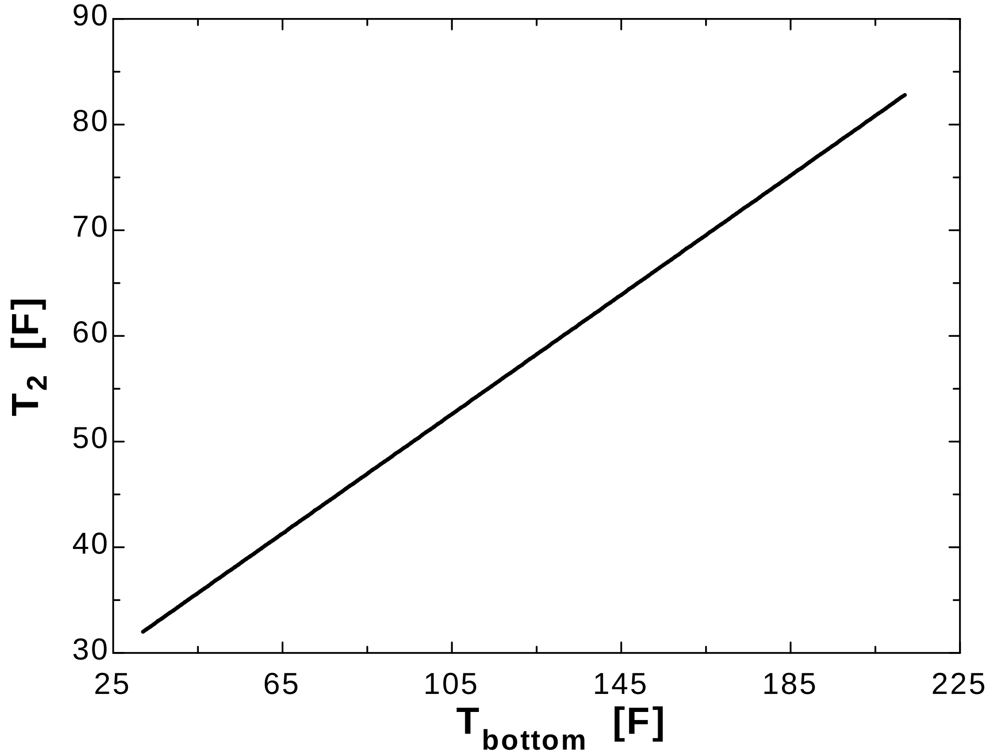
"GIVEN"
T_top=32 "[F], parameter to be varied"
T_bottom=212 "[F], parameter to be varied"
DELTAx=1 "[ft]"
DELTAy=1 "[ft]"

"ANALYSIS"
l=DELTAx
T_1=T_9 "due to symmetry"
T_2=T_10 "due to symmetry"
T_3=T_11 "due to symmetry"
T_4=T_7 "due to symmetry"
T_5=T_8 "due to symmetry"
"Using the finite difference method, the six equations for the six unknown temperatures are determined to be"
"k*l/2*(T_top-T_1)/l+k*l*(T_top-T_1)/l+k*l/2*(T_2-T_1)/l=0 simplifies to for Node 1"
1/2*(T_top-T_1)+(T_top-T_1)+1/2*(T_2-T_1)=0 "Node 1"
T_1+2*T_4+T_3-4*T_2=0 "Node 2"
T_2+T_bottom+2*T_5-4*T_3=0 "Node 3"
2*T_top+T_2+T_5-4*T_4=0 "Node 4"
T_3+T_bottom+T_4+T_6-4*T_5=0 "Node 5"
T_top+T_bottom+2*T_5-4*T_6=0 "Node 6"

T_{top} [F]	T_2 [F]
32	82.81
41.47	89.61
50.95	96.41
60.42	103.2
69.89	110
79.37	116.8
88.84	123.6
98.32	130.4
107.8	137.2
117.3	144
126.7	150.8
136.2	157.6
145.7	164.4
155.2	171.2
164.6	178
174.1	184.8
183.6	191.6
193.1	198.4
202.5	205.2
212	212

T_{bottom} [F]	T_2 [F]
32	32
41.47	34.67
50.95	37.35
60.42	40.02
69.89	42.7
79.37	45.37
88.84	48.04
98.32	50.72
107.8	53.39
117.3	56.07
126.7	58.74
136.2	61.41
145.7	64.09
155.2	66.76
164.6	69.44
174.1	72.11
183.6	74.78
193.1	77.46
202.5	80.13
212	82.81





90
80
70
60
50
40
30
T2 [F]
25
65
105
145
185
225
Tbottom [F]

5-61 The top and bottom surfaces of an L-shaped long solid bar are maintained at specified temperatures while the left surface is insulated and the remaining 3 surfaces are subjected to convection. The finite difference formulation of the problem is to be obtained, and the unknown nodal temperatures are to be determined.

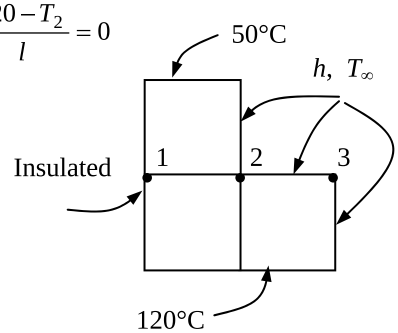
Assumptions **1** Heat transfer through the bar is given to be steady and two-dimensional. **2** There is no heat generation within the bar. **3** Thermal properties and heat transfer coefficients are constant. **4** Radiation heat transfer is negligible.

Properties The thermal conductivity is given to be k = 12 W/m·°C.

Analysis (*a*) The nodal spacing is given to be Δx=Δx=l=0.1 m, and all nodes are boundary nodes. Node 1 on the insulated boundary can be treated as an interior node for which $T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} = 0$. Using the energy balance approach and taking the direction of all heat transfer to be towards the node, the finite difference equations for the nodes are obtained to be as follows:

Node 1: $50 + 120 + 2T_2 - 4T_1 = 0$

Node 2: $hl(T_\infty - T_2) + k\dfrac{l}{2}\dfrac{50 - T_2}{l} + k\dfrac{l}{2}\dfrac{T_3 - T_2}{l} + kl\dfrac{T_1 - T_2}{l} + kl\dfrac{120 - T_2}{l} = 0$



Node 3: $hl(T_\infty - T_3) + k\dfrac{l}{2}\dfrac{T_2 - T_3}{l} + k\dfrac{l}{2}\dfrac{120 - T_3}{l} = 0$

where l = 0.1 m, k = 12 W/m·°C, h =30 W/m^2·°C, and T_∞ =25°C. This system of 3 equations with 3 unknowns constitute the finite difference formulation of the problem.

(*b*) The 3 nodal temperatures under steady conditions are determined by solving the 3 equations above simultaneously with an equation solver to be

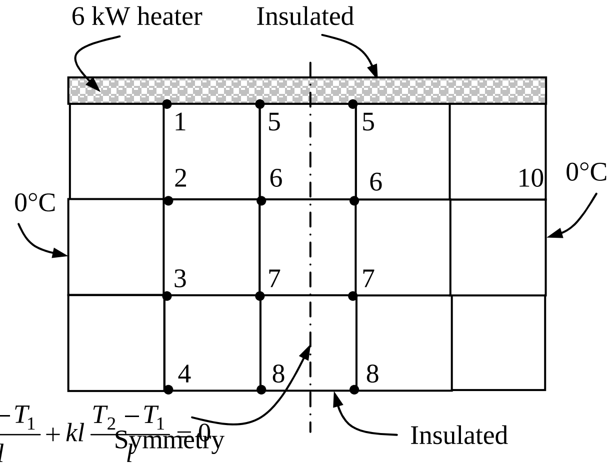
$\boldsymbol{T_1}$ **= 85.7°C,** $\boldsymbol{T_2}$ **=86.4°C,** $\boldsymbol{T_3}$ **=87.6°C**

5-62 A rectangular block is subjected to uniform heat flux at the top, and iced water at 0°C at the sides. The steady finite difference formulation of the problem is to be obtained, and the unknown nodal temperatures as well as the rate of heat transfer to the iced water are to be determined.

Assumptions **1** Heat transfer through the body is given to be steady and two-dimensional. **2** There is no heat generation within the block. **3** The heat transfer coefficient is very high so that the temperatures on both sides of the block can be taken to be 0°C. **4** Heat transfer through the bottom surface is negligible.

Properties The thermal conductivity is given to be k = 23 W/m·°C.

Analysis The nodal spacing is given to be Δx=Δx=l=0.1 m, and the general finite difference form of an interior node equation for steady 2-D heat conduction is expressed as



$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0$$

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} = 0$$

There is symmetry about a vertical line passing through the middle of the region, and we need to consider only half of the region. Note that all side surfaces are at T_0 = 0°C, and there are 8 nodes with unknown temperatures. Replacing the symmetry lines by insulation and utilizing the mirror-image concept, the finite difference equations are obtained to be as follows:

Node 1 (heat flux): $\dot{q}_0 l + k\frac{l}{2}\frac{T_0 - T_1}{l} + k\frac{l}{2}\frac{T_5 - T_1}{l} + kl\frac{T_2 - T_1}{l} = 0$

Node 2 (interior): $T_0 + T_1 + T_3 + T_6 - 4T_2 = 0$

Node 3 (interior): $T_0 + T_2 + T_4 + T_7 - 4T_3 = 0$

Node 4 (insulation): $T_0 + 2T_3 + T_8 - 4T_4 = 0$

Node 5 (heat flux): $\dot{q}_0 l + k\frac{l}{2}\frac{T_1 - T_5}{l} + kl\frac{T_6 - T_5}{l} + 0 = 0$

Node 6 (interior): $T_2 + T_5 + T_6 + T_7 - 4T_6 = 0$

Node 7 (interior): $T_3 + T_6 + T_7 + T_8 - 4T_7 = 0$

Node 8 (insulation): $T_4 + 2T_7 + T_8 - 4T_8 = 0$

where l = 0.1 m, k = 23 W/m·°C, T_0 =0°C, and $\dot{q}_0 = \dot{Q}_0 / A = (6000\text{ W})/(5\times 0.5\text{ m}^2) = 2400\text{ W/m}^2$. This system of 8 equations with 8 unknowns constitutes the finite difference formulation of the problem.

(*b*) The 8 nodal temperatures under steady conditions are determined by solving the 8 equations above simultaneously with an equation solver to be

T_1 =13.7°C, T_2 =7.4°C, T_3 =4.7°C, T_4 =3.9°C, T_5 =19.0°C, T_6 =11.3°C, T_7 =7.4°C, T_8 =6.2°C

(*c*) The rate of heat transfer from the block to the iced water is 6 kW since all the heat supplied to the block from the top must be equal to the heat transferred from the block. Therefore, $\dot{Q} = \mathbf{6\,kW}$.

Discussion The rate of heat transfer can also be determined by calculating the heat loss from the side surfaces using the heat conduction relation.

Transient Heat Conduction

5-63C The formulation of a transient heat conduction problem differs from that of a steady heat conduction problem in that the transient problem involves an *additional term* that represents the *change in the energy content* of the medium with time. This additional term $\rho A \Delta x C (T_m^{i+1} - T_m^i)/\Delta t$ represent the change in the internal energy content during Δt in the transient finite difference formulation.

5-64C The two basic methods of solution of transient problems based on finite differencing are the *explicit* and the *implicit methods*. The heat transfer terms are expressed at time step *i* in the explicit method, and at the future time step *i* + 1 in the implicit method as

Explicit method: $$\sum_{\text{All sides}} \dot{Q}^i + \dot{G}^{\text{i}}_{\text{element}} = \rho V_{\text{element}} C \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

Implicit method: $$\sum_{\text{All sides}} \dot{Q}^{i+1} + \dot{G}^{\text{i+1}}_{\text{element}} = \rho V_{\text{element}} C \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

5-65C The explicit finite difference formulation of a general interior node for transient heat conduction in a plane wall is given by $T_{m-1}^i - 2T_m^i + T_{m+1}^i + \dfrac{\dot{g}_m^i \Delta x^2}{k} = \dfrac{T_m^{i+1} - T_m^i}{\tau}$. The finite difference formulation for the steady case is obtained by simply setting $T_m^{i+1} = T_m^i$ and disregarding the time index *i*. It yields

$$T_{m-1} - 2T_m + T_{m+1} + \frac{\dot{g}_m \Delta x^2}{k} = 0$$

5-66C The explicit finite difference formulation of a general interior node for transient two-dimensional heat conduction is given by $T_{\text{node}}^{i+1} = \tau(T^{\text{i}}_{\text{left}} + T^{\text{i}}_{\text{top}} + T^{\text{i}}_{\text{right}} + T^{\text{i}}_{\text{bottom}}) + (1 - 4\tau)T^{\text{i}}_{\text{node}} + \tau \dfrac{\dot{g}^{\text{i}}_{\text{node}} l^2}{k}$. The finite difference formulation for the steady case is obtained by simply setting $T_m^{i+1} = T_m^i$ and disregarding the time index *i*. It yields

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} - 4T_{\text{node}} + \frac{\dot{g}_{\text{node}} l^2}{k} = 0$$

5-67C There is a limitation on the size of the time step Δt in the solution of transient heat conduction problems using the explicit method, but there is no such limitation in the implicit method.

5-68C The general stability criteria for the explicit method of solution of transient heat conduction problems is expressed as follows: *The coefficients of all* T_m^i *in the* T_m^{i+1} *expressions* (*called the primary coefficient*) *in the simplified expressions must be greater than or equal to zero for all nodes m.*

5-69C For transient one-dimensional heat conduction in a plane wall with both sides of the wall at specified temperatures, the stability criteria for the explicit method can be expressed in its simplest form as

$$\tau = \frac{\alpha \Delta t}{(\Delta x)^2} \leq \frac{1}{2}$$

5-70C For transient one-dimensional heat conduction in a plane wall with specified heat flux on both sides, the stability criteria for the explicit method can be expressed in its simplest form as

$$\tau = \frac{\alpha \Delta t}{(\Delta x)^2} \leq \frac{1}{2}$$

which is identical to the one for the interior nodes. This is because the heat flux boundary conditions have no effect on the stability criteria.

5-71C For transient two-dimensional heat conduction in a rectangular region with insulation or specified temperature boundary conditions, the stability criteria for the explicit method can be expressed in its simplest form as

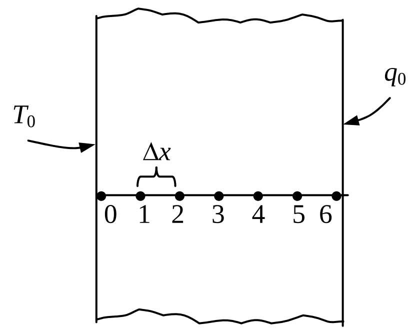
$$\tau = \frac{\alpha \Delta t}{(\Delta x)^2} \leq \frac{1}{4}$$

which is identical to the one for the interior nodes. This is because the insulation or specified temperature boundary conditions have no effect on the stability criteria.

5-72C The implicit method is unconditionally stable and thus any value of time step Δt can be used in the solution of transient heat conduction problems since there is no danger of unstability. However, using a very large value of Δt is equivalent to replacing the time derivative by a very large difference, and thus the solution will not be accurate. Therefore, we should still use the smallest time step practical to minimize the numerical error.

5-73 A plane wall with no heat generation is subjected to specified temperature at the left (node 0) and heat flux at the right boundary (node 6). The explicit transient finite difference formulation of the boundary nodes and the finite difference formulation for the total amount of heat transfer at the left boundary during the first 3 time steps are to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient, and the thermal conductivity to be constant. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** There is no heat generation in the medium.



Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *explicit* finite difference formulations become

Left boundary node: $T_0^i = T_0 = 50°\text{C}$

Right boundary node: $k\dfrac{T_5^i - T_6^i}{\Delta x} + \dot{q}_0 = \rho\dfrac{\Delta x}{2}C\dfrac{T_6^{i+1} - T_6^i}{\Delta t}$

Heat transfer at left surface: $\dot{Q}^i_{\text{left surface}} + kA\dfrac{T_1^i - T_0}{\Delta x} = \rho A\dfrac{\Delta x}{2}C\dfrac{T_6^{i+1} - T_6^i}{\Delta t}$

Noting that $Q = \dot{Q}\Delta t = \sum_i \dot{Q}^i\Delta t$, the total amount of heat transfer becomes

$$Q_{\text{left surface}} = \sum_{i=1}^{3}\dot{Q}^i_{\text{left surface}}\Delta t = \sum_{i=1}^{3}\left(kA\frac{T_0 - T_1^i}{\Delta x} + A\frac{\Delta x}{2}C\frac{T_6^{i+1} - T_6^i}{\Delta t}\right)\Delta t$$

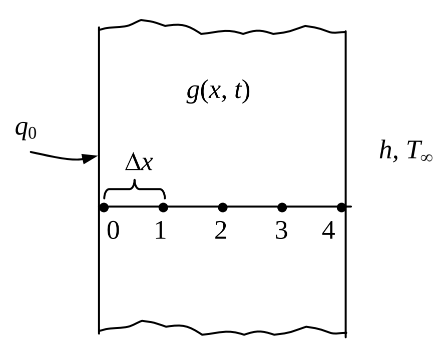
5-74 A plane wall with variable heat generation and constant thermal conductivity is subjected to uniform heat flux $\dot{q}_0$ at the left (node 0) and convection at the right boundary (node 4). The explicit transient finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient, and the thermal conductivity to be constant. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Radiation heat transfer is negligible.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *explicit* finite difference formulations become

Left boundary node:

$$kA\frac{T_1^i - T_0^i}{\Delta x} + \dot{q}_0 A + \dot{g}_0^i(A\Delta x/2) = \rho A\frac{\Delta x}{2}C\frac{T_0^{i+1} - T_0^i}{\Delta t}$$



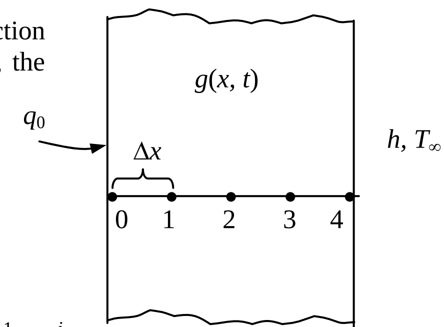
Right boundary node:

$$kA\frac{T_3^i - T_4^i}{\Delta x} + hA(T_\infty^i - T_4^i) + \dot{g}_4^i(A\Delta x/2) = \rho A\frac{\Delta x}{2}C\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

5-75 A plane wall with variable heat generation and constant thermal conductivity is subjected to uniform heat flux $\dot{q}_0$ at the left (node 0) and convection at the right boundary (node 4). The explicit transient finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient, and the thermal conductivity to be constant. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Radiation heat transfer is negligible.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *implicit* finite difference formulations become



Left boundary node:

$$kA\frac{T_1^{i+1}-T_0^{i+1}}{\Delta x}+\dot{q}_0 A+\dot{g}_0^{i+1}(A\Delta x/2)=\rho A\frac{\Delta x}{2}C\frac{T_0^{i+1}-T_0^{i}}{\Delta t}$$

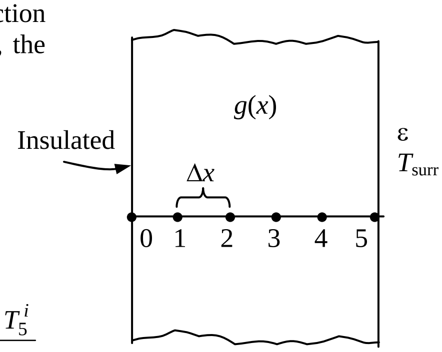
Right boundary node:

$$kA\frac{T_3^{i+1}-T_4^{i+1}}{\Delta x}+hA(T_\infty^{i+1}-T_4^{i+1})+\dot{g}_4^{i+1}(A\Delta x/2)=\rho A\frac{\Delta x}{2}C\frac{T_4^{i+1}-T_4^{i}}{\Delta t}$$

5-76 A plane wall with variable heat generation and constant thermal conductivity is subjected to insulation at the left (node 0) and radiation at the right boundary (node 5). The explicit transient finite difference formulation of the boundary nodes is to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer is negligible.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *explicit* transient finite difference formulations become



Left boundary node:

$$kA\frac{T_1^{i}-T_0^{i}}{\Delta x}+\dot{g}_0^{i}A\frac{\Delta x}{2}=\rho A\frac{\Delta x}{2}C\frac{T_0^{i+1}-T_0^{i}}{\Delta t}$$

Right boundary node:

$$\varepsilon\sigma A[(T_{\text{surr}}^{\text{i}})^4-(T_5^{i})^4]+kA\frac{T_4^{i}-T_5^{i}}{\Delta x}+\dot{g}_5^{i}A\frac{\Delta x}{2}=\rho A\frac{\Delta x}{2}C\frac{T_5^{i+1}-T_5^{i}}{\Delta t}$$

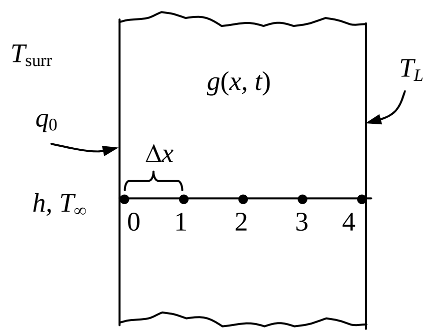
5-77 A plane wall with variable heat generation and constant thermal conductivity is subjected to combined convection, radiation, and heat flux at the left (node 0) and specified temperature at the right boundary (node 4). The explicit finite difference formulation of the left boundary and the finite difference formulation for the total amount of heat transfer at the right boundary are to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer is negligible.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *explicit* transient finite difference formulations become

Left boundary node: $$\varepsilon\sigma A[T_{surr}^4-(T_0^i)^4]+hA(T_\infty^i-T_0^i)+kA\frac{T_1^i-T_0^i}{\Delta x}+\dot{g}_0^i A\frac{\Delta x}{2}=\rho A\frac{\Delta x}{2}C\frac{T_0^{i+1}-T_0^i}{\Delta t}$$

Heat transfer at right surface: $$\dot{Q}_{\text{rightt surface}}^i+kA\frac{T_3^i-T_4^i}{\Delta x}+\dot{g}_4^i A\frac{\Delta x}{2}=\rho A\frac{\Delta x}{2}C\frac{T_4^{i+1}-T_4^i}{\Delta t}$$



Noting that $Q=\dot{Q}\Delta t=\sum_i \dot{Q}^i\Delta t$, the total amount of heat transfer becomes

$$\begin{aligned}Q_{\text{right surface}}&=\sum_{i=1}^{20}\dot{Q}_{\text{right surface}}^i\Delta t\\&=\sum_{i=1}^{20}\left(kA\frac{T_4^i-T_3^i}{\Delta x}-\dot{g}_4^i A\frac{\Delta x}{2}+\rho A\frac{\Delta x}{2}C\frac{T_4^{i+1}-T_4^i}{\Delta t}\right)\Delta t\end{aligned}$$

5-78 Starting with an energy balance on a volume element, the two-dimensional transient *explicit* finite difference equation for a general interior node in rectangular coordinates for $T(x, y, t)$ for the case of constant thermal conductivity and no heat generation is to be obtained.

Analysis (See Figure 5-49 in the text). We consider a rectangular region in which heat conduction is significant in the x and y directions, and consider a unit depth of $\Delta z = 1$ in the z direction. There is no heat generation in the medium, and the thermal conductivity k of the medium is constant. Now we divide the x-y plane of the region into a *rectangular mesh* of nodal points which are spaced Δx and Δy apart in the x and y directions, respectively, and consider a general interior node (m, n) whose coordinates are $x = m\Delta x$ and $y = n\Delta y$. Noting that the volume element centered about the general interior node (m, n) involves heat conduction from four sides (right, left, top, and bottom) and expressing them at previous time step i, the transient explicit finite difference formulation for a general interior node can be expressed as

$$k(\Delta y \times 1)\frac{T^i_{m-1,n} - T^i_{m,n}}{\Delta x} + k(\Delta x \times 1)\frac{T^i_{m,n+1} - T^i_{m,n}}{\Delta y} + k(\Delta y \times 1)\frac{T^i_{m+1,n} - T^i_{m,n}}{\Delta x} + k(\Delta x \times 1)\frac{T^i_{m,n-1} - T^i_{m,n}}{\Delta y}$$

$$= \rho(\Delta x \times \Delta y \times 1)C\frac{T^{i+1}_{m,n} - T^i_{m,n}}{\Delta t}$$

Taking a square mesh ($\Delta x = \Delta y = l$) and dividing each term by k gives, after simplifying,

$$T^i_{m-1,n} + T^i_{m+1,n} + T^i_{m,n+1} + T^i_{m,n-1} - 4T^i_{m,n} = \frac{T^{i+1}_{m,n} - T^i_{m,n}}{\tau}$$

where $\alpha = k/(\rho C)$ is the thermal diffusivity of the material and $\tau = \alpha\Delta t/l^2$ is the dimensionless mesh Fourier number. It can also be expressed in terms of the temperatures at the neighboring nodes in the following easy-to-remember form:

$$T^i_{\text{left}} + T^i_{\text{top}} + T^i_{\text{right}} + T^i_{\text{bottom}} - 4T^i_{\text{node}} = \frac{T^{i+1}_{\text{node}} - T^i_{\text{node}}}{\tau}$$

Discussion We note that setting $T^{i+1}_{\text{node}} = T^i_{\text{node}}$ gives the steady finite difference formulation.

5-79 Starting with an energy balance on a volume element, the two-dimensional transient *implicit* finite difference equation for a general interior node in rectangular coordinates for $T(x, y, t)$ for the case of constant thermal conductivity and no heat generation is to be obtained.

Analysis (See Figure 5-49 in the text). We consider a rectangular region in which heat conduction is significant in the x and y directions, and consider a unit depth of $\Delta z = 1$ in the z direction. There is no heat generation in the medium, and the thermal conductivity k of the medium is constant. Now we divide the x-y plane of the region into a *rectangular mesh* of nodal points which are spaced Δx and Δy apart in the x and y directions, respectively, and consider a general interior node (m, n) whose coordinates are $x = m\Delta x$ and $y = n\Delta y$. Noting that the volume element centered about the general interior node (m, n) involves heat conduction from four sides (right, left, top, and bottom) and expressing them at previous time step i, the transient *implicit* finite difference formulation for a general interior node can be expressed as

$$k(\Delta y \times 1)\frac{T_{m-1,n}^{i+1} - T_{m,n}^{i+1}}{\Delta x} + k(\Delta x \times 1)\frac{T_{m,n+1}^{i+1} - T_{m,n}^{i+1}}{\Delta y} + k(\Delta y \times 1)\frac{T_{m+1,n}^{i+1} - T_{m,n}^{i}}{\Delta x} + k(\Delta x \times 1)\frac{T_{m,n-1}^{i+1} - T_{m,n}^{i+1}}{\Delta y}$$
$$= \rho(\Delta x \times \Delta y \times 1)C\frac{T_{m,n}^{i+1} - T_{m,n}^{i}}{\Delta t}$$

Taking a square mesh ($\Delta x = \Delta y = l$) and dividing each term by k gives, after simplifying,

$$T_{m-1,n}^{i+1} + T_{m+1,n}^{i+1} + T_{m,n+1}^{i+1} + T_{m,n-1}^{i+1} - 4T_{m,n}^{i+1} = \frac{T_{m}^{i+1} - T_{m}^{i}}{\tau}$$

where $\alpha = k/(\rho C)$ is the thermal diffusivity of the material and $\tau = \alpha\Delta t / l^2$ is the dimensionless mesh Fourier number. It can also be expressed in terms of the temperatures at the neighboring nodes in the following easy-to-remember form:

$$T_{\text{left}}^{i+1} + T_{\text{top}}^{i+1} + T_{\text{right}}^{i+1} + T_{\text{bottom}}^{i+1} - 4T_{\text{node}}^{i+1} = \frac{T_{\text{node}}^{i+1} - T_{\text{node}}^{i}}{\tau}$$

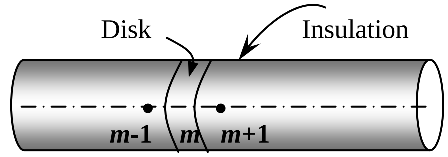
Discussion We note that setting $T_{\text{node}}^{i+1} = T_{\text{node}}^{i}$ gives the steady finite difference formulation.

5-80 Starting with an energy balance on a disk volume element, the one-dimensional transient explicit finite difference equation for a general interior node for $T(z,t)$ in a cylinder whose side surface is insulated for the case of constant thermal conductivity with uniform heat generation is to be obtained.

Analysis We consider transient one-dimensional heat conduction in the axial z direction in an insulated cylindrical rod of constant cross-sectional area A with constant heat generation $\dot{g}_0$ and constant conductivity k with a mesh size of Δz in the z direction. Noting that the volume element of a general interior node m involves heat conduction from two sides and the volume of the element is $V_{\text{element}} = A\Delta z$, the transient explicit finite difference formulation for an interior node can be expressed as

$$kA\frac{T_{m-1}^i - T_m^i}{\Delta x} + kA\frac{T_{m+1}^i - T_m^i}{\Delta x} + \dot{g}_0 A\Delta x = \rho A\Delta x C\frac{T_m^{i+1} - T_m^i}{\Delta t}$$

Canceling the surface area A and multiplying by Δx/k, it simplifies to



$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_0 \Delta x^2}{k} = \frac{(\Delta x)^2}{\alpha\Delta t}(T_m^{i+1} - T_m^i)$$

where $\alpha = k/(\rho C)$ is the *thermal diffusivity* of the wall material.

Using the definition of the dimensionless *mesh Fourier number* $\tau = \dfrac{\alpha\Delta t}{(\Delta x)^2}$, the last equation reduces to

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_0 \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

Discussion We note that setting $T_{\text{m}}^{i+1} = T_{\text{m}}^i$ gives the steady finite difference formulation.

5-81 A composite plane wall consists of two layers A and B in perfect contact at the interface where node 1 is at the interface. The wall is insulated at the left (node 0) and subjected to radiation at the right boundary (node 2). The complete transient explicit finite difference formulation of this problem is to be obtained.

Assumptions **1** Heat transfer through the wall is given to be transient and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer is negligible. **3** There is no heat generation.

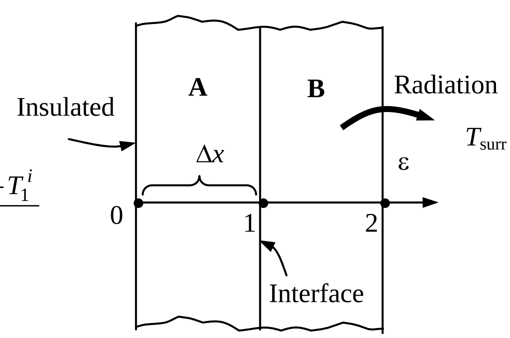
Analysis Using the energy balance approach with a unit area A = 1 and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become

Node 0 (at left boundary):

$$k_A\frac{T_1^i - T_0^i}{\Delta x} = \rho_A\frac{\Delta x}{2}C_A\frac{T_0^{i+1} - T_0^i}{\Delta t}$$

Node 1 (at interface):

$$k_A\frac{T_0^i - T_1^i}{\Delta x} + k_B\frac{T_2^i - T_1^i}{\Delta x} = \left(\rho_A\frac{\Delta x}{2}C_A + \rho_B\frac{\Delta x}{2}C_B\right)\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

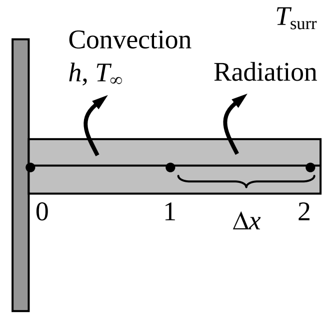


Node 2 (at right boundary):

$$\varepsilon\sigma[T_{\text{surr}}^4 - (T_2^i)^4] + k_B\frac{T_1^i - T_2^i}{\Delta x} = \rho_B\frac{\Delta x}{2}C_B\frac{T_2^{i+1} - T_2^i}{\Delta t}$$

5-82 A pin fin with negligible heat transfer from its tip is considered. The complete finite difference formulation for the determination of nodal temperatures is to be obtained.

Assumptions **1** Heat transfer through the pin fin is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer coefficient is constant and uniform. **3** Heat loss from the fin tip is given to be negligible.



Analysis The nodal network consists of 3 nodes, and the base temperature T_0 at node 0 is specified. Therefore, there are two unknowns T_1 and T_2, and we need two equations to determine them. Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the explicit transient finite difference formulations become

Node 1 (at midpoint):

$$\varepsilon\sigma p\Delta x[T_{surr}^4-(T_1^i)^4]+hp\Delta x(T_\infty-T_1^i)+kA\frac{T_2^i-T_1^i}{\Delta x}+kA\frac{T_0^i-T_1^i}{\Delta x}=\rho A\Delta xC\frac{T_1^{i+1}-T_1^i}{\Delta t}$$

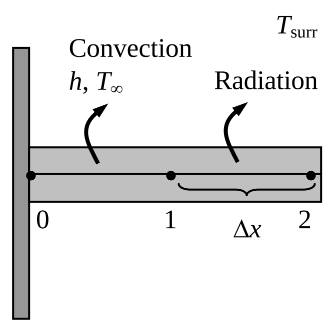
Node 2 (at fin tip):

$$\varepsilon\sigma\left(p\frac{\Delta x}{2}\right)[T_{surr}^4-(T_2^i)^4]+h\left(p\frac{\Delta x}{2}\right)(T_\infty-T_2^i)+kA\frac{T_1^i-T_2^i}{\Delta x}=\rho A\frac{\Delta x}{2}C\frac{T_2^{i+1}-T_2^i}{\Delta t}$$

where $A=\pi D^2/4$ is the cross-sectional area and $p=\pi D$ is the perimeter of the fin.

5-83 A pin fin with negligible heat transfer from its tip is considered. The complete finite difference formulation for the determination of nodal temperatures is to be obtained.

Assumptions **1** Heat transfer through the pin fin is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer coefficient is constant and uniform. **3** Heat loss from the fin tip is given to be negligible.



Analysis The nodal network consists of 3 nodes, and the base temperature T_0 at node 0 is specified. Therefore, there are two unknowns T_1 and T_2, and we need two equations to determine them. Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the implicit transient finite difference formulations become

Node 1: $$\varepsilon\sigma p\Delta x[T_{surr}^4-(T_1^{i+1})^4]+hp\Delta x(T_\infty-T_1^{i+1})+kA\frac{T_2^{i+1}-T_1^{i+1}}{\Delta x}+kA\frac{T_0^{i+1}-T_1^{i+1}}{\Delta x}=\rho A\Delta xC\frac{T_1^{i+1}-T_1^i}{\Delta t}$$

Node 2: $$\varepsilon\sigma\left(p\frac{\Delta x}{2}\right)[T_{surr}^4-(T_2^{i+1})^4]+h\left(p\frac{\Delta x}{2}\right)(T_\infty-T_2^{i+1})+kA\frac{T_1^{i+1}-T_2^{i+1}}{\Delta x}=\rho A\frac{\Delta x}{2}C\frac{T_2^{i+1}-T_2^i}{\Delta t}$$

where $A=\pi D^2/4$ is the cross-sectional area and $p=\pi D$ is the perimeter of the fin.

5-84 A uranium plate initially at a uniform temperature is subjected to insulation on one side and convection on the other. The transient finite difference formulation of this problem is to be obtained, and the nodal temperatures after 5 min and under steady conditions are to be determined.

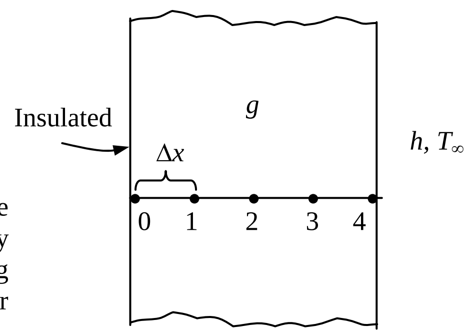
Assumptions **1** Heat transfer is one-dimensional since the plate is large relative to its thickness. **2** Thermal conductivity is constant. **3** Radiation heat transfer is negligible.

Properties The conductivity and diffusivity are given to be k = 28 W/m·°C and $\alpha = 12.5\times 10^{-6}\ \text{m}^2/\text{s}$.

Analysis The nodal spacing is given to be Δx = 0.02 m. Then the number of nodes becomes $M = L/\Delta x + 1$ = 0.08/0.02+1 = 5. This problem involves 5 unknown nodal temperatures, and thus we need to have 5 equations. Node 0 is on insulated boundary, and thus we can treat it as an interior note by using the mirror image concept. Nodes 1, 2, and 3 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

$$\rightarrow T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i + \tau\frac{\dot{g}_m^i \Delta x^2}{k}$$



The finite difference equation for node 4 on the right surface subjected to convection is obtained by applying an energy balance on the half volume element about node 4 and taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 0 (insulated)}: \quad T_0^{i+1} = \tau(T_1^i + T_1^i) + (1-2\tau)T_0^i + \tau\frac{\dot{g}_0 \Delta x^2}{k}$$

$$\text{Node 1 (interior)}: \quad T_1^{i+1} = \tau(T_0^i + T_2^i) + (1-2\tau)T_1^i + \tau\frac{\dot{g}_0 \Delta x^2}{k}$$

$$\text{Node 2 (interior)}: \quad T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i + \tau\frac{\dot{g}_0 \Delta x^2}{k}$$

$$\text{Node 3 (interior)}: \quad T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i + \tau\frac{\dot{g}_0 \Delta x^2}{k}$$

$$\text{Node 4 (convection)}: \quad h(T_\infty - T_4^i) + k\frac{T_3^i - T_4^i}{\Delta x} + \dot{g}_0\frac{\Delta x}{2} = \rho\frac{\Delta x}{2}C\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

or

$$T_4^{i+1} = \left(1 - 2\tau - 2\tau\frac{h\Delta x}{k}\right)T_4^i + 2\tau T_3^i + 2\tau\frac{h\Delta x}{k}T_\infty + \tau\frac{\dot{g}_0(\Delta x)^2}{k}$$

where $\Delta x = 0.02\text{ m}, \dot{g}_0 = 10^6\ \text{W/m}^3$, $k = 28\text{ W/m}\cdot°\text{C}$, $h = 35\text{ W/m}^2\cdot°\text{C}$, $T_\infty = 20°\text{C}$, and $\alpha = 12.5\times 10^{-6}$ m^2/s. The upper limit of the time step Δt is determined from the stability criteria that requires all primary coefficients to be greater than or equal to zero. The coefficient of T_4^i is smaller in this case, and thus the stability criteria for this problem can be expressed as

$$1 - 2\tau - 2\tau\frac{h\Delta x}{k} \ge 0 \quad \rightarrow \quad \tau \le \frac{1}{2(1 + h\Delta x/k)} \quad \rightarrow \quad \Delta t \le \frac{\Delta x^2}{2\alpha(1 + h\Delta x/k)}$$

since $\tau = \alpha\Delta t/\Delta x^2$. Substituting the given quantities, the maximum allowable the time step becomes

$$\Delta t \le \frac{(0.02\text{ m})^2}{2(12.5\times 10^{-6}\ \text{m}^2/\text{s})[1 + (35\text{ W/m}^2.°\text{C})(0.02\text{ m})/(28\text{ W/m.°C})]} = 15.6\text{ s}$$

Therefore, any time step less than 15.5 s can be used to solve this problem. For convenience, let us choose the time step to be Δt = 15 s. Then the mesh Fourier number becomes

$$\tau = \frac{\alpha \Delta t}{\Delta x^2} = \frac{(12.5 \times 10^{-6} \text{ m}^2/\text{s})(15 \text{ s})}{(0.02 \text{ m})^2} = 0.46875$$

Substituting this value of τ and other given quantities, the nodal temperatures after 5×60/15 = 20 time steps (5 min) are determined to be

After 5 min: $\boldsymbol{T_0}$ **=228.9°C,** $\boldsymbol{T_1}$ **=228.4°C,** $\boldsymbol{T_2}$ **=226.8°C,** $\boldsymbol{T_3}$ **=224.0°C, and** $\boldsymbol{T_4}$ **=219.9 °C**

(*b*) The time needed for transient operation to be established is determined by increasing the number of time steps until the nodal temperatures no longer change. In this case steady operation is established in ---- min, and the nodal temperatures under steady conditions are determined to be

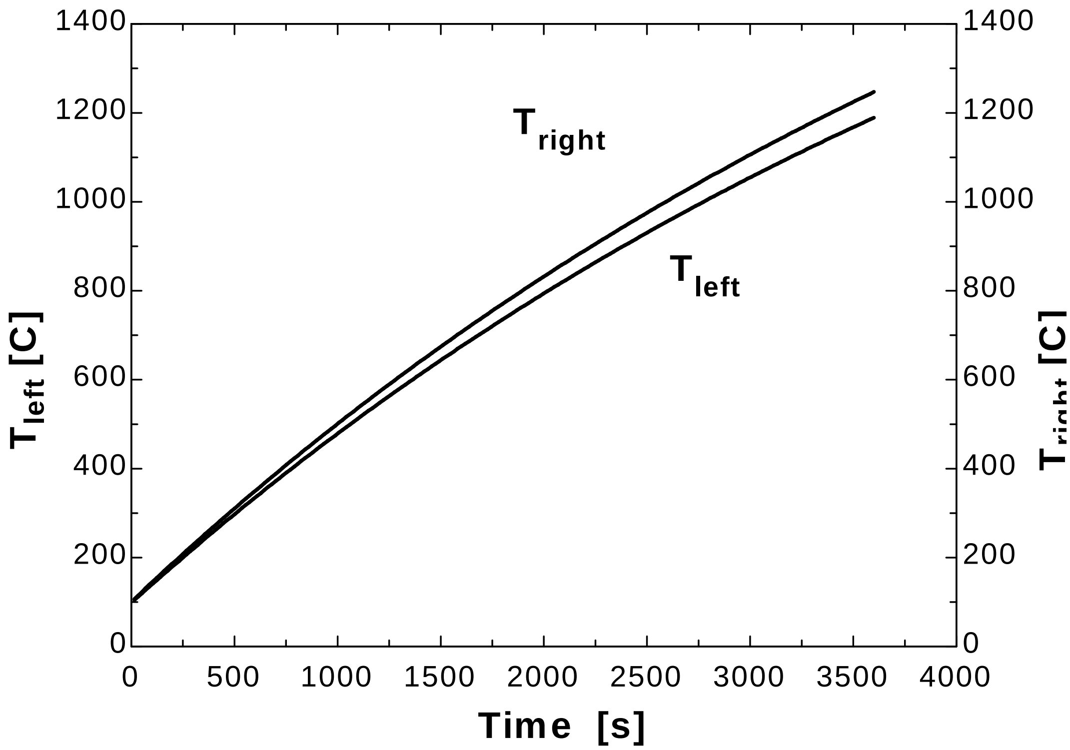
$\boldsymbol{T_0}$ **=2420°C,** $\boldsymbol{T_1}$ **=2413°C,** $\boldsymbol{T_2}$ **=2391°C,** $\boldsymbol{T_3}$ **=2356°C, and** $\boldsymbol{T_4}$ **=2306 °C**

Discussion The steady solution can be checked independently by obtaining the steady finite difference formulation, and solving the resulting equations simultaneously.

5-85 "!PROBLEM 5-85"

```
"GIVEN"
L=0.08 "[m]"
k=28 "[W/m-C]"
alpha=12.5E-6 "[m^2/s]"
T_i=100 "[C]"
g_dot=1E6 "[W/m^3]"
T_infinity=20 "[C]"
h=35 "[W/m^2-C]"
DELTAx=0.02 "[m]"
"time=300 [s], parameter to be varied"

"ANALYSIS"
M=L/DELTAx+1 "Number of nodes"
DELTAt=15 "[s]"
tau=(alpha*DELTAt)/DELTAx^2

"The technique is to store the temperatures in the parametric table and recover them (as old
temperatures)
using the variable ROW.  The first row contains the initial values so Solve Table must begin
at row 2.
Use the DUPLICATE statement to reduce the number of equations that need to be typed.
Column 1
contains the time, column 2 the value of T[1], column 3, the value of T[2], etc., and column 7
the Row."
Time=TableValue(Row-1,#Time)+DELTAt
Duplicate i=1,5
    T_old[i]=TableValue(Row-1,#T[i])
end

"Using the explicit finite difference approach, the six equations for the six unknown
temperatures are determined to be"
T[1]=tau*(T_old[2]+T_old[2])+(1-2*tau)*T_old[1]+tau*(g_dot*DELTAx^2)/k "Node 1,
insulated"
T[2]=tau*(T_old[1]+T_old[3])+(1-2*tau)*T_old[2]+tau*(g_dot*DELTAx^2)/k "Node 2"
T[3]=tau*(T_old[2]+T_old[4])+(1-2*tau)*T_old[3]+tau*(g_dot*DELTAx^2)/k "Node 3"
T[4]=tau*(T_old[3]+T_old[5])+(1-2*tau)*T_old[4]+tau*(g_dot*DELTAx^2)/k "Node 4"
T[5]=(1-2*tau-
2*tau*(h*DELTAx)/k)*T_old[5]+2*tau*T_old[4]+2*tau*(h*DELTAx)/k*T_infinity+tau*(g_dot*DE
LTAx^2)/k "Node 4, convection"
```

Time [s]	**T_1 [C]**	**T_2 [C]**	**T_3 [C]**	**T_4 [C]**	**T_5 [C]**	**Row**
0	100	100	100	100	100	1
15	106.7	106.7	106.7	106.7	104.8	2
30	113.4	113.4	113.4	112.5	111.3	3
45	120.1	120.1	119.7	119	117	4
60	126.8	126.6	126.3	125.1	123.3	5
75	133.3	133.2	132.6	131.5	129.2	6
90	139.9	139.6	139.1	137.6	135.5	7
105	146.4	146.2	145.4	144	141.5	8
120	152.9	152.6	151.8	150.2	147.7	9
135	159.3	159.1	158.1	156.5	153.7	10
…	…	…	…	…	…	…
…	…	…	…	…	…	…
3465	1217	1213	1203	1185	1160	232
3480	1220	1216	1206	1188	1163	233
3495	1223	1220	1209	1192	1167	234
3510	1227	1223	1213	1195	1170	235
3525	1230	1227	1216	1198	1173	236
3540	1234	1230	1219	1201	1176	237
3555	1237	1233	1223	1205	1179	238
3570	1240	1237	1226	1208	1183	239
3585	1244	1240	1229	1211	1186	240
3600	1247	1243	1233	1214	1189	241

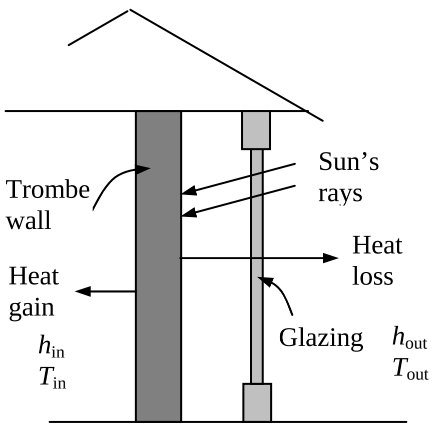


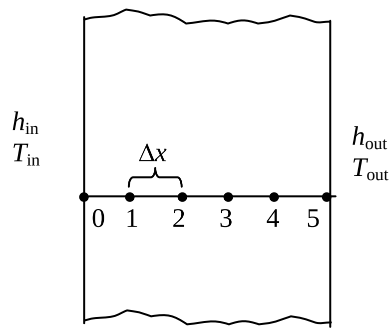
5-86 The passive solar heating of a house through a Trombe wall is studied. The temperature distribution in the wall in 12 h intervals and the amount of heat transfer during the first and second days are to be determined.

Assumptions **1** Heat transfer is one-dimensional since the exposed surface of the wall large relative to its thickness. **2** Thermal conductivity is constant. **3** The heat transfer coefficients are constant.

Properties The wall properties are given to be k = 0.70 W/m·°C, $\alpha = 0.44\times10^{-6}$ m^2/s , and $\kappa = 0.76$. The hourly variation of monthly average ambient temperature and solar heat flux incident on a vertical surface is given to be

Time of day	Ambient Temperature, °C	Solar insolation W/m^2
7am-10am	0	375
10am-1pm	4	750
1pm-4pm	6	580
4pm-7pm	1	95
7pm-10pm	-2	0
10pm-1am	-3	0
1am-4am	-4	0
4am-7am	-4	0





Analysis The nodal spacing is given to be Δx = 0.05 m, Then the number of nodes becomes $M = L/\Delta x + 1$ = 0.30/0.05+1 = 7. This problem involves 7 unknown nodal temperatures, and thus we need to have 7 equations. Nodes 1, 2, 3, 4, and 5 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \;\rightarrow\; T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i$$

The finite difference equation for boundary nodes 0 and 6 are obtained by applying an energy balance on the half volume elements and taking the direction of all heat transfers to be towards the node under consideration:

Node 0: $h_{in} A(T_{in}^i - T_0^i) + kA\dfrac{T_1^i - T_0^i}{\Delta x} = \rho A \dfrac{\Delta x}{2} C \dfrac{T_0^{i+1} - T_0^i}{\Delta t}$

or $\quad T_0^{i+1} = \left(1 - 2\tau - 2\tau\dfrac{h_{in}\Delta x}{k}\right)T_0^i + 2\tau T_1^i + 2\tau\dfrac{h_{in}\Delta x}{k}T_{in}$

Node 1 $(m=1)$: $T_1^{i+1} = \tau(T_0^i + T_2^i) + (1-2\tau)T_1^i$

Node 2 $(m=2)$: $T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i$

Node 3 $(m=3)$: $T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i$

Node 4 $(m=4)$: $T_4^{i+1} = \tau(T_3^i + T_5^i) + (1-2\tau)T_4^i$

Node 5 $(m=5)$: $T_5^{i+1} = \tau(T_4^i + T_6^i) + (1-2\tau)T_5^i$

Node 6 $$h_{\text{out}}A(T_{\text{out}}^i - T_6^i) + \kappa A\dot{q}_{\text{solar}}^i + kA\frac{T_5^i - T_6^i}{\Delta x} = \rho A\frac{\Delta x}{2}C\frac{T_6^{i+1} - T_6^i}{\Delta t}$$

or $$T_6^{i+1} = \left(1 - 2\tau - 2\tau\frac{h_{\text{out}}\Delta x}{k}\right)T_6^i + 2\tau T_5^i + 2\tau\frac{h_{\text{out}}\Delta x}{k}T_{\text{out}}^i + 2\tau\frac{\kappa\dot{q}_{\text{solar}}^i\Delta x}{k}$$

where L = 0.30 m, k = 0.70 W/m.°C, $\alpha = 0.44\times10^{-6}$ m^2/s, T_{out} and $\dot{q}_{\text{solar}}$ are as given in the table, $\kappa = 0.76$ h_{out} = 3.4 W/m^2.°C, T_{in} = 20°C, h_{in} = 9.1 W/m^2.°C, and Δx = 0.05 m.

Next we need to determine the upper limit of the time step Δt from the stability criteria since we are using the explicit method. This requires the identification of the smallest primary coefficient in the system. We know that the boundary nodes are more restrictive than the interior nodes, and thus we examine the formulations of the boundary nodes 0 and 6 only. The smallest and thus the most restrictive primary coefficient in this case is the coefficient of T_0^i in the formulation of node 0 since $h_{\text{in}} > h_{\text{out}}$, and thus

$$1 - 2\tau - 2\tau\frac{h_{\text{in}}\Delta x}{k} < 1 - 2\tau - 2\tau\frac{h_{\text{out}}\Delta x}{k}$$

Therefore, the stability criteria for this problem can be expressed as

$$1 - 2\tau - 2\tau\frac{h_{\text{in}}\Delta x}{k} \ge 0 \quad\rightarrow\quad \tau \le \frac{1}{2(1 + h_{\text{in}}\Delta x/k)} \quad\rightarrow\quad \Delta t \le \frac{\Delta x^2}{2\alpha(1 + h_{\text{in}}\Delta x/k)}$$

since $\tau = \alpha\Delta t/\Delta x^2$. Substituting the given quantities, the maximum allowable the time step becomes

$$\Delta t \le \frac{(0.05\text{ m})^2}{2(0.44\times10^{-6}\text{ m}^2/\text{s})[1 + (9.1\text{ W/m}^2.°\text{C})(0.05\text{ m})/(0.70\text{ W/m}.°\text{C})]} = 1722\text{ s}$$

Therefore, any time step less than 1722 s can be used to solve this problem. For convenience, let us choose the time step to be Δt = 900 s = 15 min. Then the mesh Fourier number becomes

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(0.44\times10^{-6}\text{ m}^2/\text{s})(900\text{ s})}{(0.05\text{ m})^2} = 0.1584$$

Initially (at 7 am or t = 0), the temperature of the wall is said to vary linearly between 20°C at node 0 and 0°C at node 6. Noting that there are 6 nodal spacing of equal length, the temperature change between two neighboring nodes is (20 - 0)°F/6 = 3.33°C. Therefore, the initial nodal temperatures are

$$T_0^0 = 20°\text{C},\ T_1^0 = 16.66°\text{C},\ T_2^0 = 13.33°\text{C},\ T_3^0 = 10°\text{C},\ T_4^0 = 6.66°\text{C},\ T_5^0 = 3.33°\text{C},\ T_6^0 = 0°\text{C}$$

Substituting the given and calculated quantities, the nodal temperatures after 6, 12, 18, 24, 30, 36, 42, and 48 h are calculated and presented in the following table and chart.

Time	Time step, i	Nodal temperatures, °C						
		T_0	T_1	T_2	T_3	T_4	T_5	T_6
0 h (7am)	0	20.0	16.7	13.3	10.0	6.66	3.33	0.0
6 h (1 pm)	24	17.5	16.1	15.9	18.1	24.8	38.8	61.5
12 h (7 pm)	48	21.4	22.9	25.8	30.2	34.6	37.2	35.8
18 h (1 am)	72	22.9	24.6	26.0	26.6	26.0	23.5	19.1
24 h (7 am)	96	21.6	22.5	22.7	22.1	20.4	17.7	13.9
30 h (1 pm)	120	21.0	21.8	23.4	26.8	34.1	47.6	68.9
36 h (7 pm)	144	24.1	27.0	31.3	36.4	41.1	43.2	40.9
42 h (1 am)	168	24.7	27.6	29.9	31.1	30.5	27.8	22.6
48 h (7 am)	192	23.0	24.6	25.5	25.2	23.7	20.7	16.3

The rate of heat transfer from the Trombe wall to the interior of the house during each time step is determined from Newton's law of cooling using the average temperature at the inner surface of the wall (node 0) as

$$Q^i_{\text{Trumbe wall}} = \dot{Q}^i_{\text{Trumbe wall}}\Delta t = h_{\text{in}} A(T^i_0 - T_{\text{in}})\Delta t = h_{\text{in}} A[(T^i_0 + T^{i-1}_0)/2 - T_{\text{in}}]\Delta t$$

Therefore, the amount of heat transfer during the first time step (i = 1) or during the first 15 min period is

$$Q^1_{\text{Trumbe wall}} = h_{\text{in}} A[(T^1_0 + T^0_0)/2 - T_{\text{in}}]\Delta t = (9.1\ \text{W/m}^2.°\text{C})(2.8\times 7\ \text{m}^2)[(68.3+70)/2 - 70°\text{F}](0.25\ \text{h}) = -96.8\ \text{Btu}$$

The negative sign indicates that heat is transferred to the Trombe wall from the air in the house which represents a heat loss. Then the total heat transfer during a specified time period is determined by adding the heat transfer amounts for each time step as

$$Q_{\text{Trumbe wall}} = \sum_{i=1}^{I} Q^i_{\text{Trumbe wall}} = \sum_{i=1}^{I} h_{\text{in}} A[(T^i_0 + T^{i-1}_0)/2 - T_{\text{in}}]\Delta t$$

where I is the total number of time intervals in the specified time period. In this case I = 48 for 12 h, 96 for 24 h, etc. Following the approach described above using a computer, the amount of heat transfer between the Trombe wall and the interior of the house is determined to be

$Q_{\text{Trombe wall}}$ = - 3421 kJ after 12 h

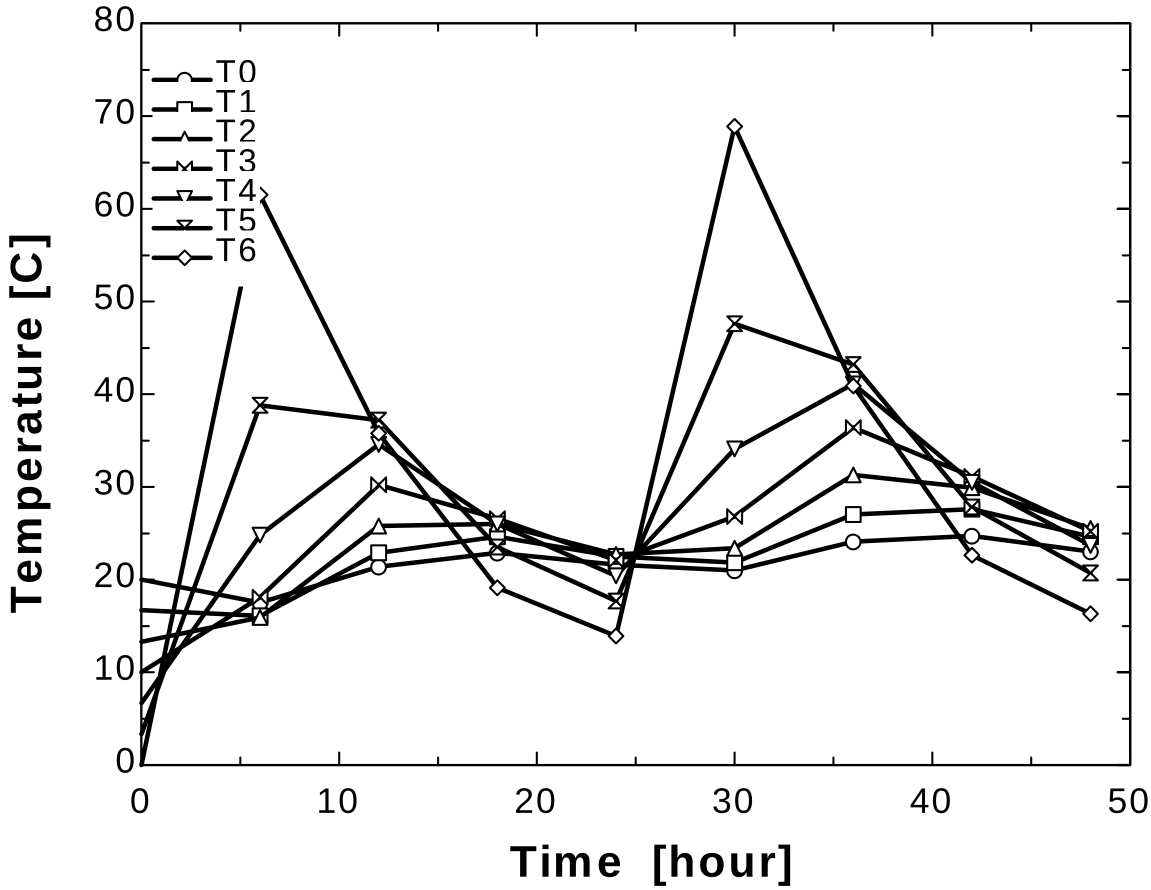
$Q_{\text{Trombe wall}}$ = 1753 kJ after 24 h

$Q_{\text{Trombe wall}}$ = 5393 kJ after 36 h

$Q_{\text{Trombe wall}}$ = 15,230 kJ after 48 h

Discussion Note that the interior temperature of the Trombe wall drops in early morning hours, but then rises as the solar energy absorbed by the exterior surface diffuses through the wall. The exterior surface temperature of the Trombe wall rises from 0 to 61.5°C in just 6 h because of the solar energy absorbed, but then drops to 13.9°C by next morning as a result of heat loss at night. Therefore, it may be worthwhile to cover the outer surface at night to minimize the heat losses.

Also the house loses 3421 kJ through the Trombe wall the 1st daytime as a result of the low start-up temperature, but delivers about 13,500 kJ of heat to the house the second day. It can be shown that the Trombe wall will deliver even more heat to the house during the 3rd day since it will start the day at a higher average temperature.

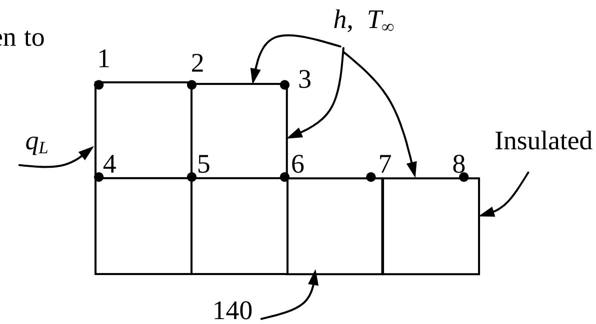


80
70
60
50
40
30
20
10
0
0
10
20
30
40
50
Temperature [C]
Time [hour]
T0
T1
T2
T3
T4
T5
T6

5-87 Heat conduction through a long L-shaped solid bar with specified boundary conditions is considered. The temperature at the top corner (node #3) of the body after 2, 5, and 30 min is to be determined with the transient explicit finite difference method.

Assumptions **1** Heat transfer through the body is given to be transient and two-dimensional. **2** Thermal conductivity is constant. **3** Heat generation is uniform.

Properties The conductivity and diffusivity are given to be k = 15 W/m·°C and $\alpha = 3.2\times10^{-6}\ \text{m}^2/\text{s}$.



Analysis The nodal spacing is given to be Δx=Δx=l=0.015 m. The explicit finite difference equations are determined on the basis of the energy balance for the transient case expressed as

$$\sum_{\text{All sides}} \dot{Q}^i + \dot{G}^{\text{i}}_{\text{element}} = \rho V_{\text{element}} C \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

The quantities h, T_∞, $\dot{g}$, and $\dot{q}_R$ do not change with time, and thus we do not need to use the superscript i for them. Also, the energy balance expressions can be simplified using the definitions of thermal diffusivity $\alpha = k/(\rho C)$ and the dimensionless mesh Fourier number $\tau = \alpha \Delta t / l^2$ where $\Delta x = \Delta y = l$. We note that all nodes are boundary nodes except node 5 that is an interior node. Therefore, we will have to rely on energy balances to obtain the finite difference equations. Using energy balances, the finite difference equations for each of the 8 nodes are obtained as follows:

Node 1: $\dot{q}_L \dfrac{l}{2} + h\dfrac{l}{2}(T_\infty - T_1^i) + k\dfrac{l}{2}\dfrac{T_2^i - T_1^i}{l} + k\dfrac{l}{2}\dfrac{T_4^i - T_1^i}{l} + \dot{g}_0 \dfrac{l^2}{4} = \rho \dfrac{l^2}{4} C \dfrac{T_1^{i+1} - T_1^i}{\Delta t}$

Node 2: $hl(T_\infty - T_2^i) + k\dfrac{l}{2}\dfrac{T_1^i - T_2^i}{l} + k\dfrac{l}{2}\dfrac{T_3^i - T_2^i}{l} + kl\dfrac{T_5^i - T_2^i}{l} + \dot{g}_0 \dfrac{l^2}{2} = \rho \dfrac{l^2}{2} C \dfrac{T_2^{i+1} - T_2^i}{\Delta t}$

Node 3: $hl(T_\infty - T_3^i) + k\dfrac{l}{2}\dfrac{T_2^i - T_3^i}{l} + k\dfrac{l}{2}\dfrac{T_6^i - T_3^i}{l} + \dot{g}_0 \dfrac{l^2}{4} = \rho \dfrac{l^2}{4} C \dfrac{T_3^{i+1} - T_3^i}{\Delta t}$

(It can be rearranged as $T_3^{i+1} = \left(1 - 4\tau - 4\tau \dfrac{hl}{k}\right) T_3^i + 2\tau \left(T_4^i + T_6^i + 2\dfrac{hl}{k} T_\infty + \dfrac{\dot{g}_3 l^2}{2k} \right)$)

Node 4: $\dot{q}_L l + k\dfrac{l}{2}\dfrac{T_1^i - T_4^i}{l} + k\dfrac{l}{2}\dfrac{140 - T_4^i}{l} + kl\dfrac{T_5^i - T_4^i}{l} + \dot{g}_0 \dfrac{l^2}{2} = \rho \dfrac{l^2}{2} C \dfrac{T_4^{i+1} - T_4^i}{\Delta t}$

Node 5 (interior): $T_5^{i+1} = (1 - 4\tau) T_5^i + \tau \left(T_2^i + T_4^i + T_6^i + 140 + \dfrac{\dot{g}_0 l^2}{k} \right)$

Node 6: $hl(T_\infty - T_6^i) + k\dfrac{l}{2}\dfrac{T_3^i - T_6^i}{l} + kl\dfrac{T_5^i - T_6^i}{l} + kl\dfrac{140 - T_6^i}{l} + k\dfrac{l}{2}\dfrac{T_7^i - T_6^i}{l} + \dot{g}_0 \dfrac{3l^2}{4} = \rho \dfrac{3l^2}{4} C \dfrac{T_6^{i+1} - T_6^i}{\Delta t}$

Node 7: $hl(T_\infty - T_7^i) + k\dfrac{l}{2}\dfrac{T_6^i - T_7^i}{l} + k\dfrac{l}{2}\dfrac{T_8^i - T_7^i}{l} + kl\dfrac{140 - T_7^i}{l} + \dot{g}_0 \dfrac{l^2}{2} = \rho \dfrac{l^2}{2} C \dfrac{T_7^{i+1} - T_7^i}{\Delta t}$

Node 8: $h\dfrac{l}{2}(T_\infty - T_8^i) + k\dfrac{l}{2}\dfrac{T_7^i - T_8^i}{l} + k\dfrac{l}{2}\dfrac{140 - T_8^i}{l} + \dot{g}_0 \dfrac{l^2}{4} = \rho \dfrac{l^2}{4} C \dfrac{T_8^{i+1} - T_8^i}{\Delta t}$

where $\dot{g}_0 = 2\times10^7\ \text{W/m}^3$, $\dot{q}_L = 8000\ \text{W/m}^2$, l = 0.015 m, k =15 W/m·°C, h = 80 W/m^2·°C, and T_∞ =25°C.

The upper limit of the time step Δt is determined from the stability criteria that requires the coefficient of T_m^i in the T_m^{i+1} expression (the primary coefficient) be greater than or equal to zero for all nodes. The smallest primary coefficient in the 8 equations above is the coefficient of T_3^i in the T_3^{i+1} expression since it is exposed to most convection per unit volume (this can be verified), and thus the stability criteria for this problem can be expressed as

$$1-4\tau-4\tau\frac{hl}{k}\geq 0 \quad\rightarrow\quad \tau\leq\frac{1}{4(1+hl/k)} \quad\rightarrow\quad \Delta t\leq\frac{l^2}{4\alpha(1+hl/k)}$$

since $\tau=\alpha\Delta t/l^2$. Substituting the given quantities, the maximum allowable value of the time step is determined to be

$$\Delta t\leq\frac{(0.015\text{ m})^2}{4(3.2\times10^{-6}\text{ m}^2/\text{s})[1+(80\text{ W/m}^2.°\text{C})(0.015\text{ m})/(15\text{ W/m}.°\text{C})]}=16.3\text{ s}$$

Therefore, any time step less than 16.3 s can be used to solve this problem. For convenience, we choose the time step to be Δt = 15 s. Then the mesh Fourier number becomes

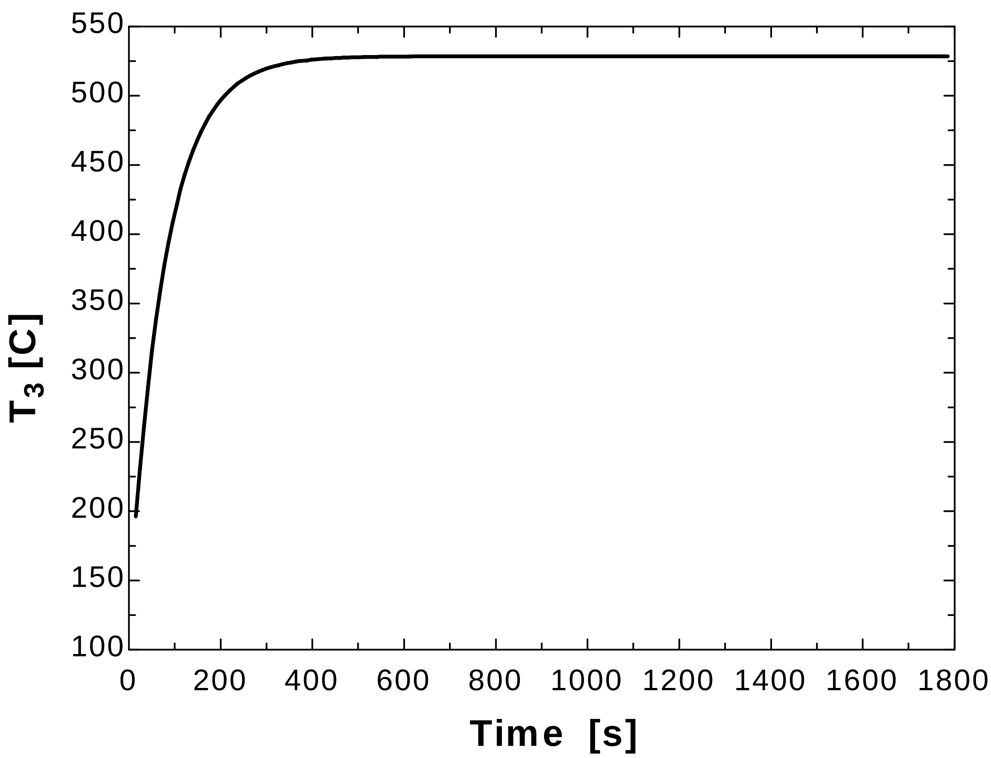
$$\tau=\frac{\alpha\Delta t}{l^2}=\frac{(3.2\times10^{-6}\text{ m}^2/\text{s})(15\text{ s})}{(0.015\text{ m})^2}=0.2133 \qquad (\text{for } \Delta t = 15\text{ s})$$

Using the specified initial condition as the solution at time t = 0 (for i = 0), sweeping through the 9 equations above will give the solution at intervals of 15 s. Using a computer, the solution at the upper corner node (node 3) is determined to be **441, 520**, and **529°C** at 2, 5, and 30 min, respectively. It can be shown that the steady state solution at node 3 is 531°C.

5-88 "!PROBLEM 5-88"

```
"GIVEN"
T_i=140 "[C]"
k=15 "[W/m-C]"
alpha=3.2E-6 "[m^2/s]"
g_dot=2E7 "[W/m^3]"
T_bottom=140 "[C]"
T_infinity=25 "[C]"
h=80 "[W/m^2-C]"
q_dot_L=8000 "[W/m^2]"
DELTAx=0.015 "[m]"
DELTAy=0.015 "[m]"
"time=120 [s], parameter to be varied"

"ANALYSIS"
l=DELTAx
DELTAt=15 "[s]"
tau=(alpha*DELTAt)/l^2
RhoC=k/alpha "RhoC=rho*C"
"The technique is to store the temperatures in the parametric table and recover them (as old
temperatures)
using the variable ROW.  The first row contains the initial values so Solve Table must begin
at row 2.
Use the DUPLICATE statement to reduce the number of equations that need to be typed.
Column 1
contains the time, column 2 the value of T[1], column 3, the value of T[2], etc., and column
10 the Row."
Time=TableValue('Table 1',Row-1,#Time)+DELTAt
Duplicate i=1,8
    T_old[i]=TableValue('Table 1',Row-1,#T[i])
end
"Using the explicit finite difference approach, the eight equations for the eight unknown
temperatures are determined to be"
q_dot_L*l/2+h*l/2*(T_infinity-T_old[1])+k*l/2*(T_old[2]-T_old[1])/l+k*l/2*(T_old[4]-
T_old[1])/l+g_dot*l^2/4=RhoC*l^2/4*(T[1]-T_old[1])/DELTAt "Node 1"
h*l*(T_infinity-T_old[2])+k*l/2*(T_old[1]-T_old[2])/l+k*l/2*(T_old[3]-T_old[2])/l+k*l*(T_old[5]-
T_old[2])/l+g_dot*l^2/2=RhoC*l^2/2*(T[2]-T_old[2])/DELTAt "Node 2"
h*l*(T_infinity-T_old[3])+k*l/2*(T_old[2]-T_old[3])/l+k*l/2*(T_old[6]-
T_old[3])/l+g_dot*l^2/4=RhoC*l^2/4*(T[3]-T_old[3])/DELTAt "Node 3"
q_dot_L*l+k*l/2*(T_old[1]-T_old[4])/l+k*l/2*(T_bottom-T_old[4])/l+k*l*(T_old[5]-
T_old[4])/l+g_dot*l^2/2=RhoC*l^2/2*(T[4]-T_old[4])/DELTAt "Node 4"
T[5]=(1-4*tau)*T_old[5]+tau*(T_old[2]+T_old[4]+T_old[6]+T_bottom+g_dot*l^2/k) "Node 5"
h*l*(T_infinity-T_old[6])+k*l/2*(T_old[3]-T_old[6])/l+k*l*(T_old[5]-T_old[6])/l+k*l*(T_bottom-
T_old[6])/l+k*l/2*(T_old[7]-T_old[6])/l+g_dot*3/4*l^2=RhoC*3/4*l^2*(T[6]-T_old[6])/DELTAt
"Node 6"
h*l*(T_infinity-T_old[7])+k*l/2*(T_old[6]-T_old[7])/l+k*l/2*(T_old[8]-T_old[7])/l+k*l*(T_bottom-
T_old[7])/l+g_dot*l^2/2=RhoC*l^2/2*(T[7]-T_old[7])/DELTAt "Node 7"
h*l/2*(T_infinity-T_old[8])+k*l/2*(T_old[7]-T_old[8])/l+k*l/2*(T_bottom-
T_old[8])/l+g_dot*l^2/4=RhoC*l^2/4*(T[8]-T_old[8])/DELTAt "Node 8"
```

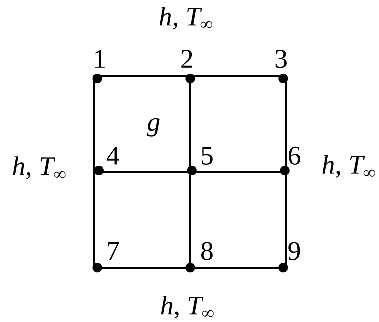
Time [s]	T_1 [C]	T_2 [C]	T_3 [C]	T_4 [C]	T_5 [C]	T_6 [C]	T_7 [C]	T_8 [C]	Row
0	140	140	140	140	140	140	140	140	1
15	203.5	200.1	196.1	207.4	204	201.4	200.1	200.1	2
30	265	259.7	252.4	258.2	253.7	243.7	232.7	232.5	3
45	319	312.7	300.3	299.9	293.5	275.7	252.4	250.1	4
60	365.5	357.4	340.3	334.6	326.4	300.7	265.2	260.4	5
75	404.6	394.9	373.2	363.6	353.5	320.6	274.1	267	6
90	437.4	426.1	400.3	387.8	375.9	336.7	280.8	271.6	7
105	464.7	451.9	422.5	407.9	394.5	349.9	286	275	8
120	487.4	473.3	440.9	424.5	409.8	360.7	290.1	277.5	9
135	506.2	491	456.1	438.4	422.5	369.6	293.4	279.6	10
…	…	…	…	…	…	…	…	…	…
…	…	…	…	…	…	…	…	…	…
1650	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	111
1665	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	112
1680	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	113
1695	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	114
1710	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	115
1725	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	116
1740	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	117
1755	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	118
1770	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	119
1785	596.3	575.7	528.5	504.6	483.1	411.9	308.8	288.9	120



5-89 A long solid bar is subjected to transient two-dimensional heat transfer. The centerline temperature of the bar after 10 min and after steady conditions are established are to be determined.

Assumptions **1** Heat transfer through the body is given to be transient and two-dimensional. **2** Heat is generated uniformly in the body. **3** The heat transfer coefficient also includes the radiation effects.

Properties The conductivity and diffusivity are given to be k = 28 W/m·°C and $\alpha = 12\times10^{-6}\ \text{m}^2/\text{s}$.



Analysis The nodal spacing is given to be Δx=Δx=l=0.1 m. The explicit finite difference equations are determined on the basis of the energy balance for the transient case expressed as

$$\sum_{\text{All sides}} \dot{Q}^i + \dot{G}^{\text{i}}_{\text{element}} = \rho V_{\text{element}} C \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

The quantities h, T_∞, and $\dot{g}_0$ do not change with time, and thus we do not need to use the superscript i for them. The general explicit finite difference form of an interior node for transient two-dimensional heat conduction is expressed as

$$T_{\text{node}}^{i+1} = \tau(T_{\text{left}}^{\text{i}} + T_{\text{top}}^{\text{i}} + T_{\text{right}}^{\text{i}} + T_{\text{bottom}}^{\text{i}}) + (1-4\tau)T_{\text{node}}^{\text{i}} + \tau\frac{\dot{g}_{\text{node}}^{\text{i}} l^2}{k}$$

There is symmetry about the vertical, horizontal, and diagonal lines passing through the center. Therefore, $T_1 = T_3 = T_7 = T_9$ and $T_2 = T_4 = T_6 = T_8$, and T_1, T_2, and T_5 are the only 3 unknown nodal temperatures, and thus we need only 3 equations to determine them uniquely. Also, we can replace the symmetry lines by insulation and utilize the mirror-image concept when writing the finite difference equations for the interior nodes. The finite difference equations for boundary nodes are obtained by applying an energy balance on the volume elements and taking the direction of all heat transfers to be towards the node under consideration:

Node 1: $hl(T_\infty - T_1^i) + k\frac{l}{2}\frac{T_2^i - T_1^i}{l} + k\frac{l}{2}\frac{T_4^i - T_1^i}{l} + \dot{g}_0\frac{l^2}{4} = \rho\frac{l^2}{4}C\frac{T_1^{i+1} - T_1^i}{\Delta t}$

Node 2: $h\frac{l}{2}(T_\infty - T_2^i) + k\frac{l}{2}\frac{T_1^i - T_2^i}{l} + k\frac{l}{2}\frac{T_5^i - T_2^i}{l} + \dot{g}_0\frac{l^2}{4} = \rho\frac{l^2}{4}C\frac{T_2^{i+1} - T_2^i}{\Delta t}$

Node 5 (interior): $T_5^{i+1} = (1-4\tau)T_5^i + \tau\left(4T_2^i + \frac{\dot{g}_0 l^2}{k}\right)$

where $\dot{g}_0 = 8\times10^5\ \text{W/m}^3$, l = 0.1 m, and k = 28 W/m·°C, h = 45 W/m^2·°C, and T_∞ =30°C.

The upper limit of the time step Δt is determined from the stability criteria that requires the coefficient of T_m^i in the T_m^{i+1} expression (the primary coefficient) be greater than or equal to zero for all nodes. The smallest primary coefficient in the 3 equations above is the coefficient of T_1^i in the T_1^{i+1} expression since it is exposed to most convection per unit volume (this can be verified), and thus the stability criteria for this problem can be expressed as

$$1 - 4\tau - 4\tau\frac{hl}{k} \ge 0 \quad \rightarrow \quad \tau \le \frac{1}{4(1+hl/k)} \quad \rightarrow \quad \Delta t \le \frac{l^2}{4\alpha(1+hl/k)}$$

since $\tau = \alpha\Delta t / l^2$. Substituting the given quantities, the maximum allowable value of the time step is determined to be

$$\Delta t \le \frac{(0.1\,\text{m})^2}{4(12\times10^{-6}\ \text{m}^2/\text{s})[1+(45\ \text{W/m}^2.°\text{C})(0.1\,\text{m})/(28\ \text{W/m}.°\text{C})]} = 179\,\text{s}$$

Therefore, any time step less than 179 s can be used to solve this problem. For convenience, we choose the time step to be Δt = 60 s. Then the mesh Fourier number becomes

$$\tau = \frac{\alpha \Delta t}{l^2} = \frac{(12\times 10^{-6}\ \text{m}^2/\text{s})(60\ \text{s})}{(0.1\ \text{m})^2} = 0.072 \qquad (\text{for} \quad \Delta t = 60\ \text{s})$$

Using the specified initial condition as the solution at time t = 0 (for i = 0), sweeping through the 3 equations above will give the solution at intervals of 1 min. Using a computer, the solution at the center node (node 5) is determined to be **217.2°C**, 302.8°C, 379.3°C, 447.7°C, 508.9°C, 612.4°C, 695.1°C, and 761.2°C at 10, 15, 20, 25, 30, 40, 50, and 60 min, respectively. Continuing in this manner, it is observed that steady conditions are reached in the medium after about 6 hours for which the temperature at the center node is **1023°C**.

5-90E A plain window glass initially at a uniform temperature is subjected to convection on both sides. The transient finite difference formulation of this problem is to be obtained, and it is to be determined how long it will take for the fog on the windows to clear up (i.e., for the inner surface temperature of the window glass to reach 54°F).

Assumptions **1** Heat transfer is one-dimensional since the window is large relative to its thickness. **2** Thermal conductivity is constant. **3** Radiation heat transfer is negligible.

Properties The conductivity and diffusivity are given to be k = 0.48 Btu/h.ft·°F and $\alpha = 4.2 \times 10^{-6}\ \text{ft}^2/\text{s}$.

Analysis The nodal spacing is given to be Δx = 0.125 in. Then the number of nodes becomes $M = L/\Delta x + 1$ = 0.375/0.125+1 = 4. This problem involves 4 unknown nodal temperatures, and thus we need to have 4 equations. Nodes 2 and 3 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \quad \rightarrow \quad T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i$$

since there is no heat generation. The finite difference equation for nodes 1 and 4 on the surfaces subjected to convection is obtained by applying an energy balance on the half volume element about the node, and taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (convection)}: \quad h_i(T_i - T_1^i) + k\frac{T_2^i - T_1^i}{\Delta x} = \rho\frac{\Delta x}{2}C\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

or

$$T_1^{i+1} = \left(1 - 2\tau - 2\tau\frac{h_i \Delta x}{k}\right)T_1^i + 2\tau T_2^i + 2\tau\frac{h_i \Delta x}{k}T_i$$

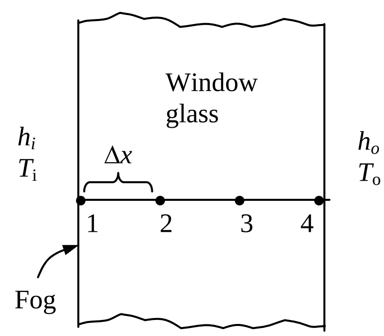
$$\text{Node 2 (interior)}: \quad T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i$$

$$\text{Node 3 (interior)}: \quad T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i$$

$$\text{Node 4 (convection)}: \quad h_o(T_o - T_4^i) + k\frac{T_3^i - T_4^i}{\Delta x} = \rho\frac{\Delta x}{2}C\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

or

$$T_4^{i+1} = \left(1 - 2\tau - 2\tau\frac{h_o \Delta x}{k}\right)T_4^i + 2\tau T_3^i + 2\tau\frac{h_o \Delta x}{k}T_o$$



where Δx = 0.125/12 ft , k = 0.48 Btu/h.ft·°F, h_i = 1.2 Btu/h.ft^2·°F, T_i =35+2*(t/60)°F (t in seconds), h_o = 2.6 Btu/h.ft^2·°F, and T_o =35°F. The upper limit of the time step Δt is determined from the stability criteria that requires all primary coefficients to be greater than or equal to zero. The coefficient of T_4^i is smaller in this case, and thus the stability criteria for this problem can be expressed as

$$1 - 2\tau - 2\tau\frac{h\Delta x}{k} \geq 0 \quad \rightarrow \quad \tau \leq \frac{1}{2(1 + h\Delta x/k)} \quad \rightarrow \quad \Delta t \leq \frac{\Delta x^2}{2\alpha(1 + h\Delta x/k)}$$

since $\tau = \alpha\Delta t/\Delta x^2$. Substituting the given quantities, the maximum allowable time step becomes

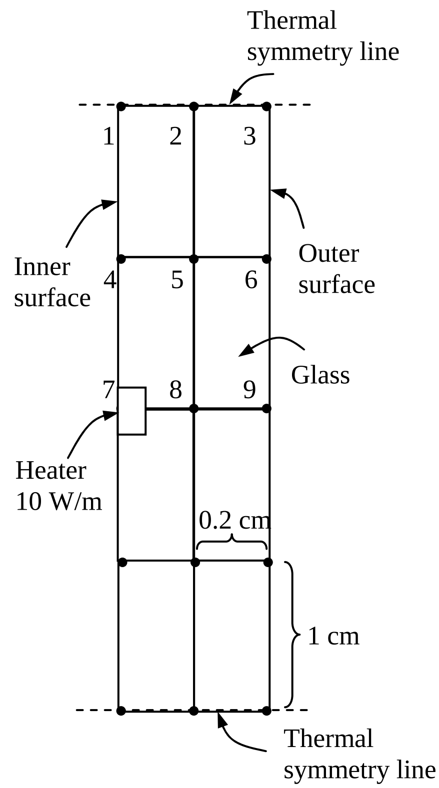
$$\Delta t \leq \frac{(0.125/12\ \text{ft})^2}{2(4.2\times10^{-6}\ \text{ft}^2/\text{s})[1 + (2.6\ \text{Btu/h.ft}^2.°\text{F})(0.125/12\ \text{m})/(0.48\ \text{Btu/h.ft.°F})]} = 12.2\ \text{s}$$

Therefore, any time step less than 12.2 s can be used to solve this problem. For convenience, let us choose the time step to be Δt = 10 s. Then the mesh Fourier number becomes

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(4.2\times10^{-6}\ \text{ft}^2/\text{s})(10\ \text{s})}{(0.125/12\ \text{ft})^2} = 0.3871$$

Substituting this value of τ and other given quantities, the time needed for the inner surface temperature of the window glass to reach 54°F to avoid fogging is determined to be *never*. This is because steady conditions are reached in about 156 min, and the inner surface temperature at that time is determined to be 48.0°F. Therefore, the window will will be fogged at all times.

5-91 The formation of fog on the glass surfaces of a car is to be prevented by attaching electric resistance heaters to the inner surfaces. The temperature distribution throughout the glass 15 min after the strip heaters are turned on and also when steady conditions are reached are to be determined using the explicit method.



Assumptions **1** Heat transfer through the glass is given to be transient and two-dimensional. **2** Thermal conductivity is constant. **3** There is heat generation only at the inner surface, which will be treated as prescribed heat flux.

Properties The conductivity and diffusivity are given to be k = 0.84 W/m·°C and $\alpha = 0.39 \times 10^{-6}\ \text{m}^2/\text{s}$.

Analysis The nodal spacing is given to be Δx = 0.2 cm and Δy = 1 cm. The explicit finite difference equations are determined on the basis of the energy balance for the transient case expressed as

$$\sum_{\text{All sides}} \dot{Q}^i + \dot{G}^{\text{i}}_{\text{element}} = \rho V_{\text{element}} C \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

We consider only 9 nodes because of symmetry. Note that we do not have a square mesh in this case, and thus we will have to rely on energy balances to obtain the finite difference equations. Using energy balances, the finite difference equations for each of the 9 nodes are obtained as follows:

Node 1: $h_i \frac{\Delta y}{2}(T_i - T_1^i) + k\frac{\Delta x}{2}\frac{T_4^i - T_1^i}{\Delta y} + k\frac{\Delta y}{2}\frac{T_2^i - T_1^i}{\Delta x} = \rho C \frac{\Delta x}{2}\frac{\Delta y}{2}\frac{T_1^{i+1} - T_1^i}{\Delta t}$

Node 2: $k\frac{\Delta y}{2}\frac{T_1^i - T_2^i}{\Delta x} + k\frac{\Delta y}{2}\frac{T_3^i - T_2^i}{\Delta x} + k\Delta x\frac{T_5^i - T_2^i}{\Delta y} = \rho C \Delta x \frac{\Delta y}{2}\frac{T_2^{i+1} - T_2^i}{\Delta t}$

Node 3: $h_o \frac{\Delta y}{2}(T_o - T_3^i) + k\frac{\Delta x}{2}\frac{T_6^i - T_3^i}{\Delta y} + k\frac{\Delta y}{2}\frac{T_2^i - T_3^i}{\Delta x} = \rho C \frac{\Delta x}{2}\frac{\Delta y}{2}\frac{T_3^{i+1} - T_3^i}{\Delta t}$

Node 4: $h_i \Delta y(T_i - T_4^i) + k\frac{\Delta x}{2}\frac{T_1^i - T_4^i}{\Delta y} + k\frac{\Delta x}{2}\frac{T_7^i - T_4^i}{\Delta y} + k\Delta y\frac{T_5^i - T_4^i}{\Delta x} = \rho C \Delta y \frac{\Delta x}{2}\frac{T_4^{i+1} - T_4^i}{\Delta t}$

Node 5: $k\Delta y\frac{T_4^i - T_5^i}{\Delta x} + k\Delta y\frac{T_6^i - T_5^i}{\Delta x} + k\Delta x\frac{T_8^i - T_5^i}{\Delta y} + k\Delta x\frac{T_2^i - T_5^i}{\Delta y} = \rho C \Delta x \Delta y \frac{T_5^{i+1} - T_5^i}{\Delta t}$

Node 6: $h_o \Delta y(T_i - T_6^i) + k\frac{\Delta x}{2}\frac{T_3^i - T_6^i}{\Delta y} + k\frac{\Delta x}{2}\frac{T_9^i - T_6^i}{\Delta y} + k\Delta y\frac{T_5^i - T_6^i}{\Delta x} = \rho C \Delta y \frac{\Delta x}{2}\frac{T_6^{i+1} - T_6^i}{\Delta t}$

Node 7: $5\ \text{W} + h_i \frac{\Delta y}{2}(T_i - T_7^i) + k\frac{\Delta x}{2}\frac{T_4^i - T_7^i}{\Delta y} + k\frac{\Delta y}{2}\frac{T_8^i - T_7^i}{\Delta x} = \rho C \frac{\Delta x}{2}\frac{\Delta y}{2}\frac{T_7^{i+1} - T_7^i}{\Delta t}$

Node 8: $k\frac{\Delta y}{2}\frac{T_7^i - T_8^i}{\Delta x} + k\frac{\Delta y}{2}\frac{T_9^i - T_8^i}{\Delta x} + k\Delta x\frac{T_5^i - T_8^i}{\Delta y} = \rho C \Delta x \frac{\Delta y}{2}\frac{T_8^{i+1} - T_8^i}{\Delta t}$

Node 9: $h_o \frac{\Delta y}{2}(T_o - T_9^i) + k\frac{\Delta x}{2}\frac{T_6^i - T_9^i}{\Delta y} + k\frac{\Delta y}{2}\frac{T_8^i - T_9^i}{\Delta x} = \rho C \frac{\Delta x}{2}\frac{\Delta y}{2}\frac{T_9^{i+1} - T_9^i}{\Delta t}$

where k = 0.84 W/m.°C, $\alpha = k/\rho C = 0.39\times10^{-6}\ \text{m}^2/\text{s}$, $T_i = T_o$ = -3°C h_i = 6 W/m^2.°C, h_o = 20 W/m^2.°C, Δx = 0.002 m, and Δy = 0.01 m.

The upper limit of the time step Δt is determined from the stability criteria that requires the coefficient of T_m^i in the T_m^{i+1} expression (the primary coefficient) be greater than or equal to zero for all nodes. The smallest primary coefficient in the 9 equations above is the coefficient of T_9^i in the T_6^{i+1} expression since it is exposed to most convection per unit volume (this can be verified). The equation for node 6 can be rearranged as

$$T_6^{i+1} = \left[1-2\alpha\Delta t\left(\frac{h_o}{k\Delta x}+\frac{1}{\Delta y^2}+\frac{1}{\Delta x^2}\right)\right]T_6^i + 2\alpha\Delta t\left(\frac{h_o}{k\Delta x}T_0 + \frac{T_3^i+T_9^i}{\Delta y^2}+\frac{T_5^i}{\Delta x^2}\right)$$

Therefore, the stability criteria for this problem can be expressed as

$$1-2\alpha\Delta t\left(\frac{h_o}{k\Delta x}+\frac{1}{\Delta y^2}+\frac{1}{\Delta x^2}\right)\ge 0 \;\rightarrow\; \Delta \text{t} \le \frac{1}{2\alpha\left(\dfrac{h_o}{k\Delta x}+\dfrac{1}{\Delta y^2}+\dfrac{1}{\Delta x^2}\right)}$$

Substituting the given quantities, the maximum allowable value of the time step is determined to be

or, $$\Delta \text{t} \le \frac{1}{2\times(0.39\times10^{6}\ \text{m}^2/s)\left(\dfrac{20\ \text{W/m}^2\cdot°\text{C}}{(0.84\ \text{W/m}\cdot°\text{C})(0.002\ \text{m})}+\dfrac{1}{(0.002\ \text{m})^2}+\dfrac{1}{(0.01\ \text{m})^2}\right)} = 4.8\ \text{s}$$

Therefore, any time step less than 4.8 s can be used to solve this problem. For convenience, we choose the time step to be Δt = 4 s. Then the temperature distribution throughout the glass 15 min after the strip heaters are turned on and when steady conditions are reached are determined to be (from the EES solutions disk)

15 min: T_1 = -2.4°C, T_2 = -2.4°C, T_3 = -2.5°C, T_4 = -1.8°C, T_5 = -2.0°C,

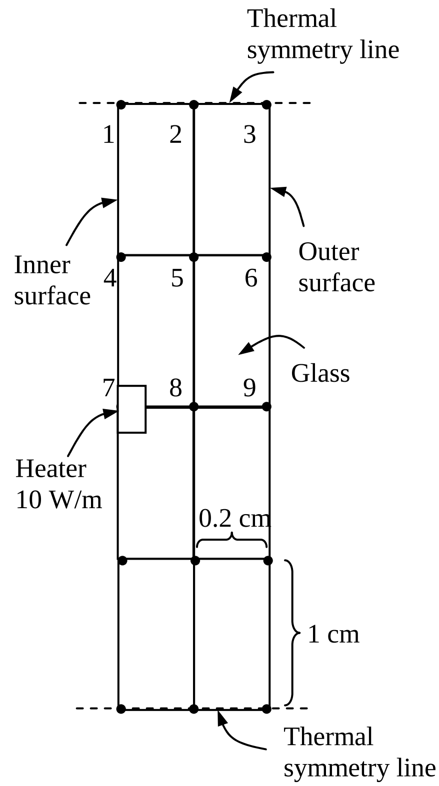
T_6 = -2.7°C, T_7 = 12.3°C, T_8 = 10.7°C, T_9 = 9.6°C

Steady-state: T_1 = -2.4°C, T_2 = -2.4°C, T_3 = -2.5°C, T_4 = -1.8°C, T_5 = -2.0°C,

T_6 = -2.7°C, T_7 = 12.3°C, T_8 = 10.7°C, T_9 = 9.6°C

Discussion Steady operating conditions are reached in about 8 min.

5-92 The formation of fog on the glass surfaces of a car is to be prevented by attaching electric resistance heaters to the inner surfaces. The temperature distribution throughout the glass 15 min after the strip heaters are turned on and also when steady conditions are reached are to be determined using the implicit method with a time step of Δt = 1 min.



Assumptions **1** Heat transfer through the glass is given to be transient and two-dimensional. **2** Thermal conductivity is constant. **3** There is heat generation only at the inner surface, which will be treated as prescribed heat flux.

Properties The conductivity and diffusivity are given to be k = 0.84 W/m·°C and $\alpha = 0.39 \times 10^{-6}\ \text{m}^2/\text{s}$.

Analysis The nodal spacing is given to be Δx = 0.2 cm and Δy = 1 cm. The implicit finite difference equations are determined on the basis of the energy balance for the transient case expressed as

$$\sum_{\text{All sides}} \dot{Q}^{i+1} + \dot{G}^{i+1}_{\text{element}} = \rho V_{\text{element}} C \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

We consider only 9 nodes because of symmetry. Note that we do not have a square mesh in this case, and thus we will have to rely on energy balances to obtain the finite difference equations. Using energy balances, the finite difference equations for each of the 9 nodes are obtained as follows:

Node 1: $h_i \dfrac{\Delta y}{2}(T_i - T_1^{i+1}) + k\dfrac{\Delta x}{2}\dfrac{T_4^{i+1} - T_1^{i+1}}{\Delta y} + k\dfrac{\Delta y}{2}\dfrac{T_2^{i+1} - T_1^{i+1}}{\Delta x} = \rho C \dfrac{\Delta x}{2}\dfrac{\Delta y}{2}\dfrac{T_1^{i+1} - T_1^i}{\Delta t}$

Node 2: $k\dfrac{\Delta y}{2}\dfrac{T_1^{i+1} - T_2^{i+1}}{\Delta x} + k\dfrac{\Delta y}{2}\dfrac{T_3^{i+1} - T_2^{i+1}}{\Delta x} + k\Delta x\dfrac{T_5^{i+1} - T_2^{i+1}}{\Delta y} = \rho C \Delta x\dfrac{\Delta y}{2}\dfrac{T_2^{i+1} - T_2^i}{\Delta t}$

Node 3: $h_o \dfrac{\Delta y}{2}(T_o - T_3^{i+1}) + k\dfrac{\Delta x}{2}\dfrac{T_6^{i+1} - T_3^{i+1}}{\Delta y} + k\dfrac{\Delta y}{2}\dfrac{T_2^{i+1} - T_3^{i+1}}{\Delta x} = \rho C \dfrac{\Delta x}{2}\dfrac{\Delta y}{2}\dfrac{T_3^{i+1} - T_3^i}{\Delta t}$

N4: $h_i \Delta y(T_i - T_4^{i+1}) + k\dfrac{\Delta x}{2}\dfrac{T_1^{i+1} - T_4^{i+1}}{\Delta y} + k\dfrac{\Delta x}{2}\dfrac{T_7^{i+1} - T_4^{i+1}}{\Delta y} + k\Delta y\dfrac{T_5^{i+1} - T_4^{i+1}}{\Delta x} = \rho C \Delta y\dfrac{\Delta x}{2}\dfrac{T_4^{i+1} - T_4^i}{\Delta t}$

Node 5: $k\Delta y\dfrac{T_4^{i+1} - T_5^{i+1}}{\Delta x} + k\Delta y\dfrac{T_6^{i+1} - T_5^{i+1}}{\Delta x} + k\Delta x\dfrac{T_8^{i+1} - T_5^{i+1}}{\Delta y} + k\Delta x\dfrac{T_2^{i+1} - T_5^{i+1}}{\Delta y} = \rho C \Delta x \Delta y\dfrac{T_5^{i+1} - T_5^i}{\Delta t}$

N6: $h_o \Delta y(T_i - T_6^{i+1}) + k\dfrac{\Delta x}{2}\dfrac{T_3^{i+1} - T_6^{i+1}}{\Delta y} + k\dfrac{\Delta x}{2}\dfrac{T_9^{i+1} - T_6^{i+1}}{\Delta y} + k\Delta y\dfrac{T_5^{i+1} - T_6^{i+1}}{\Delta x} = \rho C \Delta y\dfrac{\Delta x}{2}\dfrac{T_6^{i+1} - T_6^i}{\Delta t}$

Node 7: $5\ \text{W} + h_i \dfrac{\Delta y}{2}(T_i - T_7^{i+1}) + k\dfrac{\Delta x}{2}\dfrac{T_4^{i+1} - T_7^{i+1}}{\Delta y} + k\dfrac{\Delta y}{2}\dfrac{T_8^{i+1} - T_7^{i+1}}{\Delta x} = \rho C \dfrac{\Delta x}{2}\dfrac{\Delta y}{2}\dfrac{T_7^{i+1} - T_7^i}{\Delta t}$

Node 8: $k\dfrac{\Delta y}{2}\dfrac{T_7^{i+1} - T_8^{i+1}}{\Delta x} + k\dfrac{\Delta y}{2}\dfrac{T_9^{i+1} - T_8^{i+1}}{\Delta x} + k\Delta x\dfrac{T_5^{i+1} - T_8^{i+1}}{\Delta y} = \rho C \Delta x\dfrac{\Delta y}{2}\dfrac{T_8^{i+1} - T_8^i}{\Delta t}$

Node 9: $h_o \dfrac{\Delta y}{2}(T_o - T_9^{i+1}) + k\dfrac{\Delta x}{2}\dfrac{T_6^{i+1} - T_9^{i+1}}{\Delta y} + k\dfrac{\Delta y}{2}\dfrac{T_8^{i+1} - T_9^{i+1}}{\Delta x} = \rho C \dfrac{\Delta x}{2}\dfrac{\Delta y}{2}\dfrac{T_9^{i+1} - T_9^i}{\Delta t}$

where k = 0.84 W/m.°C, $\alpha = k/\rho C = 0.39\times10^{-6}\ \text{m}^2/\text{s}$, $T_i = T_o$ = -3°C h_i = 6 W/m^2.°C, h_o = 20 W/m^2.°C, Δx = 0.002 m, and Δy = 0.01 m. Taking time step to be Δt = 1 min, the temperature distribution throughout the glass 15 min after the strip heaters are turned on and when steady conditions are reached are determined to be (from the EES solutions disk)

15 min: T_1 = -2.4°C, T_2 = -2.4°C, T_3 = -2.5°C, T_4 = -1.8°C, T_5 = -2.0°C,

T_6 = -2.7°C, T_7 = 12.3°C, T_8 = 10.7°C, T_9 = 9.6°C

Steady-state: T_1 = -2.4°C, T_2 = -2.4°C, T_3 = -2.5°C, T_4 = -1.8°C, T_5 = -2.0°C,

T_6 = -2.7°C, T_7 = 12.3°C, T_8 = 10.7°C, T_9 = 9.6°C

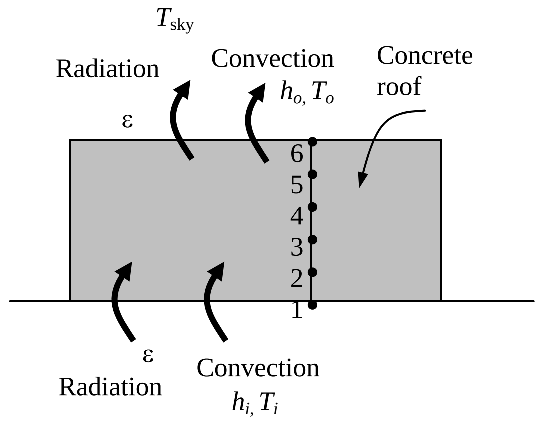
Discussion Steady operating conditions are reached in about 8 min.

5-93 The roof of a house initially at a uniform temperature is subjected to convection and radiation on both sides. The temperatures of the inner and outer surfaces of the roof at 6 am in the morning as well as the average rate of heat transfer through the roof during that night are to be determined.

Assumptions **1** Heat transfer is one-dimensional. **2** Thermal properties, heat transfer coefficients, and the indoor and outdoor temperatures are constant. **3** Radiation heat transfer is significant.

Properties The conductivity and diffusivity are given to be k = 1.4 W/m.°C and $\alpha = 0.69\times 10^{-6}\ \text{m}^2/\text{s}$. The emissivity of both surfaces of the concrete roof is 0.9.

Analysis The nodal spacing is given to be Δx = 0.03 m. Then the number of nodes becomes $M = L/\Delta x + 1$ = 0.15/0.03+1 = 6. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations. Nodes 2, 3, 4, and 5 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as



$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

$$\rightarrow\ T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i + \tau\frac{\dot{g}_m^i \Delta x^2}{k}$$

The finite difference equations for nodes 1 and 6 subjected to convection and radiation are obtained from an energy balance by taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (convection)}:\quad h_i(T_i - T_1^i) + k\frac{T_2^i - T_1^i}{\Delta x} + \varepsilon\sigma\left[T_{wall}^4 - (T_1^i + 273)^4\right] = \rho\frac{\Delta x}{2}C\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

$$\text{Node 2 (interior)}:\quad T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i$$

$$\text{Node 3 (interior)}:\quad T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i$$

$$\text{Node 4 (interior)}:\quad T_4^{i+1} = \tau(T_3^i + T_5^i) + (1-2\tau)T_4^i$$

$$\text{Node 5 (interior)}:\quad T_5^{i+1} = \tau(T_4^i + T_6^i) + (1-2\tau)T_5^i$$

$$\text{Node 6 (convection)}:\quad h_o(T_0 - T_6^i) + k\frac{T_5^i - T_6^i}{\Delta x} + \varepsilon\sigma\left[T_{sky}^4 - (T_6^i + 273)^4\right] = \rho\frac{\Delta x}{2}C\frac{T_6^{i+1} - T_6^i}{\Delta t}$$

where k = 1.4 W/m.°C, $\alpha = k/\rho C = 0.69\times 10^{-6}\ \text{m}^2/\text{s}$, T_i = 20°C, T_{wall} = 293 K, T_o = 6°C, T_{sky} =260 K, h_i = 5 W/m^2.°C, h_o = 12 W/m^2.°C, Δx = 0.03 m, and Δt = 5 min. Also, the mesh Fourier number is

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(0.69\times 10^{-6}\ \text{m}^2/\text{s})(300\ \text{s})}{(0.03\ \text{m})^2} = 0.230$$

Substituting this value of τ and other given quantities, the inner and outer surface temperatures of the roof after 12×(60/5) = 144 time steps (12 h) are determined to be **T_1 = 10.3°C and T_6 = -0.97°C.**

(*b*) The average temperature of the inner surface of the roof can be taken to be

$$T_{1,ave} = \frac{T_{1@6\,\text{PM}} + T_{1@6\,\text{AM}}}{2} = \frac{18 + 10.3}{2} = 14.15°\text{C}$$

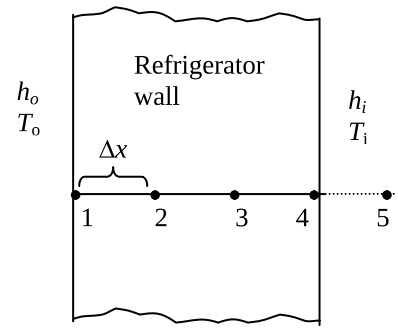
Then the average rate of heat loss through the roof that night becomes

$$\begin{aligned}\dot{Q}_{ave} &= h_i A_s(T_i - T_{1,ave}) + \varepsilon\sigma A_s\left[T_{wall}^4 - (T_1^i + 273)^4\right]\\ &= (5\ \text{W/m}^2\cdot°\text{C})(20\times 20\ \text{m}^2)(20 - 14.15)°\text{C} + 0.9(20\times 20\ \text{m}^2)(5.67\times 10^{-8}\ \text{W/m}^2\cdot\text{K}^4)[(293\ \text{K})^4 - (14.15 + 273\ \text{K})^4]\\ &= \mathbf{23{,}360\ W}\end{aligned}$$

5-94 A refrigerator whose walls are constructed of 3-cm thick urethane insulation malfunctions, and stops running for 6 h. The temperature inside the refrigerator at the end of this 6 h period is to be determined.

Assumptions **1** Heat transfer is one-dimensional since the walls are large relative to their thickness. **2** Thermal properties, heat transfer coefficients, and the outdoor temperature are constant. **3** Radiation heat transfer is negligible. **4** The temperature of the contents of the refrigerator, including the air inside, rises uniformly during this period. **5** The local atmospheric pressure is 1 atm. **6** The space occupied by food and the corner effects are negligible. **7** Heat transfer through the bottom surface of the refrigerator is negligible.

Properties The conductivity and diffusivity are given to be k = 1.4 W/m.°C and $\alpha = 0.69 \times 10^{-6}\ \text{m}^2/\text{s}$. The average specific heat of food items is given to be 3.6 kJ/kg.°C. The specific heat and density of air at 1 atm and 3°C are C_p = 1.004 kJ/kg.°C and ρ = 1.29 kg/m^3 (Table A-15.



Analysis The nodal spacing is given to be Δx = 0.01 m. Then the number of nodes becomes $M = L/\Delta x + 1$ = 0.03/0.01+1 = 4. This problem involves 4 unknown nodal temperatures, and thus we need to have 4 equations. Nodes 2 and 3 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \ \rightarrow\ T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i + \tau\frac{\dot{g}_m^i \Delta x^2}{k}$$

The finite difference equations for nodes 1 and 4 subjected to convection and radiation are obtained from an energy balance by taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (convection)}: \quad h_o(T_0 - T_1^i) + k\frac{T_2^i - T_1^i}{\Delta x} = \rho\frac{\Delta x}{2}C\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

$$\text{Node 2 (interior)}: \quad T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i$$

$$\text{Node 3 (interior)}: \quad T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i$$

$$\text{Node 4 (convection)}: \quad h_i(T_5^i - T_4^i) + k\frac{T_3^i - T_4^i}{\Delta x} = \rho\frac{\Delta x}{2}C\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

where k = 0.026 W/m.°C, $\alpha = k/\rho C = 0.36 \times 10^{-6}\ \text{m}^2/\text{s}$, $T_5 = T_i$ = 3°C (initially), T_o = 25°C, h_i = 6 W/m^2.°C, h_o = 9 W/m^2.°C, Δx = 0.01 m, and Δt = 1 min. Also, the mesh Fourier number is

$$\tau = \frac{\alpha \Delta t}{\Delta x^2} = \frac{(0.39 \times 10^{-6}\ \text{m}^2/\text{s})(60\ \text{s})}{(0.01\ \text{m})^2} = 0.216$$

The volume of the refrigerator cavity and the mass of air inside are

$$V = (1.80 - 0.03)(0.8 - 0.03)(0.7 - 0.03) = 0.913\,\text{m}^3$$

$$m_{air} = \rho V = (1.29\ \text{kg/m}^3)(0.824\ \text{m}^3) = 1.063\ \text{kg}$$

Energy balance for the air space of the refrigerator can be expressed as

$$\text{Node 5 (refrig. air)}: \quad h_i A_i (T_4^i - T_5^i) = (mC\Delta T)_{air} + (mC\Delta T)_{food}$$

or

$$h_i A_i (T_4^i - T_5^i) = \left[(mC)_{air} + (mC)_{food}\right]\frac{T_5^{i+1} - T_5^i}{\Delta t}$$

where

$$A_i = 2(1.77 \times 0.77) + 2(1.77 \times 0.67) + (0.77 \times 0.67) = 5.6135\,\text{m}^2$$

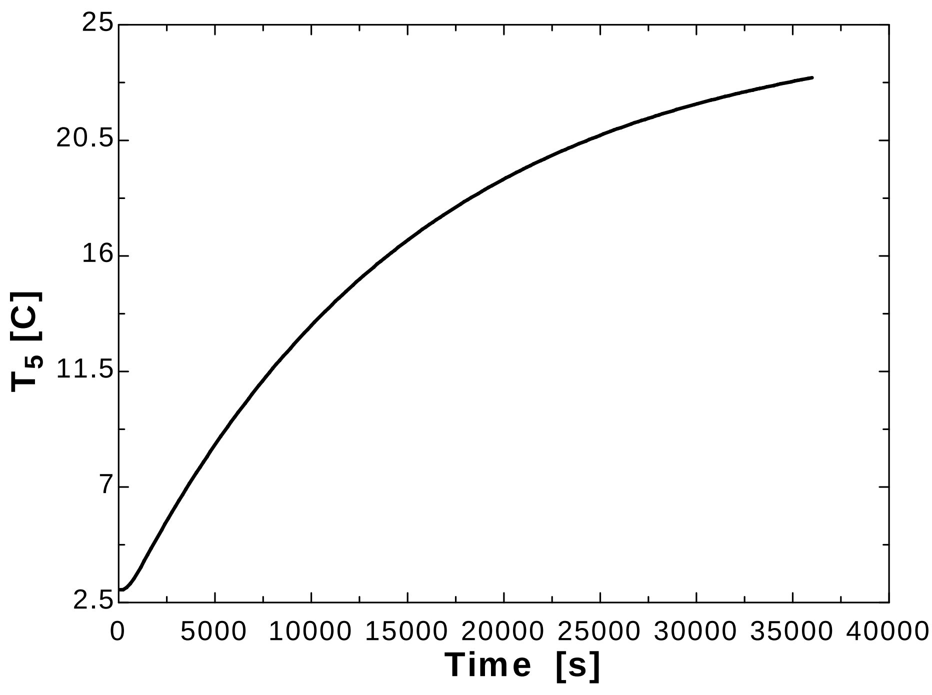
Substituting, temperatures of the refrigerated space after 6×60 = 360 time steps (6 h) is determined to be

$\boldsymbol{T_{in} = T_5}$ **=19.6°C.**

5-95 "!PROBLEM 5-95"

```
"GIVEN"
t_ins=0.03 "[m]"
k=0.026 "[W/m-C]"
alpha=0.36E-6 "[m^2/s]"
T_i=3 "[C]"
h_i=6 "[W/m^2-C]"
h_o=9 "[W/m^2-C]"
T_infinity=25 "[C]"
m_food=15 "[kg]"
C_food=3600 "[J/kg-C]"
DELTAx=0.01 "[m]"
DELTAt=60 "[s]"
"time=6*3600 [s], parameter to be varied"

"PROPERTIES"
rho_air=density(air, T=T_i, P=101.3)
C_air=CP(air, T=T_i)*Convert(kJ/kg-C, J/kg-C)

"ANALYSIS"
M=t_ins/DELTAx+1 "Number of nodes"
tau=(alpha*DELTAt)/DELTAx^2
RhoC=k/alpha "RhoC=rho*C"

"The technique is to store the temperatures in the parametric table and recover them (as old
temperatures)
using the variable ROW.  The first row contains the initial values so Solve Table must begin
at row 2.
Use the DUPLICATE statement to reduce the number of equations that need to be typed.
Column 1
contains the time, column 2 the value of T[1], column 3, the value of T[2], etc., and column 7
the Row."
Time=TableValue('Table 1',Row-1,#Time)+DELTAt
Duplicate i=1,5
    T_old[i]=TableValue('Table 1',Row-1,#T[i])
end

"Using the explicit finite difference approach, the six equations for the six unknown
temperatures are determined to be"
h_o*(T_infinity-T_old[1])+k*(T_old[2]-T_old[1])/DELTAx=RhoC*DELTAx/2*(T[1]-
T_old[1])/DELTAt "Node 1, convection"
T[2]=tau*(T_old[1]+T_old[3])+(1-2*tau)*T_old[2] "Node 2"
T[3]=tau*(T_old[2]+T_old[4])+(1-2*tau)*T_old[3] "Node 3"
h_i*(T_old[5]-T_old[4])+k*(T_old[3]-T_old[4])/DELTAx=RhoC*DELTAx/2*(T[4]-
T_old[4])/DELTAt "Node 4, convection"
h_i*A_i*(T_old[4]-T_old[5])=m_air*C_air*(T[5]-T_old[5])/DELTAt+m_food*C_food*(T[5]-
T_old[5])/DELTAt "Node 5, refrig. air"

A_i=2*(1.8-0.03)*(0.8-0.03)+2*(1.8-0.03)*(0.7-0.03)+(0.8-0.03)*(0.7-0.03)
m_air=rho_air*V_air
V_air=(1.8-0.03)*(0.8-0.03)*(0.7-0.03)
```

Time [s]	T_1 [C]	T_2 [C]	T_3 [C]	T_4 [C]	T_5 [C]	Row
0	3	3	3	3	3	1
60	35.9	3	3	3	3	2
120	5.389	10.11	3	3	3	3
180	36.75	7.552	4.535	3	3	4
240	6.563	13.21	4.855	3.663	3	5
300	37	9.968	6.402	3.517	3.024	6
360	7.374	15.04	6.549	4.272	3.042	7
420	37.04	11.55	7.891	4.03	3.087	8
480	8.021	16.27	7.847	4.758	3.122	9
540	36.97	12.67	8.998	4.461	3.182	10
…	…	…	…	…	…	…
…	…	…	…	…	…	…
35460	24.85	24.23	23.65	23.09	22.86	592
35520	24.81	24.24	23.65	23.1	22.87	593
35580	24.85	24.23	23.66	23.11	22.88	594
35640	24.81	24.24	23.67	23.12	22.88	595
35700	24.85	24.24	23.67	23.12	22.89	596
35760	24.81	24.25	23.68	23.13	22.9	597
35820	24.85	24.25	23.68	23.14	22.91	598
35880	24.81	24.26	23.69	23.15	22.92	599
35940	24.85	24.25	23.69	23.15	22.93	600
36000	24.82	24.26	23.7	23.16	22.94	601



Special Topic: Controlling the Numerical Error

5-96C The results obtained using a numerical method differ from the exact results obtained analytically because the results obtained by a numerical method are approximate. The difference between a numerical solution and the exact solution (the error) is primarily due to two sources: The *discretization error* (also called the *truncation* or *formulation* error) which is caused by the approximations used in the formulation of the numerical method, and the *round-off error* which is caused by the computers' representing a number by using a limited number of significant digits and continuously rounding (or chopping) off the digits it cannot retain.

5-97C The *discretization error* (also called the *truncation* or *formulation* error) is due to replacing the derivatives by differences in each step, or replacing the actual temperature distribution between two adjacent nodes by a straight line segment. The difference between the two solutions at each time step is called the *local discretization error*. The total discretization error at any step is called the *global* or *accumulated discretization error*. The local and global discretization errors are identical for the first time step.

5-98C Yes, the global (accumulated) discretization error be less than the local error during a step. The global discretization error usually increases with increasing number of steps, but the opposite may occur when the solution function changes direction frequently, giving rise to local discretization errors of opposite signs which tend to cancel each other.

5-99C The Taylor series expansion of the temperature at a specified nodal point *m* about time t_i is

$$T(x_m, t_i + \Delta t) = T(x_m, t_i) + \Delta t \frac{\partial T(x_m, t_i)}{\partial t} + \frac{1}{2}\Delta t^2 \frac{\partial^2 T(x_m, t_i)}{\partial t^2} + \cdots$$

The finite difference formulation of the time derivative at the same nodal point is expressed as

$$\frac{\partial T(x_m, t_i)}{\partial t} \cong \frac{T(x_m, t_i + \Delta t) - T(x_m, t_i)}{\Delta t} = \frac{T_m^{i+1} - T_m^i}{\Delta t} \quad \text{or} \quad T(x_m, t_i + \Delta t) \cong T(x_m, t_i) + \Delta t \frac{\partial T(x_m, t_i)}{\partial t}$$

which resembles the Taylor series expansion terminated after the first two terms.

5-100C The Taylor series expansion of the temperature at a specified nodal point *m* about time t_i is

$$T(x_m, t_i + \Delta t) = T(x_m, t_i) + \Delta t \frac{\partial T(x_m, t_i)}{\partial t} + \frac{1}{2}\Delta t^2 \frac{\partial^2 T(x_m, t_i)}{\partial t^2} + \cdots$$

The finite difference formulation of the time derivative at the same nodal point is expressed as

$$\frac{\partial T(x_m, t_i)}{\partial t} \cong \frac{T(x_m, t_i + \Delta t) - T(x_m, t_i)}{\Delta t} = \frac{T_m^{i+1} - T_m^i}{\Delta t} \quad \text{or} \quad T(x_m, t_i + \Delta t) \cong T(x_m, t_i) + \Delta t \frac{\partial T(x_m, t_i)}{\partial t}$$

which resembles the Taylor series expansion terminated after the first two terms. Therefore, the 3rd and following terms in the Taylor series expansion represent the error involved in the finite difference approximation. For a sufficiently small time step, these terms decay rapidly as the order of derivative increases, and their contributions become smaller and smaller. The first term neglected in the Taylor series expansion is proportional to $(\Delta t)^2$, and thus the local discretization error is also proportional to $(\Delta t)^2$.

The global discretization error is proportional to the step size to Δt itself since, at the worst case, the accumulated discretization error after I time steps during a time period t_0 is $I\Delta t^2 = (t_0 / \Delta t)\Delta t^2 = t_0 \Delta t$ which is proportional to Δt.

5-101C The *round-off error* is caused by retaining a limited number of digits during calculations. It depends on the number of calculations, the method of rounding off, the type of the computer, and even the sequence of calculations. Calculations that involve the alternate addition of small and large numbers are most susceptible to round-off error.

5-102C As the step size is decreased, the discretization error decreases but the round-off error increases.

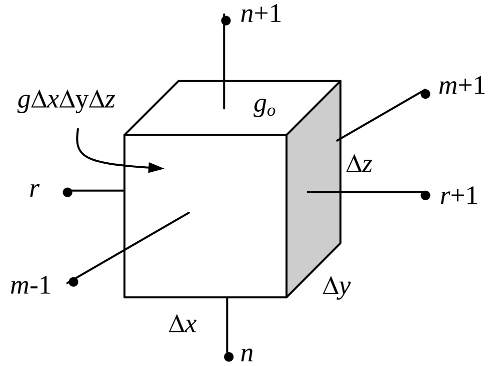
5-103C The round-off error can be reduced by avoiding extremely small mess sizes (smaller than necessary to keep the discretization error in check) and sequencing the terms in the program such that the addition of small and large numbers is avoided.

5-104C A practical way of checking if the round-off error has been significant in calculations is to repeat the calculations using double precision holding the mesh size and the size of the time step constant. If the changes are not significant, we conclude that the round-off error is not a problem.

5-105C A practical way of checking if the discretization error has been significant in calculations is to start the calculations with a reasonable mesh size Δx (and time step size Δt for transient problems), based on experience, and then to repeat the calculations using a mesh size of $\Delta x/2$. If the results obtained by halving the mesh size do not differ significantly from the results obtained with the full mesh size, we conclude that the discretization error is at an acceptable level.

Review Problems

5-106 Starting with an energy balance on a volume element, the steady three-dimensional finite difference equation for a general interior node in rectangular coordinates for $T(x, y, z)$ for the case of constant thermal conductivity and uniform heat generation is to be obtained.



Analysis We consider a *volume element* of size $\Delta x \times \Delta y \times \Delta z$ centered about a general interior node (m, n, r) in a region in which heat is generated at a constant rate of $\dot{g}_0$ and the thermal conductivity k is variable. Assuming the direction of heat conduction to be *towards* the node under consideration at all surfaces, the energy balance on the volume element can be expressed as

$$\dot{Q}_{\text{cond, left}} + \dot{Q}_{\text{cond, top}} + \dot{Q}_{\text{cond, right}} + \dot{Q}_{\text{cond, bottom}} + \dot{Q}_{\text{cond, front}} + \dot{Q}_{\text{cond, back}} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t} = 0$$

for the *steady* case. Again assuming the temperatures between the adjacent nodes to vary linearly, the energy balance relation above becomes

$$k(\Delta y \times \Delta z)\frac{T_{m-1,n,r} - T_{m,n,r}}{\Delta x} + k(\Delta x \times \Delta z)\frac{T_{m,n+1,r} - T_{m,n,r}}{\Delta y}$$
$$+ k(\Delta y \times \Delta z)\frac{T_{m+1,n,r} - T_{m,n,r}}{\Delta x} + k(\Delta x \times \Delta z)\frac{T_{m,n-1,r} - T_{m,n,r}}{\Delta y}$$
$$+ k(\Delta x \times \Delta y)\frac{T_{m,n,r-1} - T_{m,n,r}}{\Delta z} + k(\Delta x \times \Delta y)\frac{T_{m,n,r+1} - T_{m,n,r}}{\Delta z} + \dot{g}_0(\Delta x \times \Delta y \times \Delta z) = 0$$

Dividing each term by $k\,\Delta x \times \Delta y \times \Delta z$ and simplifying gives

$$\frac{T_{m-1,n,r} - 2T_{m,n,r} + T_{m+1,n,r}}{\Delta x^2} + \frac{T_{m,n-1,r} - 2T_{m,n,r} + T_{m,n+1,r}}{\Delta y^2} + \frac{T_{m,n,r-1} - 2T_{m,n,r} + T_{m,n,r+1}}{\Delta z^2} + \frac{\dot{g}_0}{k} = 0$$

For a cubic mesh with $\Delta x = \Delta y = \Delta z = l$, and the relation above simplifies to

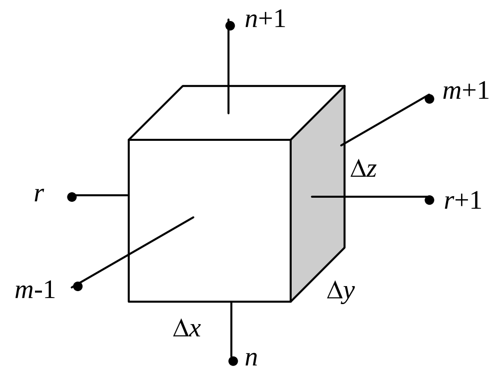
$$T_{m-1,n,r} + T_{m+1,n,r} + T_{m,n-1,r} + T_{m,n-1,r} + T_{m,n,r-1} + T_{m,n,r+1} - 6T_{m,n,r} + \frac{\dot{g}_0 l^2}{k} = 0$$

It can also be expressed in the following easy-to-remember form:

$$T_{\text{left}} + T_{\text{top}} + T_{\text{right}} + T_{\text{bottom}} + T_{\text{front}} + T_{\text{back}} - 6T_{\text{node}} + \frac{\dot{g}_0 l^2}{k} = 0$$

5-107 Starting with an energy balance on a volume element, the three-dimensional transient *explicit* finite difference equation for a general interior node in rectangular coordinates for $T(x, y, z, t)$ for the case of constant thermal conductivity k and no heat generation is to be obtained.

Analysis We consider a rectangular region in which heat conduction is significant in the x and y directions. There is no heat generation in the medium, and the thermal conductivity k of the medium is constant. Now we divide the x-y-z region into a *mesh* of nodal points which are spaced Δx, Δy, and Δz apart in the x, y, and z directions, respectively, and consider a general interior node (m, n, r) whose coordinates are $x = m\Delta x$, $y = n\Delta y$, are $z = r\Delta z$. Noting that the volume element centered about the general interior node (m, n, r) involves heat conduction from six sides (right, left, front, rear, top, and bottom) and expressing them at previous time step i, the transient explicit finite difference formulation for a general interior node can be expressed as



$$k(\Delta y\times\Delta z)\frac{T^i_{m-1,n,r}-T^i_{m,n,r}}{\Delta x}+k(\Delta x\times\Delta z)\frac{T^i_{m,n+1,r}-T^i_{m,n,r}}{\Delta y}+k(\Delta y\times\Delta z)\frac{T^i_{m+1,n,r}-T^i_{m,n,r}}{\Delta x}$$
$$+k(\Delta x\times\Delta z)\frac{T^i_{m,n-1,r}-T^i_{m,n,r}}{\Delta y}+k(\Delta x\times\Delta y)\frac{T^i_{m,n,r-1}-T^i_{m,n,r}}{\Delta z}+k(\Delta x\times\Delta y)\frac{T^i_{m,n,r+1}-T^i_{m,n,r}}{\Delta z}$$
$$=\rho(\Delta x\times\Delta y\times 1)C\frac{T^{i+1}_{m,n}-T^i_{m,n}}{\Delta t}$$

Taking a cubic mesh ($\Delta x = \Delta y = \Delta z = l$) and dividing each term by k gives, after simplifying,

$$T^i_{m-1,n,r}+T^i_{m+1,n,r}+T^i_{m,n+1,r}+T^i_{m,n-1,r}+T^i_{m,n,r-1}+T^i_{m,n,r+1}-6T^i_{m,n,r}=\frac{T^{i+1}_{m,n,r}-T^i_{m,n,r}}{\tau}$$

where $\alpha = k/(\rho C)$ is the thermal diffusivity of the material and $\tau = \alpha\Delta t/l^2$ is the dimensionless mesh Fourier number. It can also be expressed in terms of the temperatures at the neighboring nodes in the following easy-to-remember form:

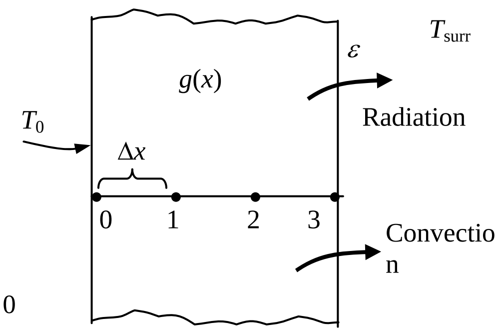
$$T^i_{\text{left}}+T^i_{\text{top}}+T^i_{\text{right}}+T^i_{\text{bottom}}+T^i_{\text{front}}+T^i_{\text{back}}-6T^i_{\text{node}}=\frac{T^{i+1}_{\text{node}}-T^i_{\text{node}}}{\tau}$$

Discussion We note that setting $T^{i+1}_{\text{node}}=T^i_{\text{node}}$ gives the steady finite difference formulation.

5-108 A plane wall with variable heat generation and constant thermal conductivity is subjected to combined convection and radiation at the right (node 3) and specified temperature at the left boundary (node 0). The finite difference formulation of the right boundary node (node 3) and the finite difference formulation for the rate of heat transfer at the left boundary (node 0) are to be determined.

Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional. **2** The thermal conductivity is given to be constant.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become



Right boundary node (all temperatures are in K):

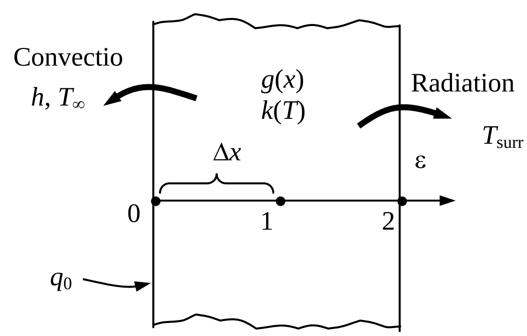
$$\varepsilon\sigma A(T_{surr}^4 - T_3^4) + hA(T_\infty - T_3) + kA\frac{T_2 - T_3}{\Delta x} + \dot{g}_3(A\Delta x/2) = 0$$

Heat transfer at left surface: $\dot{Q}_{\text{left surface}} + kA\frac{T_1 - T_0}{\Delta x} + \dot{g}_0(A\Delta x/2) = 0$

5-109 A plane wall with variable heat generation and variable thermal conductivity is subjected to uniform heat flux $\dot{q}_0$ and convection at the left (node 0) and radiation at the right boundary (node 2). The explicit transient finite difference formulation of the problem using the energy balance approach method is to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient, and the thermal conductivity and heat generation to be variables. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Radiation from the left surface, and convection from the right surface are negligible.

Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *explicit* finite difference formulations become



Left boundary node (node 0): $k_0^i A\frac{T_1^i - T_0^i}{\Delta x} + \dot{q}_0 A + hA(T_\infty - T_0^i) + \dot{g}_0^i(A\Delta x/2) = \rho A\frac{\Delta x}{2}C\frac{T_0^{i+1} - T_0^i}{\Delta t}$

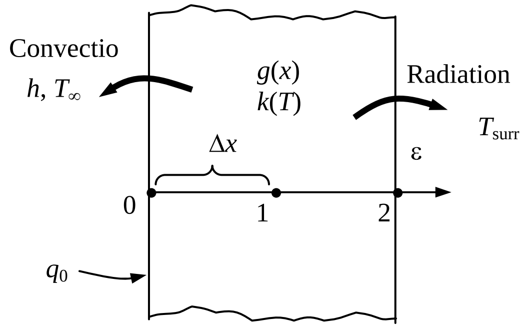
Interior node (node 1): $k_1^i A\frac{T_0^i - T_1^i}{\Delta x} + k_1^i A\frac{T_2^i - T_1^i}{\Delta x} + \dot{g}_0^i(A\Delta x) = \rho A\Delta x C\frac{T_1^{i+1} - T_1^i}{\Delta t}$

Right boundary node (node 2):

$$k_2^i A\frac{T_1^i - T_2^i}{\Delta x} + \varepsilon\sigma A[(T_{surr}^i + 273)^4 - (T_2^i + 273)^4] + \dot{g}_2^i(A\Delta x/2) = \rho A\frac{\Delta x}{2}C\frac{T_2^{i+1} - T_2^i}{\Delta t}$$

5-110 A plane wall with variable heat generation and variable thermal conductivity is subjected to uniform heat flux $\dot{q}_0$ and convection at the left (node 0) and radiation at the right boundary (node 2). The implicit transient finite difference formulation of the problem using the energy balance approach method is to be determined.

Assumptions **1** Heat transfer through the wall is given to be transient, and the thermal conductivity and heat generation to be variables. **2** Heat transfer is one-dimensional since the plate is large relative to its thickness. **3** Radiation from the left surface, and convection from the right surface are negligible.



Analysis Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the *implicit* finite difference formulations become

Left boundary node (node 0):

$$k_0^{i+1}A\frac{T_1^{i+1}-T_0^{i+1}}{\Delta x}+\dot{q}_0A+hA(T_\infty-T_0^{i+1})+\dot{g}_0^{i+1}(A\Delta x/2)=\rho A\frac{\Delta x}{2}C\frac{T_0^{i+1}-T_0^i}{\Delta t}$$

Interior node (node 1): $$k_1^{i+1}A\frac{T_0^{i+1}-T_1^{i+1}}{\Delta x}+k_1^{i+1}A\frac{T_2^{i+1}-T_1^{i+1}}{\Delta x}+\dot{g}_0^{i+1}(A\Delta x)=\rho A\Delta xC\frac{T_1^{i+1}-T_1^i}{\Delta t}$$

Right boundary node (node 2):

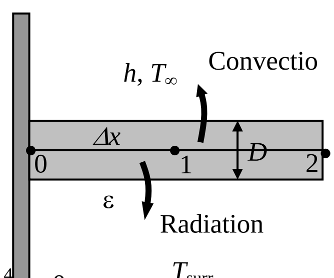
$$k_2^{i+1}A\frac{T_1^{i+1}-T_2^{i+1}}{\Delta x}+\varepsilon\sigma A[(T_{surr}^{i+1}+273)^4-(T_2^{i+1}+273)^4]+\dot{g}_2^{i+1}(A\Delta x/2)=\rho A\frac{\Delta x}{2}C\frac{T_2^{i+1}-T_2^i}{\Delta t}$$

5-111 A pin fin with convection and radiation heat transfer from its tip is considered. The complete finite difference formulation for the determination of nodal temperatures is to be obtained.

Assumptions **1** Heat transfer through the pin fin is given to be steady and one-dimensional, and the thermal conductivity to be constant. **2** Convection heat transfer coefficient and emissivity are constant and uniform.

Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional, and the thermal conductivity and heat generation to be variable. **2** Convection heat transfer at the right surface is negligible.

Analysis The nodal network consists of 3 nodes, and the base temperature T_0 at node 0 is specified. Therefore, there are two unknowns T_1 and T_2, and we need two equations to determine them. Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the finite difference formulations become



Node 1 (at midpoint):

$$kA\frac{T_0-T_1}{\Delta x}+kA\frac{T_2-T_1}{\Delta x}+h(p\Delta x/2)(T_\infty-T_1)+\varepsilon\sigma A[T_{\text{surr}}^4-(T_1+273)^4]=0$$

Node 2 (at fin tip):

$$kA\frac{T_1-T_2}{\Delta x}+h(p\Delta x/2+A)(T_\infty-T_2)+\varepsilon\sigma(p\Delta x/2+A)[T_{\text{surr}}^4-(T_2+273)^4]=0$$

where $A=\pi D^2/4$ is the cross-sectional area and $p=\pi D$ is the perimeter of the fin.

5-112 Starting with an energy balance on a volume element, the two-dimensional transient *explicit* finite difference equation for a general interior node in rectangular coordinates for $T(x, y, t)$ for the case of constant thermal conductivity k and uniform heat generation $\dot{g}_0$ is to be obtained.

Analysis (See Figure 5-24 in the text)**.** We consider a rectangular region in which heat conduction is significant in the x and y directions, and consider a unit depth of $\Delta z = 1$ in the z direction. There is uniform heat generation in the medium, and the thermal conductivity k of the medium is constant. Now we divide the x-y plane of the region into a *rectangular mesh* of nodal points which are spaced Δx and Δy apart in the x and y directions, respectively, and consider a general interior node (m, n) whose coordinates are $x = m\Delta x$ and $y = n\Delta y$. Noting that the volume element centered about the general interior node (m, n) involves heat conduction from four sides (right, left, top, and bottom) and expressing them at previous time step i, the transient explicit finite difference formulation for a general interior node can be expressed as

$$k(\Delta y \times 1)\frac{T^i_{m-1,n} - T^i_{m,n}}{\Delta x} + k(\Delta x \times 1)\frac{T^i_{m,n+1} - T^i_{m,n}}{\Delta y} + k(\Delta y \times 1)\frac{T^i_{m+1,n} - T^i_{m,n}}{\Delta x} + k(\Delta x \times 1)\frac{T^i_{m,n-1} - T^i_{m,n}}{\Delta y}$$

$$+ \dot{g}_0(\Delta x \times \Delta y \times 1) = \rho(\Delta x \times \Delta y \times 1)C\frac{T^{i+1}_{m,n} - T^i_{m,n}}{\Delta t}$$

Taking a square mesh ($\Delta x = \Delta y = l$) and dividing each term by k gives, after simplifying,

$$T^i_{m-1,n} + T^i_{m+1,n} + T^i_{m,n+1} + T^i_{m,n-1} - 4T^i_{m,n} + \frac{\dot{g}_0 l^2}{k} = \frac{T^{i+1}_{m,n} - T^i_{m,n}}{\tau}$$

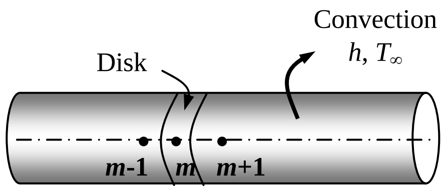
where $\alpha = k/(\rho C)$ is the thermal diffusivity of the material and $\tau = \alpha\Delta t / l^2$ is the dimensionless mesh Fourier number. It can also be expressed in terms of the temperatures at the neighboring nodes in the following easy-to-remember form:

$$T^i_{\text{left}} + T^i_{\text{top}} + T^i_{\text{right}} + T^i_{\text{bottom}} - 4T^i_{\text{node}} + \frac{\dot{g}_0 l^2}{k} = \frac{T^{i+1}_{\text{node}} - T^i_{\text{node}}}{\tau}$$

Discussion We note that setting $T^{i+1}_{\text{node}} = T^i_{\text{node}}$ gives the steady finite difference formulation.

5-113 Starting with an energy balance on a disk volume element, the one-dimensional transient explicit finite difference equation for a general interior node for $T(z,t)$ in a cylinder whose side surface is subjected to convection with a convection coefficient of h and an ambient temperature of T_∞ for the case of constant thermal conductivity with uniform heat generation is to be obtained.

Analysis We consider transient one-dimensional heat conduction in the axial z direction in a cylindrical rod of constant cross-sectional area A with constant heat generation $\dot{g}_0$ and constant conductivity k with a mesh size of Δz in the z direction. Noting that the volume element of a general interior node m involves heat conduction from two sides, convection from its lateral surface, and the volume of the element is $V_{\text{element}} = A\Delta z$, the transient explicit finite difference formulation for an interior node can be expressed as



$$hA(T_\infty - T_m^i) + kA\frac{T_{m-1}^i - T_m^i}{\Delta x} + kA\frac{T_{m+1}^i - T_m^i}{\Delta x} + \dot{g}_0 A\Delta x = \rho A\Delta x C\frac{T_m^{i+1} - T_m^i}{\Delta t}$$

Canceling the surface area A and multiplying by Δx/k, it simplifies to

$$T_{m-1}^i - (2 + h\Delta x / k)T_m^i + T_{m+1}^i + \frac{h\Delta x}{k}T_\infty + \frac{\dot{g}_0 \Delta x^2}{k} = \frac{(\Delta x)^2}{\alpha\Delta t}(T_m^{i+1} - T_m^i)$$

where $\alpha = k/(\rho C)$ is the *thermal diffusivity* of the wall material. Using the definition of the dimensionless *mesh Fourier number* $\tau = \dfrac{\alpha\Delta t}{(\Delta x)^2}$ the last equation reduces to

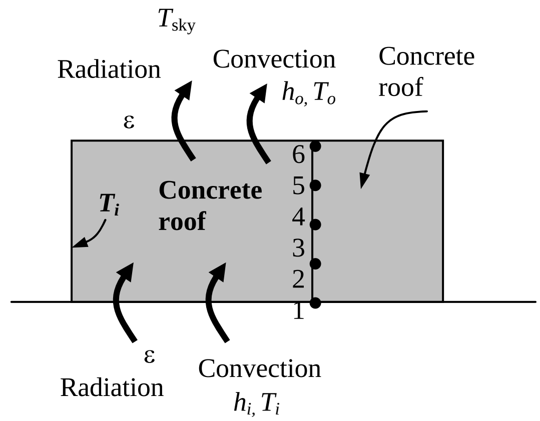
$$T_{m-1}^i - (2 + h\Delta x / k)T_m^i + T_{m+1}^i + \frac{h\Delta x}{k}T_\infty + \frac{\dot{g}_0 \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

Discussion We note that setting $T_{\text{m}}^{i+1} = T_{\text{m}}^i$ gives the steady finite difference formulation.

5-114E The roof of a house initially at a uniform temperature is subjected to convection and radiation on both sides. The temperatures of the inner and outer surfaces of the roof at 6 am in the morning as well as the average rate of heat transfer through the roof during that night are to be determined.

Assumptions **1** Heat transfer is one-dimensional since the roof is large relative to its thickness. **2** Thermal properties, heat transfer coefficients, and the indoor temperatures are constant. **3** Radiation heat transfer is significant. **4** The outdoor temperature remains constant in the 4-h blocks. **5** The given time step Δt = 5 min is less than the critical time step so that the stability criteria is satisfied.

Properties The conductivity and diffusivity are given to k = 0.81 Btu/h.ft.°F and $\alpha = 7.4\times10^{-6}\ \text{ft}^2/\text{s}$. The emissivity of both surfaces of the concrete roof is 0.9.



Analysis The nodal spacing is given to be Δx = 1 in. Then the number of nodes becomes $M = L/\Delta x + 1$ = 5/1+1 = 6. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations. Nodes 2, 3, 4, and 5 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

$$\rightarrow\ T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i + \tau\frac{\dot{g}_m^i \Delta x^2}{k}$$

The finite difference equations for nodes 1 and 6 subjected to convection and radiation are obtained from an energy balance by taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (convection)}:\quad h_i(T_i - T_1^i) + k\frac{T_2^i - T_1^i}{\Delta x} + \varepsilon\sigma\left[T_{wall}^4 - (T_1^i + 273)^4\right] = \rho\frac{\Delta x}{2}C\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

$$\text{Node 2 (interior)}:\quad T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i$$

$$\text{Node 3 (interior)}:\quad T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i$$

$$\text{Node 4 (interior)}:\quad T_4^{i+1} = \tau(T_3^i + T_5^i) + (1-2\tau)T_4^i$$

$$\text{Node 5 (interior)}:\quad T_5^{i+1} = \tau(T_4^i + T_6^i) + (1-2\tau)T_5^i$$

$$\text{Node 6 (convection)}:\quad h_o(T_0 - T_6^i) + k\frac{T_5^i - T_6^i}{\Delta x} + \varepsilon\sigma\left[T_{sky}^4 - (T_6^i + 273)^4\right] = \rho\frac{\Delta x}{2}C\frac{T_6^{i+1} - T_6^i}{\Delta t}$$

where k = 0.81 Btu/h.ft.°F, $\alpha = k/\rho C = 7.4\times10^{-6}\ \text{ft}^2/\text{s}$, T_i = 70°F, T_{wall} = 530 R, T_{sky} =445 R, h_i = 0.9 Btu/h.ft^2.°F, h_o = 2.1 Btu/h.ft^2.°F, Δx = 1/12 ft, and Δt = 5 min. Also, T_o = 50°F from 6 PM to 10 PM, 42°F from 10 PM to 2 AM, and 38°F from 2 AM to 6 AM. The mesh Fourier number is

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(7.4\times10^{-6}\ \text{ft}^2/\text{s})(300\ \text{s})}{(1/12\ \text{ft})^2} = 0.320$$

Substituting this value of τ and other given quantities, the inner and outer surface temperatures of the roof after 12×(60/5) = 144 time steps (12 h) are determined to be $\boldsymbol{T_1}$ **= 54.75°C and** $\boldsymbol{T_6}$ **= 40.18°C**

(b) The average temperature of the inner surface of the roof can be taken to be

$$T_{1,ave} = \frac{T_{1\ @\ 6\ \text{PM}} + T_{1\ @\ 6\ \text{AM}}}{2} = \frac{70 + 54.75}{2} = 62.38°\ \text{F}$$

Then the average rate of heat loss through the roof that night is determined to be

$$\begin{aligned}\dot{Q}_{ave} &= h_i A(T_i - T_{1,ave}) + \varepsilon\sigma A\left[T_{wall}^4 - (T_1^i + 273)^4\right]\\ &= (0.9\ \text{Btu/h.ft}^2\cdot\text{°F})(45\times55\ \text{ft}^2)(70 - 62.38)\text{°F}\\ &\quad + 0.9(45\times55\ \text{ft}^2)(0.1714\times10^{-8}\ \text{Btu/h.ft}^2\cdot\text{R}^4)[(530\ \text{R})^4 - (62.38 + 460\ \text{R})^4]\\ &= \mathbf{33{,}950\ Btu/h}\end{aligned}$$

5-115 A large pond is initially at a uniform temperature. Solar energy is incident on the pond surface at for 4 h The temperature distribution in the pond under the most favorable conditions is to be determined.

Assumptions **1** Heat transfer is one-dimensional since the pond is large relative to its depth. **2** Thermal properties, heat transfer coefficients, and the indoor temperatures are constant. **3** Radiation heat transfer is significant. **4** There are no convection currents in the water. **5** The given time step Δt = 15 min is less than the critical time step so that the stability criteria is satisfied. **6** All heat losses from the pond are negligible. **7** Heat generation due to absorption of radiation is uniform in each layer.

Properties The conductivity and diffusivity are given to be k = 0.61 W/m.°C and $\alpha = 0.15\times10^{-6}\ \text{m}^2/\text{s}$. The volumetric absorption coefficients of water are as given in the problem.

Analysis The nodal spacing is given to be Δx = 0.25 m. Then the number of nodes becomes $M = L/\Delta x + 1$ = 1/0.25+1 = 4. This problem involves 5 unknown nodal temperatures, and thus we need to have 5 equations. Nodes 2, 3, and 4 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \;\rightarrow\; T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i + \tau\frac{\dot{g}_m^i \Delta x^2}{k}$$

Node 0 can also be treated as an interior node by using the mirror image concept. The finite difference equation for node 4 subjected to heat flux is obtained from an energy balance by taking the direction of all heat transfers to be towards the node:

Node 0 (insulation): $T_0^{i+1} = \tau(T_1^i + T_1^i) + (1-2\tau)T_0^i + \tau\dot{g}_0(\Delta x)^2/k$

Node 0 (insulation): $T_1^{i+1} = \tau(T_0^i + T_2^i) + (1-2\tau)T_1^i + \tau\dot{g}_1(\Delta x)^2/k$

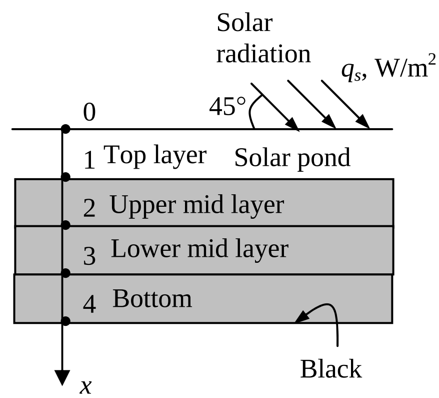
Node 2 (interior): $T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i + \tau\dot{g}_2(\Delta x)^2/k$

Node 3 (interior): $T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i + \tau\dot{g}_3(\Delta x)^2/k$

Node 6 (convection): $\dot{q}_b + k\dfrac{T_3^i - T_4^i}{\Delta x} + \tau\dot{g}_4(\Delta x)^2/k = \rho\dfrac{\Delta x}{2}C\dfrac{T_4^{i+1} - T_4^i}{\Delta t}$

where k = 0.61 W/m.°C, $\alpha = k/\rho C = 0.15\times10^{-6}\ \text{m}^2/\text{s}$, Δx = 0.25 m, and Δt = 15 min = 900 s. Also, the mesh Fourier number is

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(0.15\times10^{-6}\ \text{m}^2/\text{s})(900\ \text{s})}{(0.25\ \text{m})^2} = 0.002160$$



The values of heat generation rates at the nodal points are determined as follows:

$$\dot{g}_0 = \frac{\dot{G}_0}{\text{Volume}} = \frac{0.473\times500\ \text{W}}{(1\,\text{m}^2)(0.25\,\text{m})} = 946\ \text{W/m}^3$$

$$\dot{g}_1 = \frac{\dot{G}_1}{\text{Volume}} = \frac{[(0.473+0.061)/2]\times500\ \text{W}}{(1\,\text{m}^2)(0.25\,\text{m})} = 534\ \text{W/m}^3$$

$$\dot{g}_4 = \frac{\dot{G}_4}{\text{Volume}} = \frac{0.024\times500\ \text{W}}{(1\,\text{m}^2)(0.25\,\text{m})} = 48\ \text{W/m}^3$$

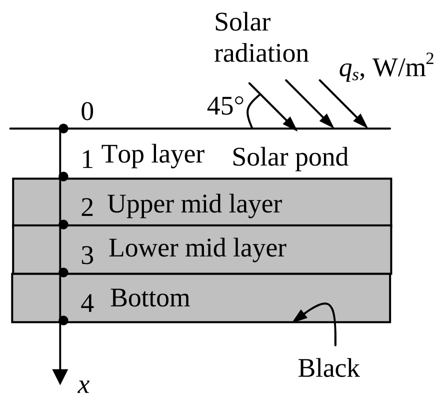
Also, the heat flux at the bottom surface is $\dot{q}_b = 0.379\times500\ \text{W/m}^2 = 4189.5\ \text{W/m}^2$. Substituting these values, the nodal temperatures in the pond after 4×(60/15) = 16 time steps (4 h) are determined to be

T_0 = 18.3°C, T_1 = 16.9°C, T_2 = 15.4°C, T_3 = 15.3°C, and T_4 = 20.2°C.

5-116 A large 1-m deep pond is initially at a uniform temperature of 15°C throughout. Solar energy is incident on the pond surface at 45° at an average rate of 500 W/m^2 for a period of 4 h The temperature distribution in the pond under the most favorable conditions is to be determined.

Assumptions **1** Heat transfer is one-dimensional since the pond is large relative to its depth. **2** Thermal properties, heat transfer coefficients, and the indoor temperatures are constant. **3** Radiation heat transfer is significant. **4** There are no convection currents in the water. **5** The given time step Δt = 15 min is less than the critical time step so that the stability criteria is satisfied. **6** All heat losses from the pond are negligible. **7** Heat generation due to absorption of radiation is uniform in each layer.

Properties The conductivity and diffusivity are given to be k = 0.61 W/m.°C and $\alpha = 0.15\times10^{-6}\ \text{m}^2/\text{s}$. The volumetric absorption coefficients of water are as given in the problem.



Analysis The nodal spacing is given to be Δx = 0.25 m. Then the number of nodes becomes $M = L/\Delta x + 1$ = 1/0.25+1 = 4. This problem involves 5 unknown nodal temperatures, and thus we need to have 5 equations. Nodes 2, 3, and 4 are interior nodes, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

$$\rightarrow\ T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i + \tau\frac{\dot{g}_m^i \Delta x^2}{k}$$

Node 0 can also be treated as an interior node by using the mirror image concept. The finite difference equation for node 4 subjected to heat flux is obtained from an energy balance by taking the direction of all heat transfers to be towards the node:

$$\text{Node 0 (insulation)}: \quad T_0^{i+1} = \tau(T_1^i + T_1^i) + (1-2\tau)T_0^i + \tau\dot{g}_0(\Delta x)^2/k$$

$$\text{Node 0 (insulation)}: \quad T_1^{i+1} = \tau(T_0^i + T_2^i) + (1-2\tau)T_1^i + \tau\dot{g}_1(\Delta x)^2/k$$

$$\text{Node 2 (interior)}: \quad T_2^{i+1} = \tau(T_1^i + T_3^i) + (1-2\tau)T_2^i + \tau\dot{g}_2(\Delta x)^2/k$$

$$\text{Node 3 (interior)}: \quad T_3^{i+1} = \tau(T_2^i + T_4^i) + (1-2\tau)T_3^i + \tau\dot{g}_3(\Delta x)^2/k$$

$$\text{Node 6 (convection)}: \quad \dot{q}_\text{b} + k\frac{T_3^i - T_4^i}{\Delta x} + \tau\dot{g}_4(\Delta x)^2/k = \rho\frac{\Delta x}{2}C\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

where k = 0.61 W/m.°C, $\alpha = k/\rho C = 0.15\times10^{-6}\ \text{m}^2/\text{s}$, Δx = 0.25 m, and Δt = 15 min = 900 s. Also, the mesh Fourier number is

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(0.15\times10^{-6}\ \text{m}^2/\text{s})(900\ \text{s})}{(0.25\ \text{ft})^2} = 0.002160$$

The absorption of solar radiation is given to be $\dot{g}(x) = \dot{q}_s(0.859 - 3.415x + 6.704x^2 - 6.339x^3 + 2.278x^4)$ where $\dot{q}_s$ is the solar flux incident on the surface of the pond in W/m^2, and x is the distance form the free surface of the pond, in m. Then the values of heat generation rates at the nodal points are determined to be

Node 0 (x = 0): $\dot{g}_0 = 500(0.859 - 3.415\times0 + 6.704\times0^2 - 6.339\times0^3 + 2.278\times0^4) = 429.5\ \text{W/m}^3$

Node1(x=.25): $\dot{g}_1 = 500(0.859 - 3.415\times0.25 + 6.704\times0.25^2 - 6.339\times0.25^3 + 2.278\times0.25^4) = 167.1\ \text{W/m}^3$

Node 2 (x=0.50): $\dot{g}_2 = 500(0.859 - 3.415\times0.5 + 6.704\times0.5^2 - 6.339\times0.5^3 + 2.278\times0.5^4) = 88.8\ \text{W/m}^3$

Node3(x=.75): $\dot{g}_3 = 500(0.859 - 3.415\times0.75 + 6.704\times0.75^2 - 6.339\times0.75^3 + 2.278\times0.75^4) = 57.6\ \text{W/m}^3$

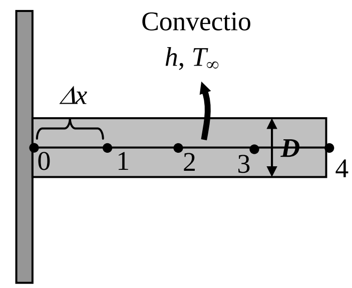
Node 4 (x = 1.00): $\dot{g}_4 = 500(0.859 - 3.415\times1 + 6.704\times1^2 - 6.339\times1^3 + 2.278\times1^4) = 43.5\ \text{W/m}^3$

Also, the heat flux at the bottom surface is $\dot{q}_b = 0.379\times500\ \text{W/m}^2 = 4189.5\ \text{W/m}^2$. Substituting these values, the nodal temperatures in the pond after 4×(60/15) = 16 time steps (4 h) are determined to be

T_0 = 16.5°C, T_1 = 15.6°C, T_2 = 15.3°C, T_3 = 15.3°C, and T_4 = 20.2°C.

5-117 A hot surface is to be cooled by aluminum pin fins. The nodal temperatures after 5 min are to be determined using the explicit finite difference method. Also to be determined is the time it takes for steady conditions to be reached.

Assumptions **1** Heat transfer through the pin fin is given to be one-dimensional. **2** The thermal properties of the fin are constant. **3** Convection heat transfer coefficient is constant and uniform. **4** Radiation heat transfer is negligible. **5** Heat loss from the fin tip is considered.



Analysis The nodal network of this problem consists of 5 nodes, and the base temperature T_0 at node 0 is specified. Therefore, there are 4 unknown nodal temperatures, and we need 4 equations to determine them. Using the energy balance approach and taking the direction of all heat transfers to be towards the node under consideration, the explicit transient finite difference formulations become

Node 1 (interior): $$hp\Delta x(T_\infty - T_1^i) + kA\frac{T_2^i - T_1^i}{\Delta x} + kA\frac{T_0 - T_1^i}{\Delta x} = \rho A\Delta xC\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

Node 2 (interior): $$hp\Delta x(T_\infty - T_2^i) + kA\frac{T_3^i - T_2^i}{\Delta x} + kA\frac{T_1^i - T_2^i}{\Delta x} = \rho A\Delta xC\frac{T_2^{i+1} - T_2^i}{\Delta t}$$

Node 3 (interior): $$hp\Delta x(T_\infty - T_3^i) + kA\frac{T_4^i - T_3^i}{\Delta x} + kA\frac{T_2^i - T_3^i}{\Delta x} = \rho A\Delta xC\frac{T_3^{i+1} - T_3^i}{\Delta t}$$

Node 4 (fin tip): $$h(p\Delta x/2 + A)(T_\infty - T_4^i) + kA\frac{T_3^i - T_4^i}{\Delta x} = \rho A(\Delta x/2)C\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

where $A = \pi D^2/4$ is the cross-sectional area and $p = \pi D$ is the perimeter of the fin. Also, D = 0.008 m, k = 237 W/m.°C, $\alpha = k/\rho C = 97.1\times10^{-6}\ \text{m}^2/\text{s}$, Δx = 0.02 m, T_∞ = 30°C, T_0 = 120°C h_o = 35 W/m^2.°C, and Δt = 1 s. Also, the mesh Fourier number is

$$\tau = \frac{\alpha\Delta t}{\Delta x^2} = \frac{(97.1\times10^{-6}\ \text{m}^2/\text{s})(1\ \text{s})}{(0.02\ \text{m})^2} = 0.24275$$

Substituting these values, the nodal temperatures along the fin after 5×60 = 300 time steps (4 h) are determined to be

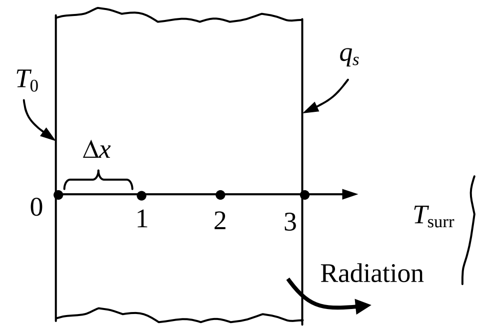
T_0 = 120°C, T_1 = 110.6°C, T_2 = 103.9°C, T_3 = 100.0°C, and T_4 = 98.5°C.

Printing the temperatures after each time step and examining them, we observe that the nodal temperatures stop changing after about 3.8 min. Thus we conclude that steady conditions are reached after **3.8 min.**

5-118E A plane wall in space is subjected to specified temperature on one side and radiation and heat flux on the other. The finite difference formulation of this problem is to be obtained, and the nodal temperatures under steady conditions are to be determined.

Assumptions **1** Heat transfer through the wall is given to be steady and one-dimensional. **2** Thermal conductivity is constant. **3** There is no heat generation. **4** There is no convection in space.

Properties The properties of the wall are given to be k=1.2 W/m·°C, ε = 0.80, and α_s = 0.45.



Analysis The nodal spacing is given to be Δx = 0.1 ft. Then the number of nodes becomes $M = L/\Delta x + 1$ = 0.3/0.1+1 = 4. The left surface temperature is given to be T_0 = 520 R = 60°F. This problem involves 3 unknown nodal temperatures, and thus we need to have 3 equations to determine them uniquely. Nodes 1 and 2 are interior nodes, and thus for them we can use the general finite difference relation expressed as

$$\frac{T_{m-1} - 2T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{g}_m}{k} = 0 \rightarrow T_{m-1} - 2T_m + T_{m+1} = 0 \quad (\text{since } \dot{g} = 0), \quad \text{for } m = 1 \text{ and } 2$$

The finite difference equation for node 3 on the right surface subjected to convection and solar heat flux is obtained by applying an energy balance on the half volume element about node 3 and taking the direction of all heat transfers to be towards the node under consideration:

$$\text{Node 1 (interior)}: \qquad T_0 - 2T_1 + T_2 = 0$$

$$\text{Node 2 (interior)}: \qquad T_1 - 2T_2 + T_3 = 0$$

$$\text{Node 3 (right surface)}: \qquad \alpha_s \dot{q}_s + \varepsilon\sigma[T_{\text{space}}^4 - (T_3 + 460)^4] + k\frac{T_2 - T_3}{\Delta x} = 0$$

where k = 1.2 Btu/h.ft.°F, ε = 0.80, α_s = 0.45, $\dot{q}_s = 300 \text{ Btu/h.ft}^2$, T_{space} = 0 R, and σ = 0.1714 Btu/h.ft^2.R^4 The system of 3 equations with 3 unknown temperatures constitute the finite difference formulation of the problem.

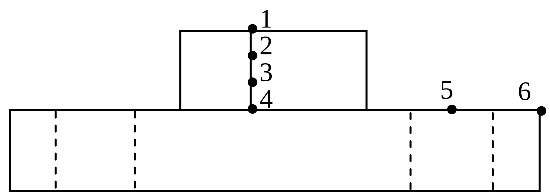
(*b*) The nodal temperatures under steady conditions are determined by solving the 3 equations above simultaneously with an equation solver to be

T_1 = 62.4°F = 522.4 R, T_2 = 64.8°F = 524.8 R, and T_3 = 67.3°F = 527.3 R

5-119 Frozen steaks are to be defrosted by placing them on a black-anodized circular aluminum plate. Using the explicit method, the time it takes to defrost the steaks is to be determined.

Assumptions **1** Heat transfer in both the steaks and the defrosting plate is one-dimensional since heat transfer from the lateral surfaces is negligible. **2** Thermal properties, heat transfer coefficients, and the surrounding air and surface temperatures remain constant during defrosting. **3** Heat transfer through the bottom surface of the plate is negligible. **4** The thermal contact resistance between the steaks and the plate is negligible. **5** Evaporation from the steaks and thus evaporative cooling is negligible. **6** The heat storage capacity of the plate is small relative to the amount of total heat transferred to the steak, and thus the heat transferred to the plate can be assumed to be transferred to the steak.

Properties The thermal properties of the steaks are ρ = 970 kg/m^3, C_p = 1.55 kJ/kg.°C, k = 1.40 W/m.°C, $\alpha = 0.93\times10^{-6}$ m^2/s, ε = 0.95, and h_{if} = 187 kJ/kg. The thermal properties of the defrosting plate are k = 237 W/m.°C, $\alpha = 97.1\times10^{-6}$ m^2/s, and ε = 0.90. The ρC_p (volumetric specific heat) values of the steaks and of the defrosting plate are



$$(\rho C_p)_{\text{plate}} = \frac{k}{\alpha} = \frac{237\ \text{W/m}\cdot{}^\circ\text{C}}{97.1\times10^{-6}\ \text{m}^2/s} = 2441\,\text{kW/m}^3\cdot{}^\circ\text{C}$$

$$(\rho C_p)_{\text{steak}} = (970\ \text{kg/m}^3)(1.55\ \text{kJ/kg}\cdot{}^\circ\text{C}) = 1504\ \text{kW/m}^3\cdot{}^\circ\text{C}$$

Analysis The nodal spacing is given to be Δx = 0.005 m in the steaks, and Δr = 0.0375 m in the plate. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations. Nodes 2 and 3 are interior nodes in a plain wall, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \ \rightarrow\ T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i$$

The finite difference equations for other nodes are obtained from an energy balance by taking the direction of all heat transfers to be towards the node under consideration:

Node 1: $$h(T_\infty - T_1^i) + \varepsilon_{\text{steak}}\sigma[(T_\infty + 273)^4 - (T_1^i + 273)^4] + k_{\text{steak}}\frac{T_2^i - T_1^i}{\Delta x} = (\rho C)_{\text{steak}}\frac{\Delta x}{2}\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

Node 2 (interior): $$T_2^{i+1} = \tau_{\text{steak}}(T_1^i + T_3^i) + (1-2\tau_{\text{steak}})T_2^i$$

Node 3 (interior): $$T_3^{i+1} = \tau_{\text{steak}}(T_2^i + T_4^i) + (1-2\tau_{\text{steak}})T_3^i$$

Node 4:

$$\pi(r_{45}^2 - r_4^2)\{h(T_\infty - T_4^i) + \varepsilon_{\text{plate}}\sigma[(T_\infty + 273)^4 - (T_4^i + 273)^4]\} + k_{\text{steak}}(\pi r_4^2)\frac{T_3^i - T_4^i}{\Delta x}$$
$$+ k_{\text{plate}}(2\pi r_{45}\delta)\frac{T_5^i - T_4^i}{\Delta r} = [(\rho C)_{\text{steak}}(\pi r_4^2 \Delta x/2) + (\rho C)_{\text{plate}}(\pi r_{45}^2\delta)]\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

Node 5:

$$2\pi r_5 \Delta r\{h(T_\infty - T_5^i) + \varepsilon_{\text{plate}}\sigma[(T_\infty + 273)^4 - (T_5^i + 273)^4]\}$$
$$+ k_{\text{plate}}(2\pi r_{56}\delta)\frac{T_6^i - T_5^i}{\Delta r} = (\rho C)_{\text{plate}}(\pi r_5^2\delta)]\frac{T_5^{i+1} - T_5^i}{\Delta t}$$

Node 6:

$$2\pi[(r_{56}+r_6)/2](\Delta r/2)\{h(T_\infty - T_6^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_6^i+273)^4]\}$$
$$+k_{\text{plate}}(2\pi r_{56}\delta)\frac{T_5^i-T_6^i}{\Delta r}=(\rho C)_{\text{plate}}[2\pi(r_{56}+r_6)/2](\Delta r/2)\delta\frac{T_6^{i+1}-T_6^i}{\Delta t}$$

where $(\rho C_p)_{\text{plate}}=2441\,\text{kW/m}^3\cdot°\text{C}$, $(\rho C_p)_{\text{steak}}=1504\,\text{kW/m}^3\cdot°\text{C}$, k_{steak} = 1.40 W/m.°C, , $\varepsilon_{\text{steak}}$ = 0.95, $\alpha_{\text{steak}}=0.93\times10^{-6}\ \text{m}^2/\text{s}$, h_{if} = 187 kJ/kg, k_{plate} = 237 W/m.°C, $\alpha_{\text{plate}}=97.1\times10^{-6}\ \text{m}^2/\text{s}$, and $\varepsilon_{\text{plate}}$ = 0.90, T_∞ = 20°C, h =12 W/m^2.°C, δ = 0.01 m, Δx = 0.005 m, Δr = 0.0375 m, and Δt = 5 s. Also, the mesh Fourier number for the steaks is

$$\tau_{\text{steak}}=\frac{\alpha\Delta t}{\Delta x^2}=\frac{(0.93\times10^{-6}\ \text{m}^2/\text{s})(5\ \text{s})}{(0.005\ \text{m})^2}=0.186$$

The various radii are r_4 =0.075 m, r_5 =0.1125 m, r_6 =0.15 m, r_{45} = (0.075+0.1125)/2 m, and r_{56} = (0.1125+0.15)/2 m.

The total amount of heat transfer needed to defrost the steaks is

$$m_{\text{steak}}=\rho V=(970\ \text{kg/m}^3)[\pi(0.075\ \text{m})^2(0.015\ \text{m})]=0.257\ \text{kg}$$

$$Q_{\text{total, steak}}=Q_{\text{sensible}}+Q_{\text{latent}}=(mC\Delta T)_{\text{steak}}+(mh_{\text{if}})_{\text{steak}}$$
$$=(0.257\ \text{kg})(1.55\ \text{kJ/kg.°C})[0-(-18°\text{C})]+(0.257\ \text{kg})(187\ \text{kJ/kg})=55.2\ \text{kJ}$$

The amount of heat transfer to the steak during a time step i is the sum of the heat transferred to the steak directly from its top surface, and indirectly through the plate, and is expressed as

$$\begin{aligned}Q_{\text{steak}}^i&=2\pi r_5\Delta r\{h(T_\infty-T_5^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_5^i+273)^4]\}\\&+2\pi[(r_{56}+r_6)/2](\Delta r/2)\{h(T_\infty-T_6^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_6^i+273)^4]\}\\&+\pi(r_{45}^2-r_4^2)\{h(T_\infty-T_4^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_4^i+273)^4]\}\\&+\pi r_1^2\{h(T_\infty-T_1^i)+\varepsilon_{\text{steak}}\sigma[(T_\infty+273)^4-(T_1^i+273)^4]\}\end{aligned}$$

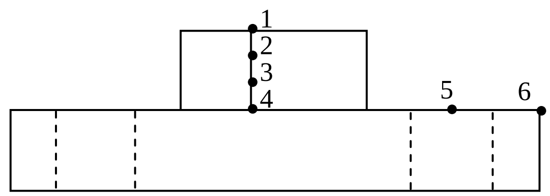
The defrosting time is determined by finding the amount of heat transfer during each time step, and adding them up until we obtain 55.2 kJ. Multiplying the number of time steps N by the time step Δt = 5 s will give the defrosting time. In this case it is determined to be

$\Delta t_{\text{defrost}}$ = NΔt = 44(5 s) = **220 s**

5-120 Frozen steaks at -18°C are to be defrosted by placing them on a 1-cm thick black-anodized circular copper defrosting plate. Using the explicit finite difference method, the time it takes to defrost the steaks is to be determined.

Assumptions **1** Heat transfer in both the steaks and the defrosting plate is one-dimensional since heat transfer from the lateral surfaces is negligible. **2** Thermal properties, heat transfer coefficients, and the surrounding air and surface temperatures remain constant during defrosting. **3** Heat transfer through the bottom surface of the plate is negligible. **4** The thermal contact resistance between the steaks and the plate is negligible. **5** Evaporation from the steaks and thus evaporative cooling is negligible. **6** The heat storage capacity of the plate is small relative to the amount of total heat transferred to the steak, and thus the heat transferred to the plate can be assumed to be transferred to the steak.

Properties The thermal properties of the steaks are ρ = 970 kg/m^3, C_p = 1.55 kJ/kg.°C, k = 1.40 W/m.°C, $\alpha = 0.93\times10^{-6}$ m^2/s, ε = 0.95, and h_{if} = 187 kJ/kg. The thermal properties of the defrosting plate are k = 401 W/m.°C, $\alpha = 117\times10^{-6}$ m^2/s, and ε = 0.90 (Table A-3). The ρC_p (volumetric specific heat) values of the steaks and of the defrosting plate are



$$(\rho C_p)_{\text{plate}} = (8933\text{ kg/m}^3)(0.385\text{ kJ/kg}\cdot{}^\circ\text{C}) = 3439\text{ kW/m}^3\cdot{}^\circ\text{C}$$
$$(\rho C_p)_{\text{steak}} = (970\text{ kg/m}^3)(1.55\text{ kJ/kg}\cdot{}^\circ\text{C}) = 1504\text{ kW/m}^3\cdot{}^\circ\text{C}$$

Analysis The nodal spacing is given to be Δx = 0.005 m in the steaks, and Δr = 0.0375 m in the plate. This problem involves 6 unknown nodal temperatures, and thus we need to have 6 equations. Nodes 2 and 3 are interior nodes in a plain wall, and thus for them we can use the general explicit finite difference relation expressed as

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{g}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \;\rightarrow\; T_m^{i+1} = \tau(T_{m-1}^i + T_{m+1}^i) + (1-2\tau)T_m^i$$

The finite difference equations for other nodes are obtained from an energy balance by taking the direction of all heat transfers to be towards the node under consideration:

Node 1: $$h(T_\infty - T_1^i) + \varepsilon_{\text{steak}}\sigma[(T_\infty + 273)^4 - (T_1^i + 273)^4] + k_{\text{steak}}\frac{T_2^i - T_1^i}{\Delta x} = (\rho C)_{\text{steak}}\frac{\Delta x}{2}\frac{T_1^{i+1} - T_1^i}{\Delta t}$$

$$\text{Node 2 (interior)}: \quad T_2^{i+1} = \tau_{\text{steak}}(T_1^i + T_3^i) + (1-2\tau_{\text{steak}})T_2^i$$
$$\text{Node 3 (interior)}: \quad T_3^{i+1} = \tau_{\text{steak}}(T_2^i + T_4^i) + (1-2\tau_{\text{steak}})T_3^i$$

Node 4:

$$\pi(r_{45}^2 - r_4^2)\{h(T_\infty - T_4^i) + \varepsilon_{\text{plate}}\sigma[(T_\infty + 273)^4 - (T_4^i + 273)^4]\} + k_{\text{steak}}(\pi r_4^2)\frac{T_3^i - T_4^i}{\Delta x}$$
$$+ k_{\text{plate}}(2\pi r_{45}\delta)\frac{T_5^i - T_4^i}{\Delta r} = [(\rho C)_{\text{steak}}(\pi r_4^2 \Delta x/2) + (\rho C)_{\text{plate}}(\pi r_{45}^2\delta)]\frac{T_4^{i+1} - T_4^i}{\Delta t}$$

Node 5:

$$2\pi r_5 \Delta r\{h(T_\infty - T_5^i) + \varepsilon_{\text{plate}}\sigma[(T_\infty + 273)^4 - (T_5^i + 273)^4]\}$$
$$+ k_{\text{plate}}(2\pi r_{56}\delta)\frac{T_6^i - T_5^i}{\Delta r} = (\rho C)_{\text{plate}}(\pi r_5^2\delta)]\frac{T_5^{i+1} - T_5^i}{\Delta t}$$

Node 6:

$$2\pi[(r_{56}+r_6)/2](\Delta r/2)\{h(T_\infty - T_6^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_6^i+273)^4]\}$$
$$+k_{\text{plate}}(2\pi r_{56}\delta)\frac{T_5^i-T_6^i}{\Delta r}=(\rho C)_{\text{plate}}[2\pi(r_{56}+r_6)/2](\Delta r/2)\delta\frac{T_6^{i+1}-T_6^i}{\Delta t}$$

where $(\rho C_p)_{\text{plate}}=3439\ \text{kW/m}^3\cdot°\text{C}$, $(\rho C_p)_{\text{steak}}=1504\ \text{kW/m}^3\cdot°\text{C}$, k_{steak} = 1.40 W/m.°C, $\varepsilon_{\text{steak}}$ = 0.95, $\alpha_{\text{steak}}=0.93\times10^{-6}\ \text{m}^2/\text{s}$, h_{if} = 187 kJ/kg, k_{plate} = 401 W/m.°C, $\alpha_{\text{plate}}=117\times10^{-6}\ \text{m}^2/\text{s}$, and $\varepsilon_{\text{plate}}$ = 0.90, T_∞ = 20°C, h =12 W/m^2.°C, δ = 0.01 m, Δx = 0.005 m, Δr = 0.0375 m, and Δt = 5 s. Also, the mesh Fourier number for the steaks is

$$\tau_{\text{steak}}=\frac{\alpha\Delta t}{\Delta x^2}=\frac{(0.93\times10^{-6}\ \text{m}^2/\text{s})(5\ \text{s})}{(0.005\ \text{m})^2}=0.186$$

The various radii are r_4 =0.075 m, r_5 =0.1125 m, r_6 =0.15 m, r_{45} = (0.075+0.1125)/2 m, and r_{56} = (0.1125+0.15)/2 m.

The total amount of heat transfer needed to defrost the steaks is

$$m_{\text{steak}}=\rho V=(970\ \text{kg/m}^3)[\pi(0.075\ \text{m})^2(0.015\ \text{m})]=0.257\ \text{kg}$$

$$Q_{\text{total, steak}}=Q_{\text{sensible}}+Q_{\text{latent}}=(mC\Delta T)_{\text{steak}}+(mh_{\text{if}})_{\text{steak}}$$
$$=(0.257\ \text{kg})(1.55\ \text{kJ/kg.°C})[0-(-18°\text{C})]+(0.257\ \text{kg})(187\ \text{kJ/kg})=55.2\ \text{kJ}$$

The amount of heat transfer to the steak during a time step i is the sum of the heat transferred to the steak directly from its top surface, and indirectly through the plate, and is expressed as

$$Q_{\text{steak}}^i=2\pi r_5\Delta r\{h(T_\infty-T_5^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_5^i+273)^4]\}$$
$$+2\pi[(r_{56}+r_6)/2](\Delta r/2)\{h(T_\infty-T_6^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_6^i+273)^4]\}$$
$$+\pi(r_{45}^2-r_4^2)\{h(T_\infty-T_4^i)+\varepsilon_{\text{plate}}\sigma[(T_\infty+273)^4-(T_4^i+273)^4]\}$$
$$+\pi r_1^2\{h(T_\infty-T_1^i)+\varepsilon_{\text{steak}}\sigma[(T_\infty+273)^4-(T_1^i+273)^4]\}$$

The defrosting time is determined by finding the amount of heat transfer during each time step, and adding them up until we obtain 55.2 kJ. Multiplying the number of time steps N by the time step Δt = 5 s will give the defrosting time. In this case it is determined to be

$\Delta t_{\text{defrost}}$ = NΔt = 47(5 s) = **235 s**

5-121 ….. 5-124 Design and Essay Problems
